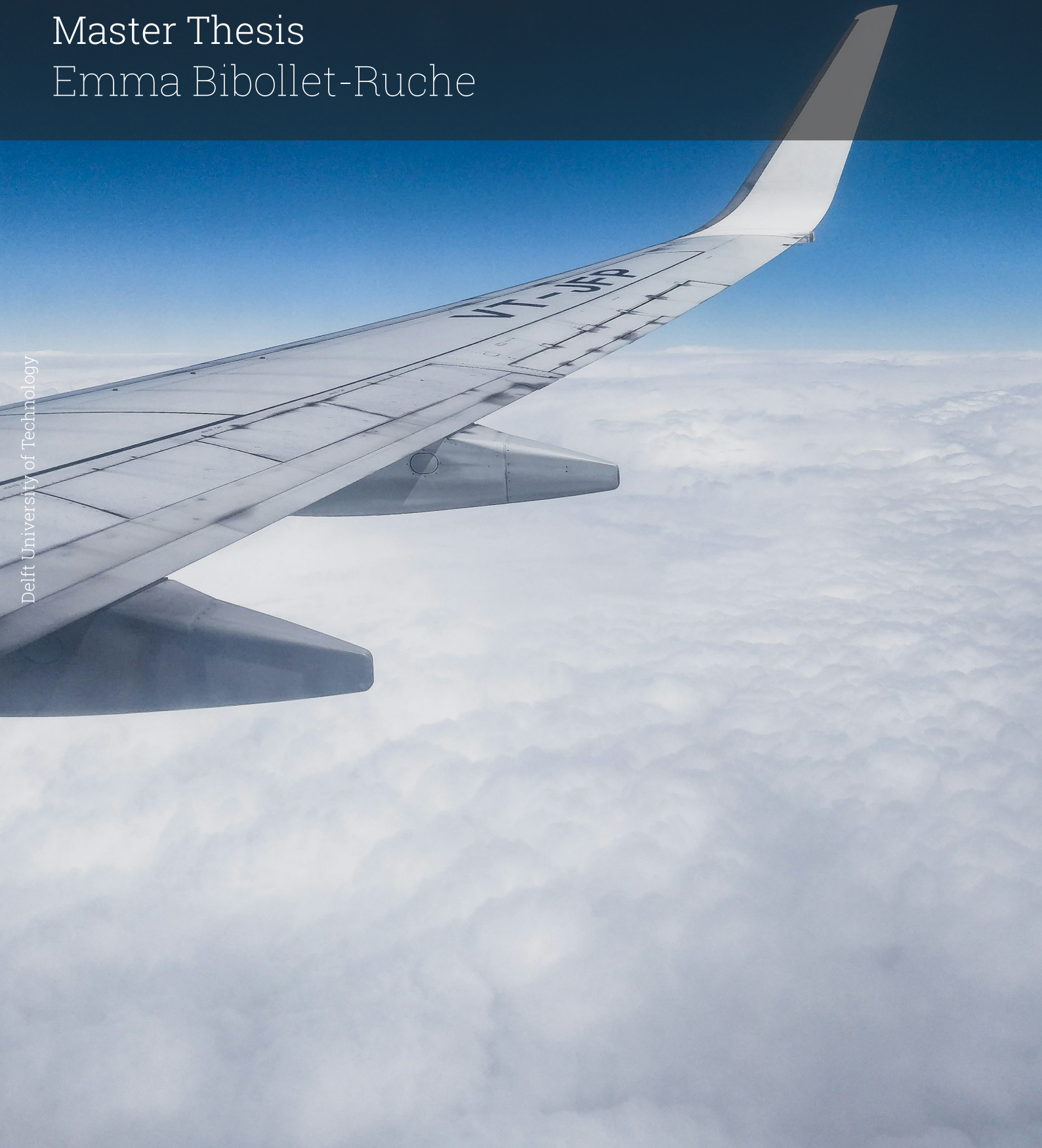


Wing Flutter Control Using Passive Devices

Master Thesis

Emma Bibollet-Ruche



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by

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Preface

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*Emma Bibollet-Ruche
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Abstract

The Flap-NES aims to control flutter by using a portion of the existing flap as a nonlinear energy sink (NES) and facilitate Targeted Energy Transfer (TET). TET is a one-way irreversible energy channelling from the primary oscillating wing structure into the Flap-NES. There are three possible TET mechanisms: recurring burstout and suppression, intermediate suppression and complete elimination.

The objective of this work is to evaluate the Flap-NES, possessing both linear and cubic stiffness, in controlling flutter of a 2-Dimensional (2D) linear typical section under linear unsteady aerodynamics.

The system is modelled as a 2D 3-Degree of freedom (DOF) nonlinear aeroelastic system featuring cubic stiffness in the flap and subjected to unsteady aerodynamics (Theodorsen). Initial linear stability analysis uses flutter diagrams to identify underlying flutter mechanisms and Hopf bifurcation airspeeds. The post-flutter dynamic response is obtained through the Harmonic Balance Method combined with numerical continuation, with stability assessed via the Hill's method.

Deviating from traditional NES tuning strategies, the Flap-NES's linear stiffness cannot be zero. As the Flap-NES is continuously submitted to aerodynamic loading, it requires a non-zero linear stiffness.

Although flutter suppression was not sustained, the successful initiation of intermediate suppression and recurring burstout and suppression TET was achieved when the natural frequency of the Flap-NES was greater than the heave and pitch modes, establishing a viable design foundation for future iterations.

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Nomenclature

Abbreviations

Abbreviation	Definition
2D	Two-Dimensional
3D	Three-Dimensional
AFT	Alternating Frequency/Time
BP	Branch Point
CG	Center of Gravity
DOF	Degrees of Freedom
DFT	Discrete Fourier Transform
EA	Elastic Axis
FFT	Fast Fourier Transform
HBM	Harmonic Balance Method
ISAE	Institut Supérieur de l'Aéronautique et de l'Espace
LCO	Limit Cycle Oscillation
LP	Limit Point
LUVLM	Linearised Unsteady Vortex Lattice Method
MATCONT	Numerical Continuation Software
MDOF	Multi-Degree of Freedom
NACA	National Advisory Committee for Aeronautics
NES	Non-Linear Energy Sink
NLTVA	Nonlinear Tuned Vibration Absorber
NS	Neimark-Sacker
NSGA	Non-dominated Sorting Genetic Algorithm
ODE	Ordinary Differential Equation
PD	Period Doubling
PRC	Permanent Resonance Capture
RC	Resonance Capture
RMS	Root Mean Square
SDOF	Single Degree of Freedom
TET	Targeted Energy Transfer
TMD	Tuned Mass Damper
TRC	Transient Resonance Capture
URANS	Unsteady Reynolds-Averaged Navier-Stokes
UVLM	Unsteady Vortex Lattice Method
WT	Wavelet Transform

Symbol	Definition	Unit
A	Structural mass matrix	[-]
A*	Non-dimensional structural mass matrix	[-]
a_h	Distance coeff EA to midchord	[-]
B	Aerodynamic mass matrix	[-]
B*	Non-dimensional aerodynamic mass matrix	[-]
b	Semi-chord	[m]
c	Chord	[m]
c_h	Distance coeff EA to flap hinge	[-]
C, C_i	Damping matrix & coefficients	[-]
C	Non-dimensional damping Matrix	[-]
C_L	Aerodynamic lift coefficient	[-]
$C_{M\alpha}, C_{M\beta}$	Aerodynamic moment coefficients	[-]
c_β^*	Non-dimensional damping Flap-NES	[-]
$C(k), F(k), G(k)$	Theodorsen function & its real/imaginary parts	[-]
D	Aerodynamic damping matrix	[-]
dU^*	Airspeed continuation step increment	[-]
dX	Displacement continuation step increment	[-]
E, K_i	Linear stiffness matrix & coefficients	[-], [N/m]
E*	Non-dimensional linear stiffness matrix	[-]
$\mathbf{E}_j^T$	Phase constraint vector	[-]
F	Aerodynamic stiffness matrix	[-]
F*	Non-dimensional aerodynamic stiffness matrix	[-]
F_{nl}, f_{nl}	Nonlinear force vectors (freq. & time domain)	[-]
g	Initial condition excitation vector	[-]
H	Hill's matrix	[-]
H	Number of included harmonics	[-]
h, ξ	Heave & Non-dimensional heave	[m], [-]
I, S	Second and first moments of inertia	[kg·m ²], [kg·m]
I_n	Identity matrix of size n	[-]
K_{nl}, K_i^{nl}	Nonlinear stiffness matrix & coefficients	[-], [N/m]
K_{nl}*	Non-dimensional nonlinear stiffness matrix	[-]
k	Reduced frequency / Iteration index	[-]
L, M_i	Aerodynamic lift & moments ($i = \alpha, \beta$)	[N], [N·m]
L^c	Circulatory lift	[N]
M	Mass matrix	[-]
M	Monodromy matrix	[-]
m	Mass	[kg]
N	Time samples	[-]
n	Number of DOFs	[-]
P	Excitation force vector	[-]
Q	Linear state matrix	[-]
R	Residual error vector	[-]
r_i	Radius of gyration ($i = \alpha, \beta$)	[-]
T	Trigonometric basis vector	[-]
T	Theodorsen coeffs.	[-]
t	Tangent predictor vector	[-]
t	Time	[s]
U, U^*	Airspeed (dimensional, non-dimensional)	[m/s], [-]
U_f^*	Non-dimensional Hopf bifurcation airspeed	[-]
$U_{f,1}^*, U_{f,2}^*$	First and second non-dimensional Hopf bifurcation airspeeds	[-]

Symbol	Definition	Unit
U_{NS}^*	Non-dimensional Neimark-Sacker bifurcation airspeed	[-]
$\mathbf{W}$	Aerodynamic State Influence Matrix	[-]
$\mathbf{W}^*$	Non-dimensional aerodynamic state influ- ence matrix	[-]
w, w_i	Aerodynamic downwash & lag states ($i =$ $1, \dots, 6$)	[-]
$\mathbf{X}$	Displacement vector	[-]
$\mathbf{x}$	State vector	[-]
x_i	Static Unbalance ($i = \alpha, \beta$)	[-]
$\mathbf{x}_p$	Known periodic solution	[-]
$\mathbf{y}$	Displacement vector	[-]
$\mathbf{y}^*$	Non-dimensional displacement vector	[-]
$\mathbf{Z}$	Frequency-domain dynamic stiffness matrix	[-]
α	Pitch	[rad]
β	Flap	[rad]
$\mathbf{\Gamma}$	DFT matrix	[-]
Γ_β	Non-dimensional Flap-NES nonlinear stiff- ness	[-]
Δs	Continuation step size	[-]
$\delta \mathbf{X}$	Amplitude correction	[-]
ϵ_{NR}	Newton-Raphson convergence tolerance	[-]
ϵ_{stab}	Stability tolerance	[-]
$\epsilon_{1,2}, \Psi_{1,2}$	Jones approximation coefficients	[-]
ζ, σ	Damping ratio & Real part of the eigenvalue	[-]
κ_i	Non-dimensional nonlinear stiffness ratio ($i =$ ξ, α)	[-]
λ_i	Floquet exponents	[-]
μ	Flap hinge angle parameter ($\arccos c_h$)	[-]
μ_i	Floquet multipliers	[-]
∇	Block-diagonal derivative operator	[-]
Ω_1, Ω_2	Non-dimensional frequency ratios (heave/pitch, flap/pitch)	[-]
ω	Frequency	[Hz]
ρ	Atmospheric density	[kg/m ³]
τ	Time (non-dim)	[-]
τ_h	Hopf test function	[-]
θ	Discretized angle for AFT method	[rad]
$\phi(t)$	Wagner function	[-]

1

Introduction

Modern aircraft are becoming more lightweight and more flexible. This increased flexibility modifies the aircraft's dynamics, producing instabilities that can aggravate its performance. Such phenomena are due to the mutual interaction between the inertial, elastic and aerodynamic forces, also known as aeroelasticity.

Aeroelastic flutter is a dynamic instability characterised by self-excited oscillations due to the coupling of the structure with the fluid flow. Beyond the linear flutter speed, structural nonlinearities associated with increased flexibility lead to the emergence of limit cycle oscillations (LCOs). LCOs represent a steady-state response of the system where the oscillation maintains a periodic amplitude at a given airspeed. As the airspeed increases, LCO amplitudes grow, leading to material fatigue, decreased flight performance, and potential structural failure.

Flutter suppression techniques are generally categorised into two schools: active and passive solutions. Active solutions rely on sensors and actuators, such as morphing wings or moving control surfaces, to counteract instability. Passive solutions do not depend on external sensors or actuators, but rather rely on inherent damping and structural properties.

The Flap-NES is a passive control solution in which a portion of the control surface functions as a Nonlinear Energy Sink (NES), characterised by its cubic stiffness, to facilitate Targeted Energy Transfer (TET). TET is the one-way, irreversible channeling of energy from the primary oscillating wing to the NES. The NES absorbs, traps and dissipates energy, preventing it from returning to the wing. This dissipation manifests itself through distinct mechanisms which must be defined for the Flap-NES.

The primary objective of this master's thesis is to evaluate the Flap-NES's capacity to mitigate self-excited flutter when its linear stiffness contribution is included. This involves analysing the airfoil's linear stability to characterise the flutter mode and linear flutter speed (Hopf bifurcation). Post-flutter nonlinear dynamics are then assessed via bifurcation diagrams to track the LCO amplitude of each degree of freedom (DOF) against airspeed.

The scope is restricted to a 2D airfoil in linear unsteady aerodynamics (Theodorsen's theory), where the Flap-NES spans the full control surface.

Chapter 2: Theoretical Background establishes the theoretical background and introduces the Flap-NES concept. **Chapter 3 - Literature Review** examines existing literature on the passive control of flutter, identifies current research gaps, and establishes the research objectives and hypotheses. **Chapter 4 - Methodology** outlines the theoretical models and numerical tools used to assess the flutter suppression capabilities of the system. **Chapter 5 - Results** details the findings of this study. **Chapter 6 - Conclusion** provides a conclusion and an overview of the outcomes. Finally, **Chapter 7 - Perspectives** offers guidelines and recommendations for future work on this topic.

Theoretical Background

This chapter establishes the theoretical foundation of flutter in nonlinear aeroelastic systems (Sections 2.1 to 2.3) and presents existing active and passive flutter control solutions (Section 2.4). This progression provides the necessary framework to introduce the Flap-NES as a passive flutter control device.

2.1. Aeroelasticity

Aeroelasticity studies the interactions between aerodynamic flows and elastic structures. While these interactions cause both static and dynamic stability problems, this thesis focuses on the dynamic instability known as flutter. Dynamic aeroelasticity involves the coupling of elastic, aerodynamic, and inertial forces, often represented by Collar's triangle (Figure 2.1). Consequently, aeroelasticity remains a critical constraint in aerospace design [9, 13].

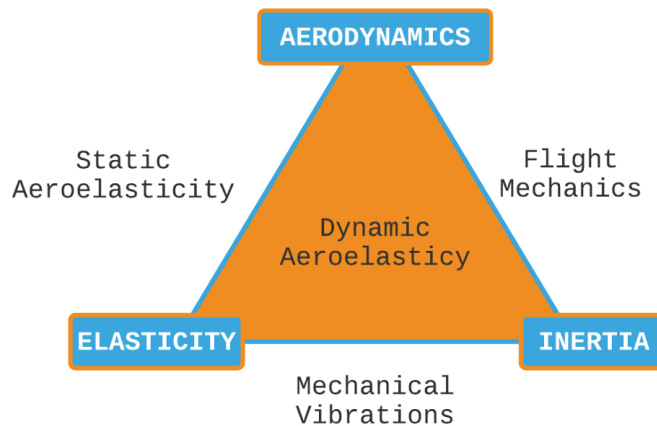


Figure 2.1: Collar's triangle [9]

2.2. Flutter

Flutter is characterised by self-sustained or divergent oscillations that occur when the aerodynamic flow injects more energy into a structure than it can dissipate. As illustrated in Figure 2.2, the flow does work on the structure when lift and vertical velocity act in the same direction. Conversely, energy is extracted when lift opposes motion. The system becomes unstable when the average energy transfer over a time period from the flow to the structure is positive [13].

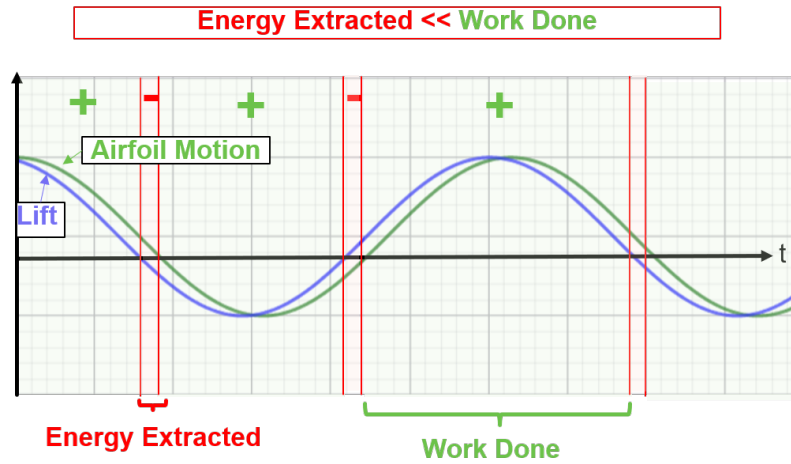


Figure 2.2: Energy balance condition for flutter

2.2.1. Classical Flutter: 2-DOF

Flutter diagrams identify the onset of aeroelastic instability and characterise the underlying mechanism by plotting modal damping and frequency against airspeed. The aerodynamic stiffness and damping are proportional to airspeed and airspeed squared, respectively (Section 4.2), therefore the aeroelastic eigenfrequencies change with airspeed [13].

The frequency plot reveals frequency coalescence: the convergence of two modal frequencies as aerodynamic forces increase. This modal coupling drives the energy exchange mechanism described previously. However, the frequency plot must be analysed alongside the damping plot, as flutter only occurs when a mode's damping ratio, ζ , becomes negative¹ [13].

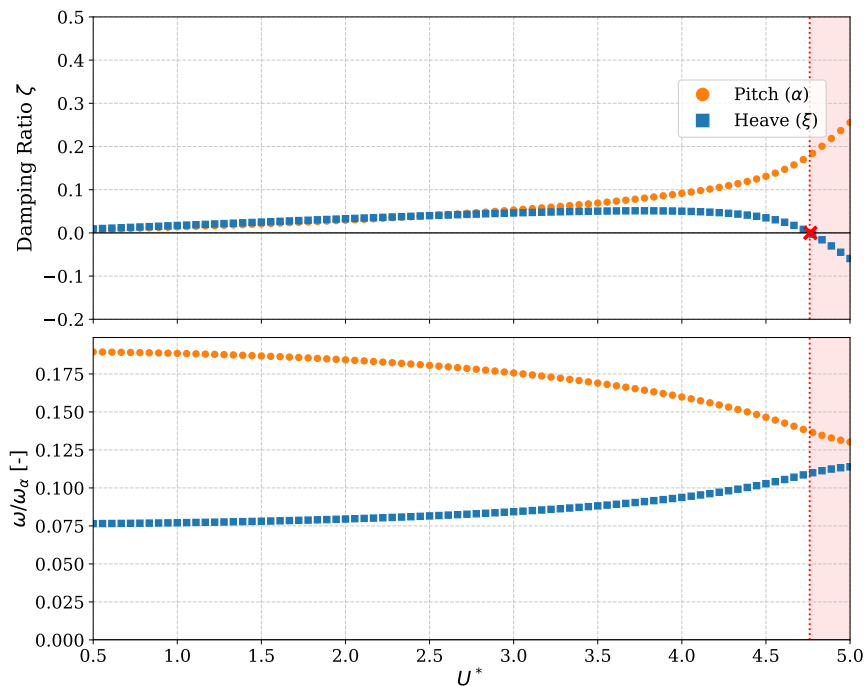


Figure 2.3: Flutter diagram: 2-DOF system exhibiting classical flutter

Parameters: $\mu = 100$, $a_h = -0.5$, $x_\alpha = 0.25$, $r_\alpha = 0.5$, $\Omega_1 = 0.5$, $\zeta_h = \zeta_\alpha = 0.002$.

¹The damping ratio is defined as $\zeta = -\frac{\sigma}{\omega_n}$, where σ is the real part of the complex eigenvalue

2.2.2. Single-DOF (SDOF) Flutter

While classical flutter arises from the coupling of two structural modes, non-classical SDOF flutter is driven by unsteady aerodynamic effects [2]. In an unsteady flow regime, the continuous shedding of vorticity creates a time-varying wake trailing behind the oscillating airfoil (Figure 2.4). As these shed vortices travel downstream, their induced velocity field interferes with the wing, causing an amplitude reduction and, critically, a temporal phase lag in the resultant aerodynamic forces [13].

This phase lag shifts the aerodynamic forces into alignment with the airfoil motion. Consequently, the airstream performs net positive work on the structure over a complete oscillation cycle (Figure 2.2), driving an instability. The flutter speed corresponds to the airspeed when the mode's damping ratio becomes negative [13].

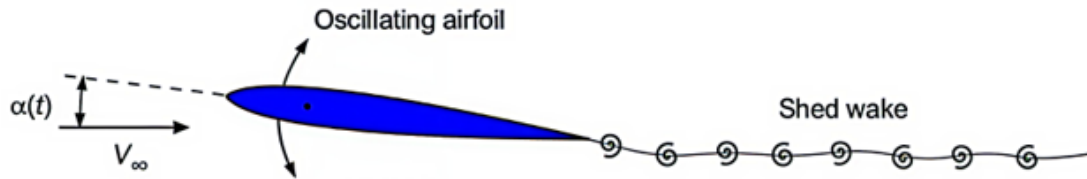


Figure 2.4: Unsteady aerodynamics: Effect of the shed wake under a changing angle of attack [28]

2.3. Flutter in Nonlinear Aeroelastic Systems

Driven by the need for fuel efficiency, modern aircraft design favors lightweight, highly flexible, high-aspect-ratio wings (Figure 2.5 and 2.6) [1, 9, 10].

Unlike traditional stiff wings that experience small deflections, high-aspect-ratio wings undergo significant deformation. The structure no longer behaves linearly; instead, it exhibits a hardening structural nonlinearity where stiffness increases with deflection (Figure 2.9). This behaviour is mathematically modelled using a cubic stiffness term [9, 10].



Figure 2.5: Helios high-altitude flying wing by NASA [21]



Figure 2.6: High aspect ratio wings on Boeing Sugar Volt

Structural nonlinearity alters the post-flutter aeroelastic response. Whereas a purely linear model predicts unbounded oscillatory growth, the introduction of cubic stiffness restricts the displacement. Consequently, the system converges to a sustained, constant-amplitude vibration known as a Limit Cycle Oscillation (LCO) (Figure 2.7) [3].

The transition from a stable static equilibrium to an LCO is defined as a Hopf bifurcation. Because this bifurcation coincides with the linear flutter airspeed, the terms are often used interchangeably in nonlinear aeroelastic literature [3].

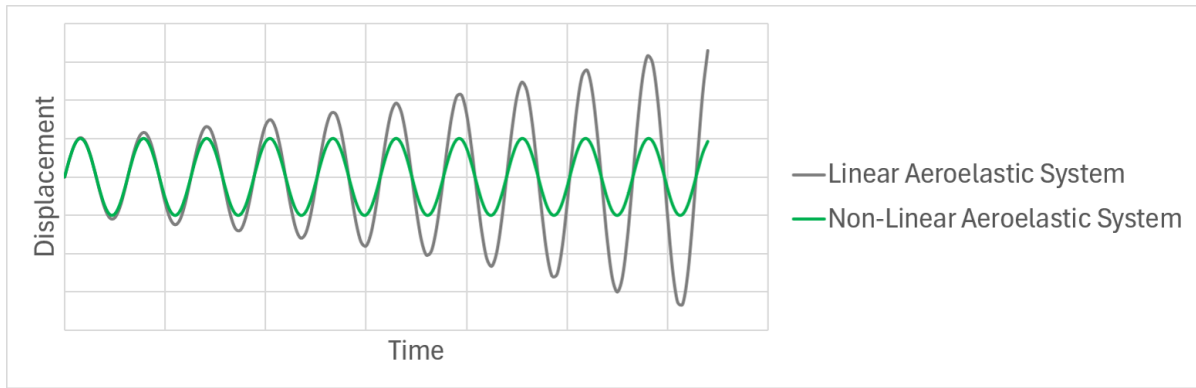


Figure 2.7: Post-Flutter: Linear vs. nonlinear response

Depending on the nature of the flutter mechanism and structural nonlinearities, the aeroelastic system will undergo one of two bifurcation types at the linear flutter boundary:

Supercritical Hopf Bifurcation: The stable focus loses stability, transitioning into an unstable focus, and a stable LCO emerges. This stable periodic branch extends forward, growing continuously in amplitude as the airspeed increases beyond the linear flutter boundary (Figure 2.8) [3].

Subcritical Hopf Bifurcation: An unstable focus becomes a stable focus, and an unstable limit cycle emerges [3]. This unstable periodic branch folds backwards into the subcritical region, coexisting with the stable aeroelastic equilibrium below the linear flutter speed (Figure 2.8). In practice, an unstable LCO cannot be sustained; instead, it acts as a boundary. If a perturbation exceeds the amplitude threshold of the unstable limit cycle branch (dashed line), the system diverges and settles onto the nearest stable branch. The primary danger of a subcritical Hopf bifurcation lies in this behaviour: the sudden jump to a high-amplitude stable state can lead to structural failure.

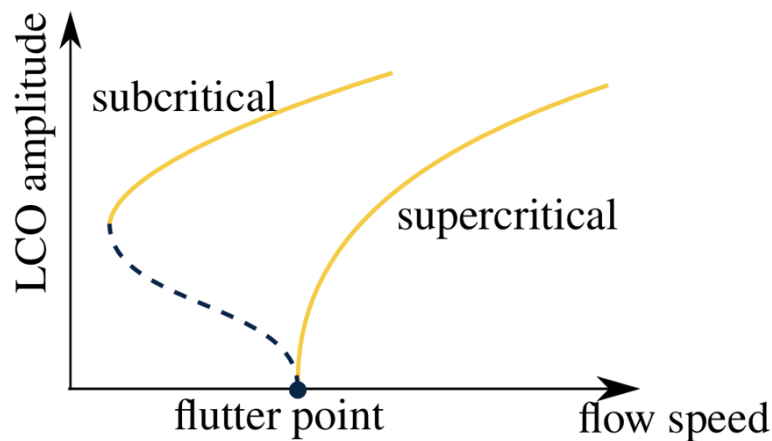


Figure 2.8: Hopf Bifurcation: Supercritical vs subcritical; Solid/dashed line: stable/unstable branch [10]

2.4. Flutter Suppression: Active and Passive Solutions

Flutter induces critical structural loads that can cause catastrophic failure, thereby restricting the operational flight envelope. Both active and passive solutions are used to mitigate these effects.

Active flutter suppression relies on sensors to measure structural accelerations or displacements, feeding a powered actuator that generates counteracting forces. This is typically implemented using existing control surfaces or morphing wing technologies. Active systems effectively delay flutter onset and limit LCO amplitudes [10].

Passive solutions require no external energy input, relying instead on the inherent damping properties of specific components. The two primary passive mechanisms are tuned mass dampers (TMDs) and nonlinear energy sinks (NESs).

A TMD consists of a linear spring, a damper, and a mass. It dissipates energy by tuning its natural frequency to match a specific structural mode and resonating out of phase to the primary system. This effectively split the original resonance peak into two smaller, mitigated peaks (Figure 2.10). Due to its linear parameters, the TMD is only effective around a single frequency.

In contrast, an NES has a nonlinear hardening spring (Figure 2.10). Lacking a fixed natural frequency, it provides broadband frequency control. The NES mitigates vibrations through Targeted Energy Transfer (TET): a mechanism of one-way, irreversible energy channelling from the primary oscillating structure to the sink, initiated only when a minimum activation energy threshold is surpassed [20] [29].

When the primary structure is excited, the NES adjusts its instantaneous frequency to match the main system, initiating a transient 1:1 resonance capture. This synchronization enables efficient energy transfer into the sink. As the NES dissipates this energy, its effective frequency drops, breaking the resonance condition. This frequency decoupling traps the energy within the NES, preventing any reversible transfer back to the primary structure [29].

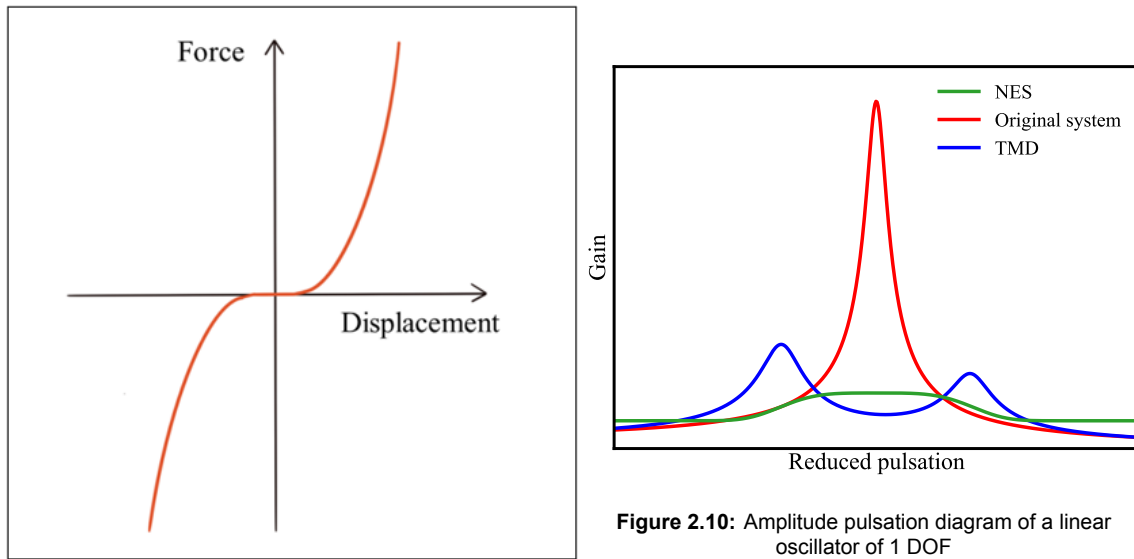


Figure 2.9: Nonlinear hardening stiffness [9]

Figure 2.10: Amplitude pulsation diagram of a linear oscillator of 1 DOF

An ungrounded NES achieves optimal energy pumping when its stiffness is purely nonlinear [29]. In practice, however, a purely nonlinear stiffness is difficult to achieve [9, 10]. With the addition of a linear stiffness component, the NES becomes a Nonlinear Tuned Vibration Absorber (NLTVA). The NLTVA does not possess an energy activation threshold but does have a zone where TET is active. The resulting energy pumping performance is weaker than that of the NES, due to the linear stiffness component. Importantly, the NLTVA can have an isolated solution, observed as isolas in the bifurcation diagram [20].

2.4.1. Flap-NES

A novel passive control device known as the Flap-NES has been proposed (Figure 2.11). This concept adapts a localised section of the wing's trailing-edge control surface to function as an NES. Governed by a cubic stiffness, the Flap-NES is designed to facilitate TET [1, 9, 10]. The Flap-NES offers two primary advantages over traditional passive strategies. First, it repurposes existing control surface mass, avoiding the weight penalty associated with internal NES configurations. Second, its external placement leverages aerodynamic damping directly from the flow [1, 9, 10].

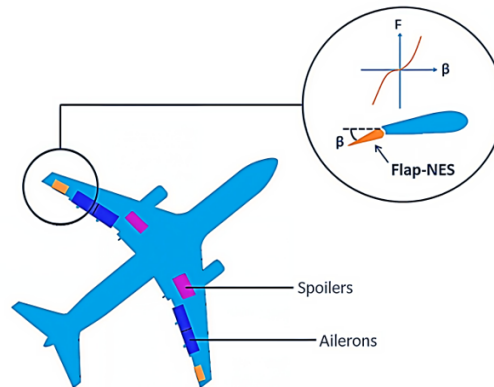


Figure 2.11: Conceptual diagram of the Flap-NES coupled to the primary wing structure [9]

Literature Review

This chapter reviews the literature on the passive control of flutter to identify critical research gaps and formulate the research objectives of this thesis.

Section 3.1 examines the state-of-the-art in flutter suppression using a classic, internal NES isolated from the aerodynamic flow. Section 3.2 details prior experimental and numerical investigations into the flutter suppression capabilities of the Flap-NES. Finally, Section 3.3 presents the research gaps, research questions and hypotheses.

3.1. Controlling Flutter Using a NES

The suppression of aeroelastic instabilities, such as self-excited flutter, using passive nonlinear energy sinks (NES) has been extensively studied since the early 2000s. The core principle is flutter mitigation via Targeted Energy Transfer (TET), defined as a one-way, irreversible energy transfer from the primary structure (the wing) to the NES.

The standard configuration, illustrated in Figure 3.1, consists of an internal NES placed within a 2-DOF airfoil. The system incorporates cubic stiffness in both pitch and heave and assumes quasi-steady aerodynamics. The NES is governed by four control parameters: mass ratio (ϵ), damping coefficient (λ), nonlinear stiffness coefficient (C), and attachment offset (δ) [19].

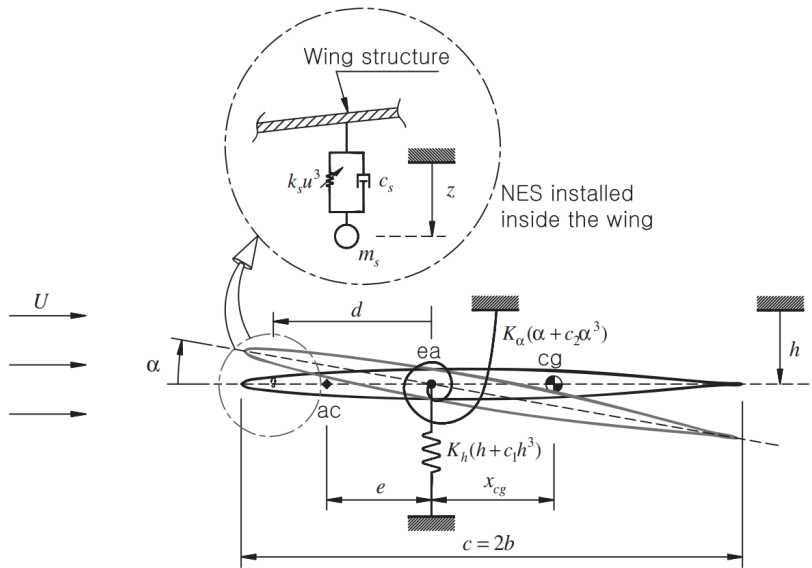


Figure 3.1: SDOF NES placed inside 2-DOF airfoil [19]

Lee et al. [19, 18, 17] and Vakakis [29] established the foundational theoretical and experimental framework for NES-driven flutter suppression, the key principles of which are detailed below. This foundation has enabled subsequent research incorporating additional system complexities, including control surfaces with freeplay [12], transonic flutter regimes [33], and hypersonic 3D wings [25].

3.1.1. TET Mechanisms for Suppressing Flutter

Lee et al. [19] identify three TET mechanisms for suppressing aeroelastic instabilities: recurring burst-out and suppression, intermediate suppression, and complete elimination. These LCO suppression mechanisms are evaluated through time-frequency analysis using Hilbert-Huang transforms.

Instantaneous energy exchanges within the system are quantified by tracking three distinct energy metrics over non-dimensional time (τ): the input energy (E^{Input}), which is the sum of the initial energy provided by the initial conditions and the nonconservative work done by the flow; the energy dissipated (E_d^{NES}) by the viscous damper of the NES; and the total instantaneous energy (E^{Total}), representing the sum of the instantaneous kinetic and potential energies of the wing and NES. The instantaneous energy balance should hold: $E^{Total}(\tau) = E^{Input}(\tau) - E_d^{NES}(\tau)$

Mechanism 1: Recurring Burstout and Suppression

This mechanism is characterized by a recurrent series of burstout and suppression of the heave and pitch modes of the wing, leading eventually to complete suppression of the aeroelastic instabilities [29].

As the wing begins to flutter, the minimum amplitude and energy viewed by the NES is reached and is therefore activated. The NES and wing are synchronized: 1:1 transient resonance capture (TRC¹) allows the NES to absorb and dissipate energy, resulting in a rest state with a total energy balance of zero. However, the aerodynamic flow is still feeding energy into the system and will disrupt the newly achieved resting state. The NES, therefore, repeatedly activates to suppress wing vibrations [19].

Figure 3.2 illustrates the instantaneous energy exchanges between the modes (upper graph) and the evolution of the total energy, the energy input fed by the aerodynamic flow, E^{Input} , and the energy dissipated by the NES, E_d^{NES} (lower graph) during TET. The variation of total energy reveals a recurrent burst-out suppression cycle: continuous aerodynamic input triggers repeated NES activations. Consequently, the instability is not eliminated, but the resulting amplitudes are drastically reduced compared to the baseline configuration (Figure 3.3) [19].

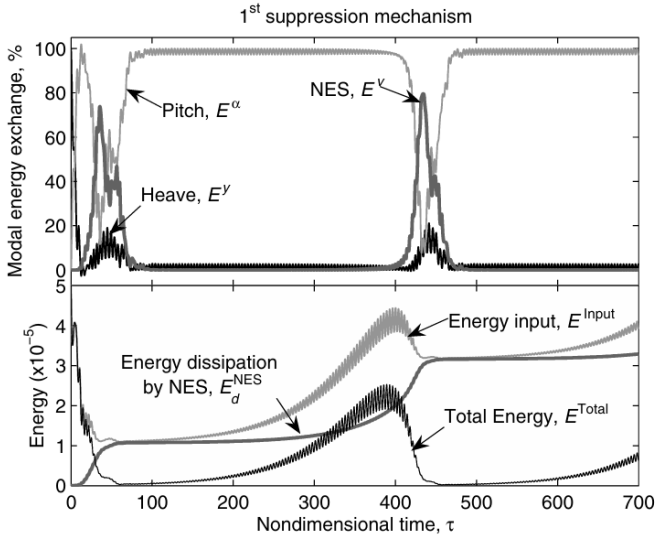


Figure 3.2: Instantaneous energy exchanges between modes (upper) and comparison of input energy from flow and energy dissipation by the NES [19]

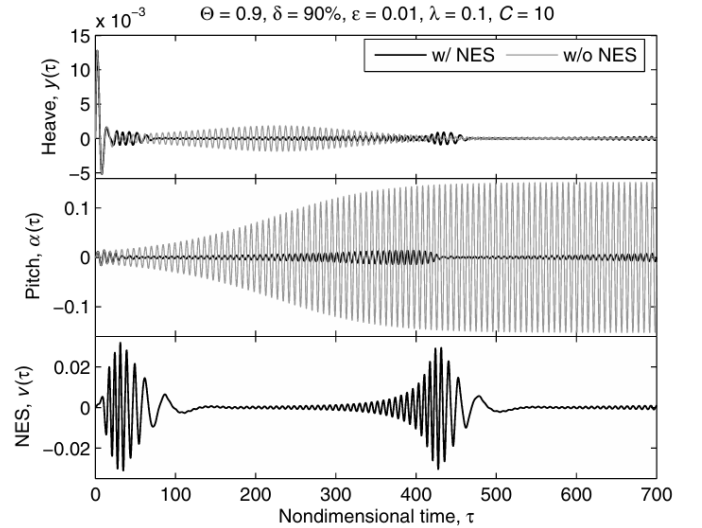


Figure 3.3: TET: Recurring burstout and suppression with NES [19]

¹Resonance captures sustained for a certain period of time with subsequent escapes of the dynamics from the resonance manifolds

The resulting TET pattern observed in Figure 3.3 corresponds to quasi-periodic motion. The quasi-periodic regime is visualised in phase space as an invariant torus (Figure 3.4). The system's response is dictated by two independent frequency bases: the primary LCO frequency (flutter) and the secondary frequency associated with the burst-suppression envelope. These two frequencies are incommensurate, meaning their ratio is an irrational number. The dynamic state never achieves a closed, periodic orbit. Instead, the system exhibits continuous quasi-periodic motion, defined by a non-repeating and drifting pattern as illustrated in Figure 3.5 [16].

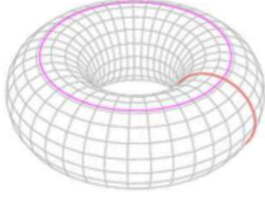


Figure 3.4: Phase space: Invariant torus. fast LCO frequency (pink) and slow modulation frequency (red).

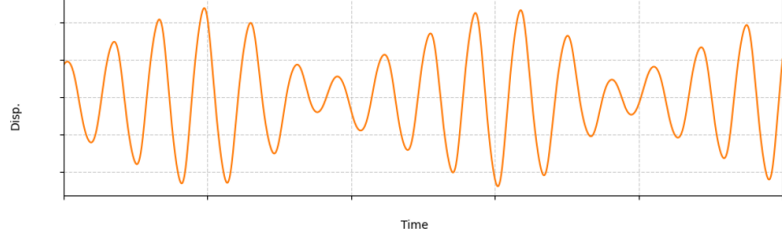


Figure 3.5: Quasi-periodic motion: Drifting and non-repeating

A Neimark-Sacker (NS) bifurcation marks the local onset of an invariant torus branch (Figure 3.6), which typically corresponds to quasi-periodic dynamics as observed in Figure 3.5 [32, 20]. This branch represents quasi-periodic rather than periodic motion, it is therefore not captured as an LCO on bifurcation diagrams. In the context of TET, identifying the NS bifurcation serves as a direct analytical criterion for detecting the onset of the recurring burst-and-suppression mechanism [19].

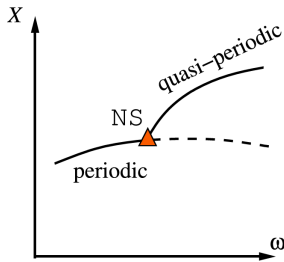


Figure 3.6: Neimark-Sacker bifurcation; Solid/dashed line: stable/unstable branch [32]

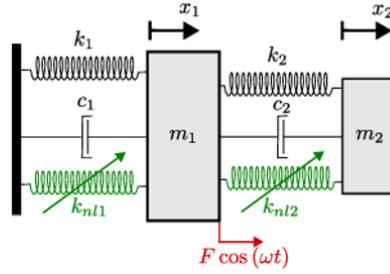


Figure 3.7: 2-DOF oscillator

A typical example is provided by a 2-DOF oscillator, as illustrated in Figure 3.7 which consists of a primary system (mass m_1 , linear damping c_1 , linear stiffness k_1 , and zero nonlinear stiffness, $k_{nl1} = 0$) coupled to a secondary subsystem representing the NES (mass m_2 , linear damping c_2 , linear stiffness k_2 , and nonlinear stiffness k_{nl2}). The main system is excited by an external force of amplitude F and frequency ω .

When the NES nonlinear stiffness is set to $k_{nl2} = 1$, the primary system reaches maximum amplitude at a resonance frequency of $\omega = 1$ rad/s (Figure 3.8). As the nonlinear stiffness is increased to $k_{nl2} = 3$, two NS bifurcations emerge, effectively bounding the amplitude of the primary mass. These two NS bifurcations delineate the frequency band in which TET occurs via recurring burst-out and suppression (Figure 3.9a). Time-integration of the system's response at $\omega = 1$ rad/s confirms that both masses exhibit quasi-periodic motion, with the primary system's maximum amplitude reduced by 40% compared to the baseline ($k_{nl2} = 1$) configuration (Figure 3.9b).

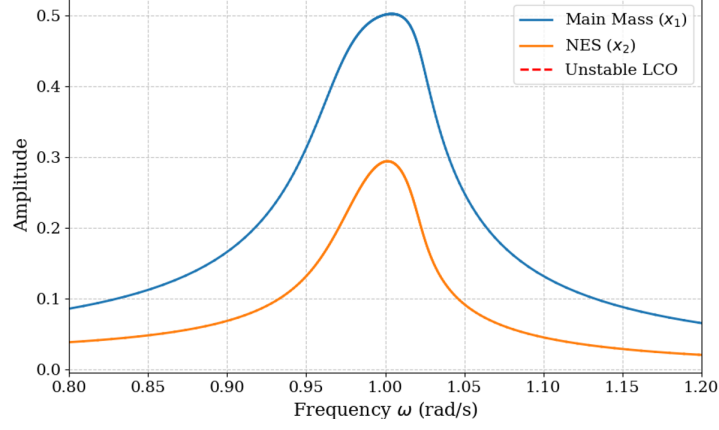
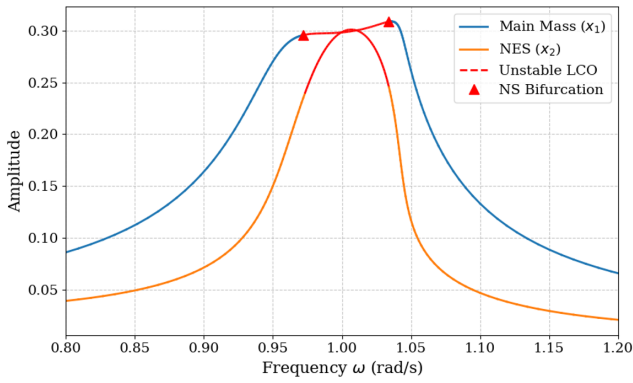
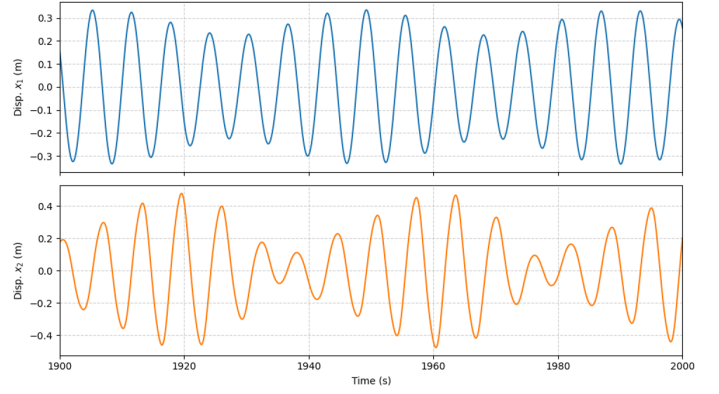


Figure 3.8: LCO amplitude (m) vs frequency response baseline configuration ($k_{nl2} = 1$).



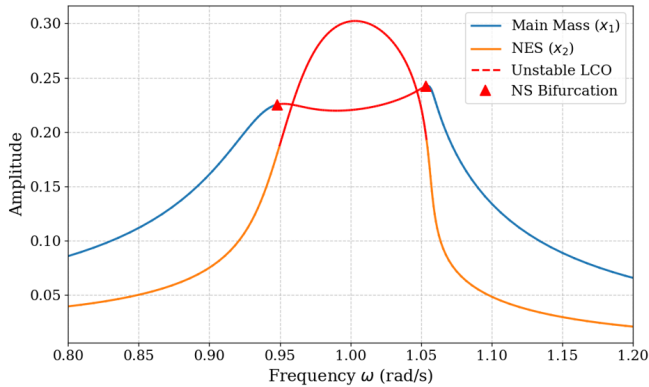
(a) LCO amplitude (m) vs frequency response



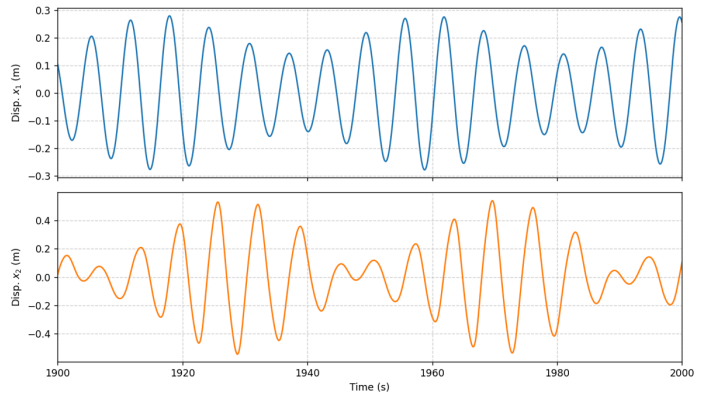
(b) Time-integration (steady-state) at $\omega = 1$ rad/s

Figure 3.9: System response for nonlinear stiffness $k_{nl2} = 3$.

Further increasing the nonlinear stiffness to $k_{nl2} = 5$ broadens the active TET frequency zone (Figure 3.10a) and reduces the maximum amplitude by 44% at $\omega = 1$ rad/s (Figure 3.10b). Ultimately, the appearance of these entrance and exit NS bifurcations strictly defines the operational boundaries for recurring burst-out and suppression TET [20].



(a) LCO amplitude (m) vs frequency response



(b) Time-integration (steady-state) at $\omega = 1$ rad/s

Figure 3.10: System response for nonlinear stiffness $k_{nl2} = 5$.

Flutter can be controlled by recurring burstout and suppression TET in a 2-DOF airfoil equipped with an internal NES. The proposed Flap-NES differs as it is the control surface which acts as the NES. The studied system evolves to a 3-DOF aeroelastic system with nonlinear stiffness in the flap. It is essential to understand how quasi-periodic motion can be induced and sustained within this system to assess if it can perform recurring burstout and suppression TET.

Dimitriadis's text, *Introduction to Nonlinear Aeroelasticity*, serves as the primary source for the study of nonlinear aeroelastic systems. Chapter 5 [4] explores quasi-periodic motion within a 3-DOF aeroelastic system featuring cubic stiffness exclusively in the flap DOF. Note that the flap also contains a linear stiffness. This configuration is analogous to the proposed Flap-NES setup. The evaluated parameters include a cubic stiffness coefficient defined as $K_\beta^3 = 100K_\beta$ and a hinge axis positioned at 80% of the chord, studied under unsteady aerodynamics. The author identifies three routes capable of inducing this quasi-periodicity: interacting limit cycles, the Hopf-Hopf bifurcation, and the Hopf-fold bifurcation.

Interacting Limit Cycles The undamped natural frequencies are $\omega_h = 1$ Hz, $\omega_\alpha = 12$ Hz, and $\omega_\beta = 7$ Hz. The flap natural frequency is therefore sandwiched between the heave and the pitch.

Figure 3.11 presents the corresponding flutter diagram. Two Hopf bifurcations are detected: $U = 46.9$ m/s, driven by a heave-flap frequency coalescence, and $U = 65.7$ m/s, driven by a flap-pitch frequency coalescence [4].

Following these Hopf Bifurcations, the LCO branch evolves into either a stable supercritical or unstable subcritical state, as dictated by the flutter diagram. Because the flap exhibits hardening stiffness, its eigenvalue increases after the first flutter mechanism activates, separating it from the heave mode therefore resulting in a stable, supercritical branch. Conversely, at the second flutter speed, the flap eigenvalue lies below that of the pitch. Crossing this second Hopf bifurcation causes the flap eigenvalue to increase and engage in a deep resonance with the pitch mode, producing an unstable, subcritical branch [4].

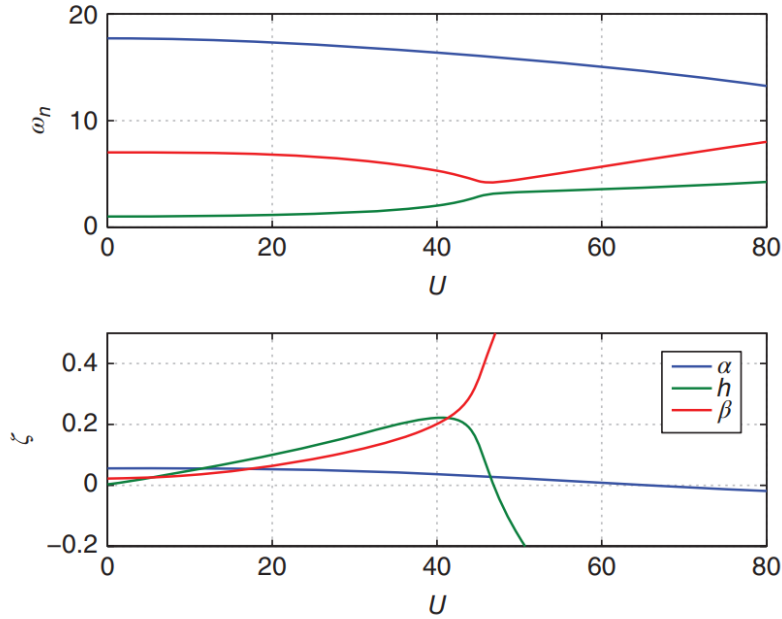


Figure 3.11: Variation of natural frequencies and damping ratios with airspeed for TORUS case [4]

In this case, the quasi-periodic motion occurs when different limit cycle branches have nearly equal amplitude but different frequencies at the same airspeed as illustrated by Figure 3.12. As such, this occurs only if the underlying linear system has at least two flutter points inside the airspeed range of interest [4].

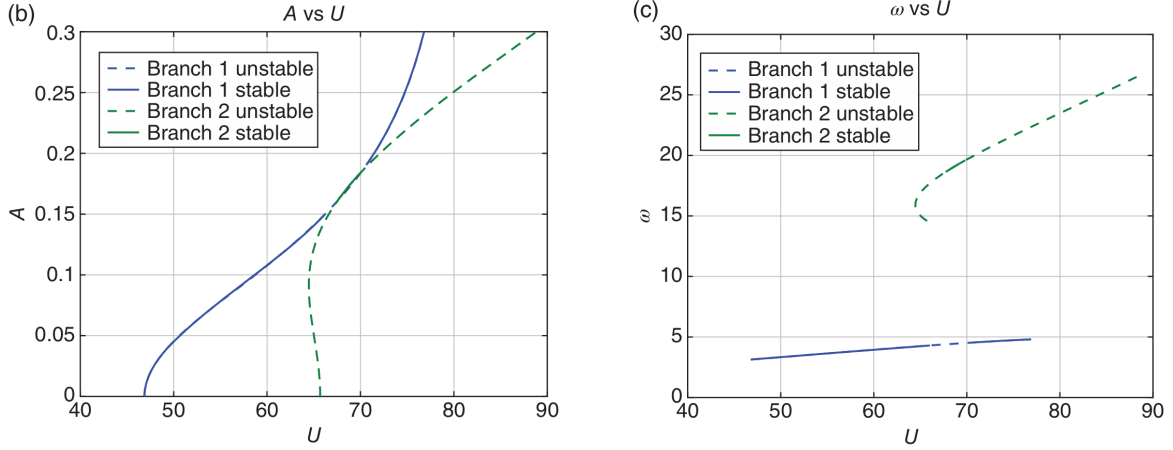


Figure 3.12: Equivalent linearisation of a 3-DOF wing with hardening flap stiffness: Torus bifurcation [4]

Hopf-Hopf Bifurcation Occurs when two distinct Hopf bifurcations emerge simultaneously. This phenomenon generates two coexisting LCO branches that exhibit similar LCO amplitudes and two LCO frequencies whose ratio is irrational in the vicinity of the bifurcation point (Figure 3.13).

To achieve the exact mathematical point of this bifurcation, the uncoupled natural frequencies must be perfectly tuned (e.g., $\omega_h = 0.9$ Hz, $\omega_\alpha = 6.85435$ Hz, and $\omega_\beta = 7$ Hz) to force two different flutter modes to occur at the exact same airspeed [4].

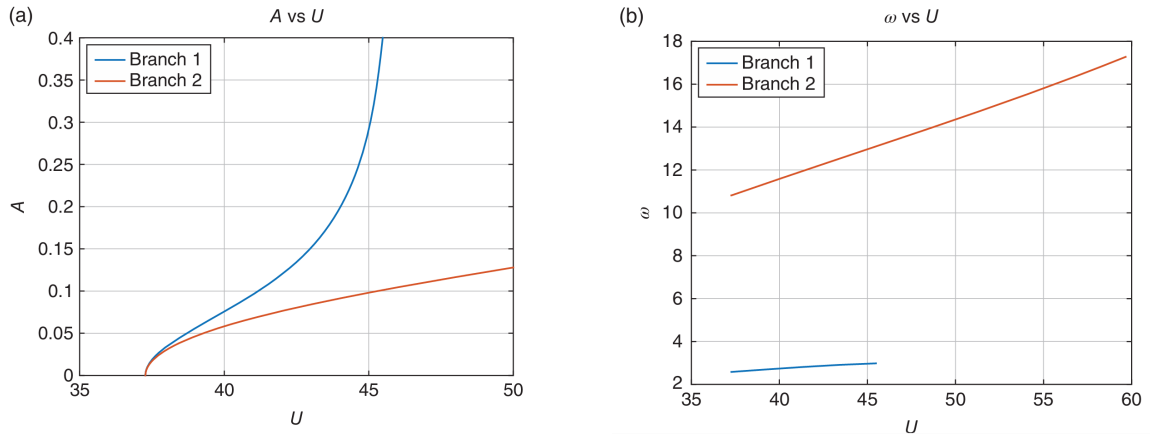


Figure 3.13: Equivalent linearisation of a 3-DOF wing with hardening flap stiffness: Hopf-Hopf bifurcation [4]

Hopf-Fold Bifurcation Occurs when a system undergoes a Hopf bifurcation and a fold bifurcation simultaneously. This arises when the parameters of the aeroelastic system are tuned such that the underlying linear system possesses both a zero eigenvalue (condition for divergence: fold) and a pair of purely imaginary eigenvalues (condition for flutter: Hopf bifurcation) around a given fixed point [4].

This route to quasi-periodic motion will not be developed further as the Flap-NES will not be beneficial under this configuration: it shows no advantage in controlling divergence.

Mechanism 2: Intermediate Suppression

The second TET mechanism, intermediate suppression, reduces the LCO amplitudes permanently. Flutter here is not eliminated but controlled. The action of the NES is non-recurring: it activates once at the beginning of the flutter onset via 1:1 TRC, as observed by the initial energy exchange in Figure 3.14. The system then reaches an equilibrium and settles into 1:1 Permanent Resonance Capture (PRC²). The energy dissipated by the Flap-NES matches the energy input, indicated by their parallel trajectories in Figure 3.14. Consequently, both the heave and pitch modes settle into constant, non-zero steady-state amplitudes that are lower than the baseline case without an NES (especially visible for the pitch mode) as illustrated by Figure 3.15.

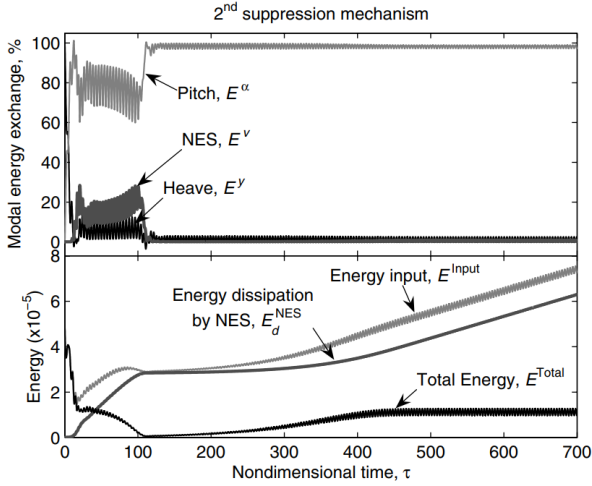


Figure 3.14: Instantaneous energy exchanges between modes (upper) and comparison of input energy from flow and energy dissipation by the NES for 2nd suppression mechanism [19]

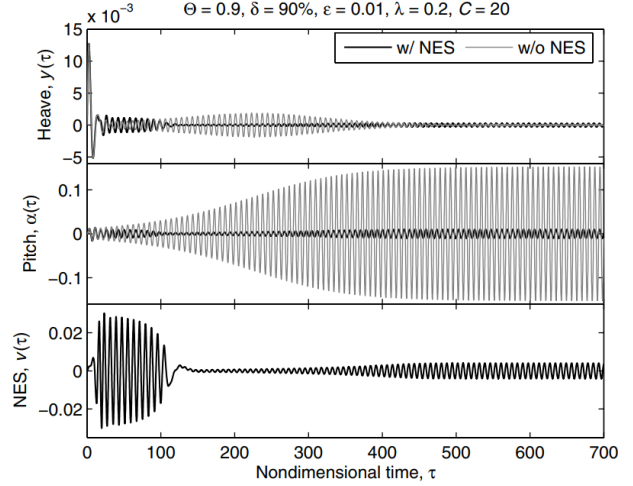


Figure 3.15: TET: Intermediate suppression [19]

Mechanism 3: Complete Elimination

The third and final TET mechanism completely eliminates the LCO. As observed in Figure 3.16, the total energy is drained sustainably as the energy dissipated by the NES balances with the energy input preventing any reappearance of LCOs. As illustrated in Figure 3.17, the heave, pitch, and Flap-NES modes exhibit decaying responses. This mechanism typically requires a larger NES mass.

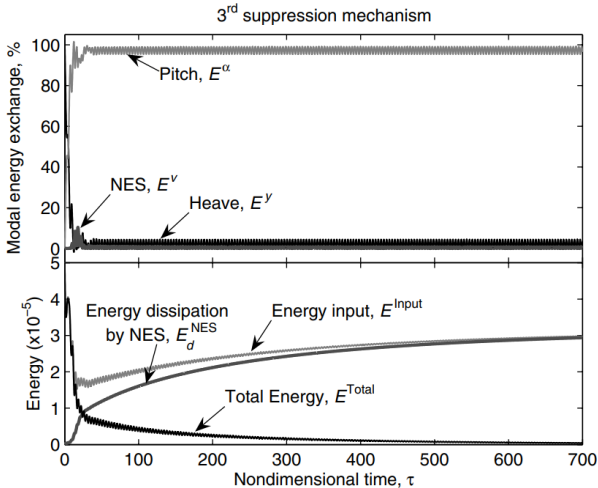


Figure 3.16: Instantaneous energy exchanges between modes (upper) and comparison of input energy from flow and energy dissipation by the NES for 3rd suppression mechanism [19]

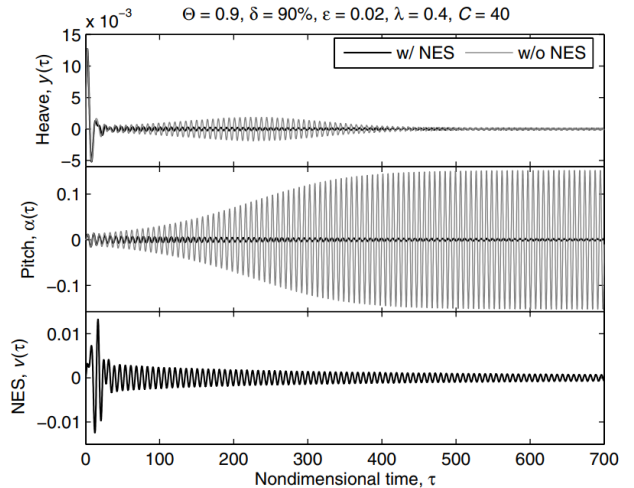


Figure 3.17: TET: Complete elimination [19]

²Resonance captures sustained permanently

No Elimination

The NES must achieve 1:1 resonance capture to effectively dissipate energy from the wing. In the case of no suppression, the system transitions from a 1:1 to a 3:1 frequency locking ratio between the heave and the pitch modes. This means that the nonlinearity fails to detune the modes, trapping the system in a regime where the pitch mode oscillates three times for every heave cycle. When these phases align, the free-stream flow injects a significant burst of aerodynamic energy directly into the structure, driving the onset of flutter (Figure 2.2) and entirely bypassing the Flap-NES. Therefore, for effective flutter control, the NES must interact with both the heave and pitch modes to disrupt this synchronization and prevent direct energy transfer from the flow to the wing.

3.1.2. NES Parameters for Effective Flutter Mitigation

Effective LCO suppression requires attaching the Flap-NES far from the wing's elastic axis. Large spatial offsets enable the Flap-NES to interact and exchange energy with both the heave and pitch modes. Specifically, aft attachments (negative offset) maximise energy dissipation and prolong nonlinear resonant interactions. Conversely, forward attachments (positive offset) attenuate energy transfer while inducing instabilities, and must therefore be avoided.

Furthermore, the mass ratio and linear damping govern the system's sensitivity to initial conditions and the complexity of post-flutter dynamics. Increasing both parameters reduces this sensitivity, facilitating complete LCO elimination. While higher damping mitigates complex post-flutter behaviours, it can also completely suppress the first TET mechanism (recurring burst-out and suppression).

3.1.3. Experimental Validation of TET-Based, Passive LCO Suppression

The theoretical framework for suppressing aeroelastic instability using a NES is validated against experimental data from Lee et al. [18] presented in the work by Vakakis et al. [29].

The experimental setup consists of a 2-DOF (pitch and heave) rigid NACA 0015 wing mounted in a low-speed wind tunnel. The NES is a low-friction carriage coupled to the wing via a viscous damper and a purely nonlinear spring (untensioned perpendicular wires). Due to hardware constraints, the NES is mounted at the wing's elastic axis (although in contradiction with previous results), and its mass ratio is artificially high (10%–12.5%). Aerodynamics are modeled as quasi-steady 2D flow. Dry friction is significant in the main wing system but negligible in the NES.

The experiments successfully verify the three theoretical TET mechanisms: recurring burst-out and suppression, intermediate suppression, and complete LCO elimination. The passive TET significantly improves the stability envelope, increasing the flutter speed from 9.5 m/s to 12 m/s.

Importantly, complete LCO suppression requires 1:1 resonance captures between the NES and both aeroelastic modes. Subharmonic (1:2) resonance captures are insufficient to break the aeroelastic internal resonance, resulting in recurrent instability burst-outs.

The experimental parametric study highlights the following design trade-offs: a high nonlinear stiffness reduces sensitivity to initial conditions, ensuring more predictable activation over varying initial states. A lower nonlinear stiffness extends the operational airspeed range for complete LCO elimination (TET mechanism 3), enabling suppression at higher velocities. A higher damping improves general LCO suppression capacity but increases sensitivity to initial conditions. A heavier mass improves TET under small perturbations but degrades performance under large perturbations by inducing complex dynamics. Finally, a lighter NES ensures robust consistent TET performance regardless of perturbation size.

Finally, the authors propose that Multi-Degree-of-Freedom (MDOF) NES designs can improve robustness and reduce the required mass ratio via simultaneous multi-modal broadband TET. A subsequent study [17] develops this concept, concluding that an optimized MDOF NES provides robust aeroelastic instability suppression by efficiently, passively, and rapidly transferring unwanted vibration energy to the furthest NES mass.

3.2. Controlling Flutter with a Flap-NES

The application of a flap as a Nonlinear Energy Sink (Flap-NES) was investigated across three consecutive PhD projects from 2017 to 2024, in collaboration between Institut Clément Ader and ISAE-SUPAERO [1, 9, 10]. This extensive body of research is divided into two primary categories: experimental and numerical.

A consensus across these three studies is that the Flap-NES operates optimally when functioning as a classical ungrounded NES. To achieve this, its linear stiffness is minimised or set to zero in favour of a purely cubic nonlinearity. This purely cubic profile provides broadband frequency control and allows Targeted Energy Transfer (TET) to engage once a minimum activation energy threshold is reached (Section 2.4).

3.2.1. Experimental

The Flap-NES is supported by extensive experimental results. First, the fundamental nonlinear behaviour introduced by the Flap-NES is characterised by Amar [1]. Second, Fernandez-Escuerdo compared the aeroelastic control capabilities with those of a Flap-Tuned Mass Damper [9]. Finally, Garcia-Perez conducted a comprehensive parametric study to evaluate the influence of individual Flap-NES design variables [10].

Non-Linear Behaviour due to Flap-NES

Amar [1] experimentally investigated a 3-DOF linear wing incorporating a Flap-NES to characterise its nonlinear aeroelastic behaviour (Figure 3.18a). The Flap-NES stiffness incorporated both linear and cubic contributions, meaning the tested NES was closer to an NLTVA. Testing was conducted at airspeeds up to 15 m/s with a turbulence intensity below 0.1%. A triangular airspeed profile (ramp-up and ramp-down as illustrated in Figure 3.18b) was employed to generate bifurcation diagrams. The resulting signals were analyzed using Fast Fourier Transforms (FFT), phase portraits, Root Mean Square (RMS) amplitudes, and Hilbert Transforms. The system's performance was benchmarked against a purely linear 3-DOF configuration (excluding only the Flap-NES cubic stiffness).

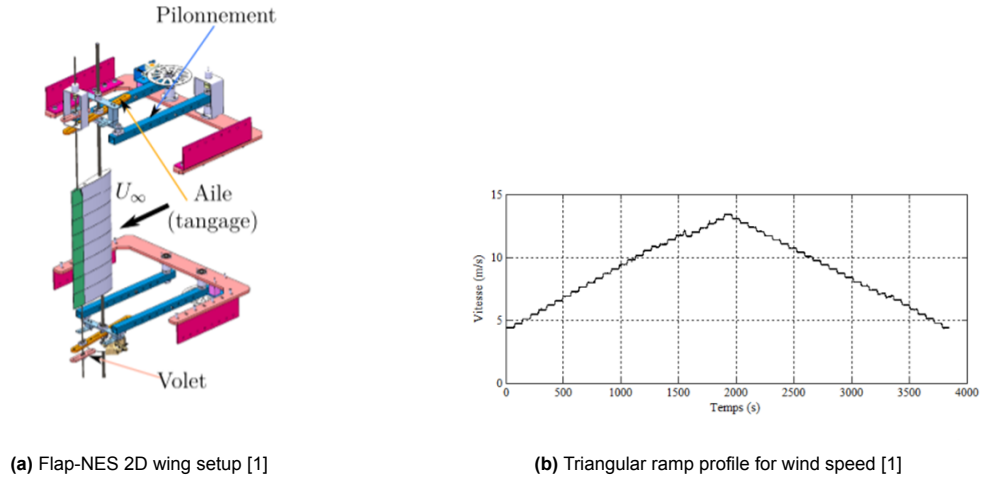


Figure 3.18: Flap-NES experimental setup [1]

Based on Figure 3.19, Amar identifies the emergence of two Hopf bifurcations at 4.4 m/s and 9.1 m/s. In comparison to the 3-DOF baseline, the linear flutter speed is therefore increased by 8.7%. However, note that the plotted velocity range begins only at $U = 4$ m/s, which calls into question whether the first Hopf bifurcation actually occurs earlier and if the resulting gain in linear flutter speed is exact.

Two distinct bifurcation branches emerge from these Hopf bifurcation points, and four LCO regimes are observed: low-frequency LCO, high-frequency LCO, period-doubling, and chaotic beating. Importantly, Amar [1] identified the first Hopf bifurcation as subcritical, as observed in Figure 3.19. As the two LCO

branches (subcritical and supercritical) co-exist, the system is largely dependent on the initial conditions and on how the system is perturbed.

Furthermore, instantaneous frequency analysis confirmed a series of resonance captures, in line with the TET mechanisms documented for internal NES configurations [19] (Section 3.1).

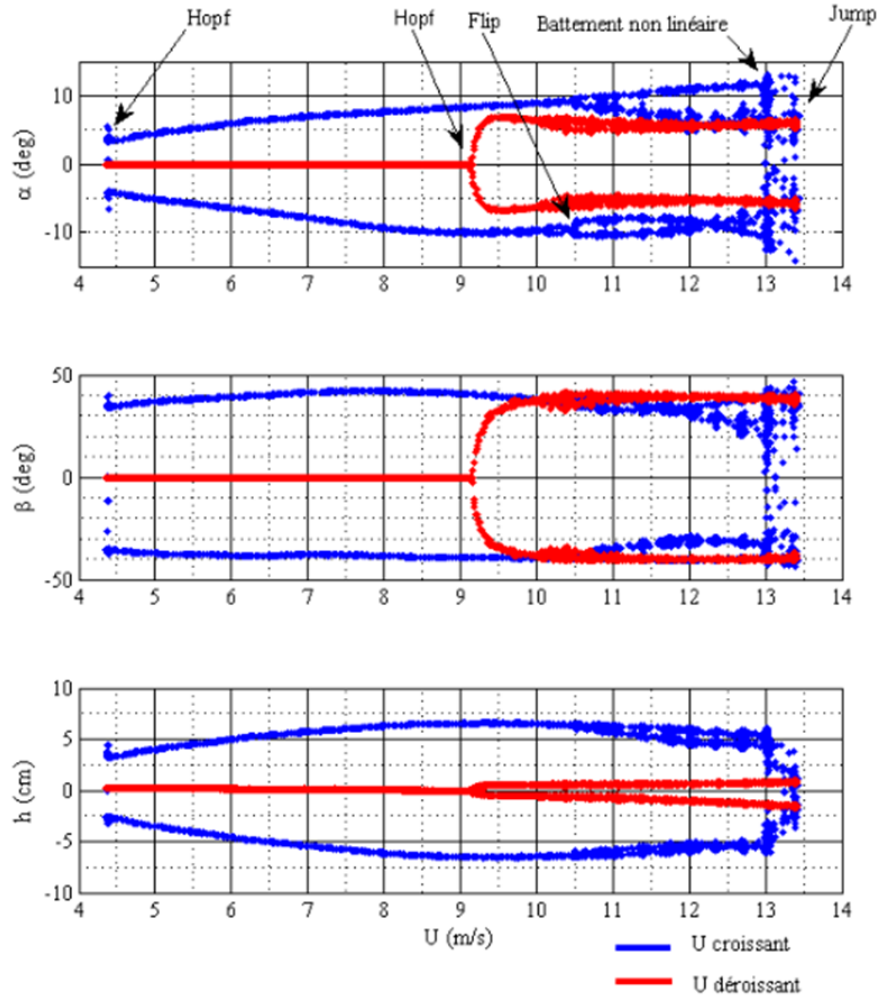


Figure 3.19: Experimental bifurcation results for Flap-NES [1]

A primary limitation of this study lies in the baseline configuration used to evaluate the Flap-NES: a 3-DOF system, including the flap's linear stiffness and damping and excluding only its cubic nonlinear stiffness. This baseline evaluates the isolated benefit of the nonlinearity. A more representative baseline would utilise a locked flap to assess the impact of the Flap-NES assembly as a whole. With a locked-flap baseline, the results may differ, especially concerning the linear flutter speed increase.

Flap-NES vs Flap-TMD

Fernandez Escudero [9] conducted an experimental campaign comparing the aeroelastic control capacities of a Flap-Tuned Mass Damper (Flap-TMD) and a Flap-NES on a 2D wing section (Figure 3.20a), tested at airspeeds up to 25 m/s with a turbulence intensity below 0.1%. The system was evaluated in three configurations: a linear 2-DOF baseline wing (with the flap locked), a Flap-TMD, and a Flap-NES (implemented using parallel traction springs as illustrated in Figure 3.20b). Because the Flap-NES retained a significant linear stiffness component equivalent to that of the Flap-TMD, it is more accurately classified as a Nonlinear Tuned Vibration Absorber (NLTVA).

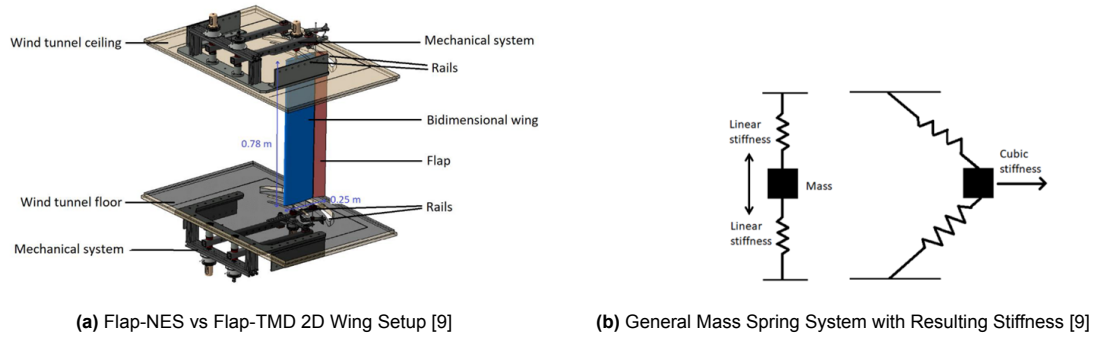


Figure 3.20: Flap-NES vs Flap TMD Experimental Setup [9]

For wind-on testing, initial conditions were prescribed via electromagnetic brakes acting on the heave and pitch degrees of freedom. However, because the flap mechanism lacked an automated braking system, its initial conditions were manually introduced during zero-wind structural characterisation. As shown in Figure 3.21, the manual inputs for the Flap-TMD and Flap-NES were non-identical. No initial conditions were introduced on the flap for post-flutter analysis.

A critical limitation of this study is that the initial displacement values for pitch and heave remain undisclosed. They could be single-valued, indicating that only a small only a small portion of the system's post-flutter response has been studied.

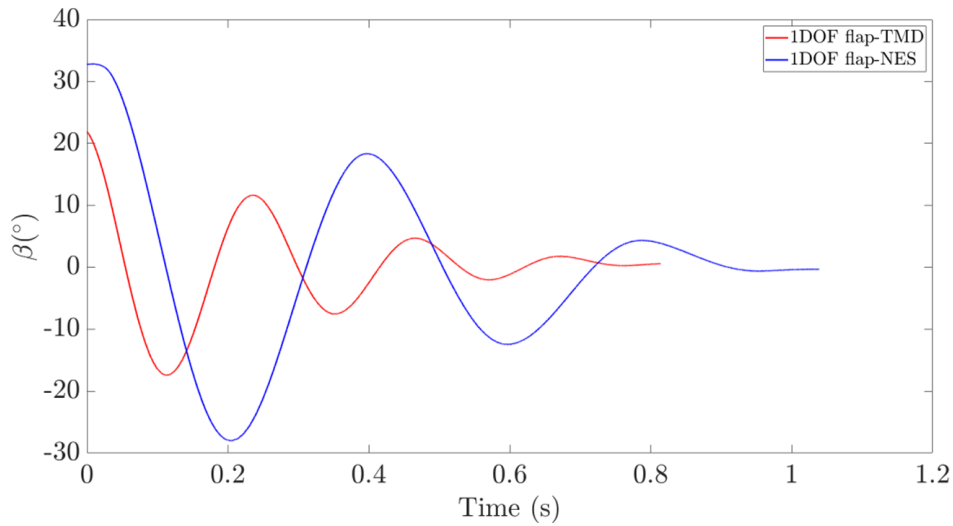


Figure 3.21: Response of the flap-TMD and flap-NES to a manually imposed initial displacement in zero wind speed [9]

The author states that flutter across all three configurations is driven by the classical modal coalescence of the heave and pitch modes, although this mechanism is not clearly apparent from Figures 3.22 and 3.23, as the threshold for modal coalescence is not defined.

Note the reduction in the pitch, α , natural frequency, which decreases from 2.75 Hz in the Flap-TMD configuration (Figure 3.22) to 2.5 Hz in the Flap-NES configuration (Figure 3.23). The author omits any discussion regarding the cause of this shift. Furthermore, the natural frequency of the Flap-NES lies above those of the heave and pitch modes (Figure 3.23). The Flap-NES with a linear stiffness seems to alter the linear dynamics of the system. This configuration directly contradicts the paper's core assertion that the Flap-NES operates optimally as an ungrounded NES with zero linear stiffness. Instead, the reported 16% increase in the linear flutter speed is achieved using a Flap-NES, which seems to rely on its linear stiffness.

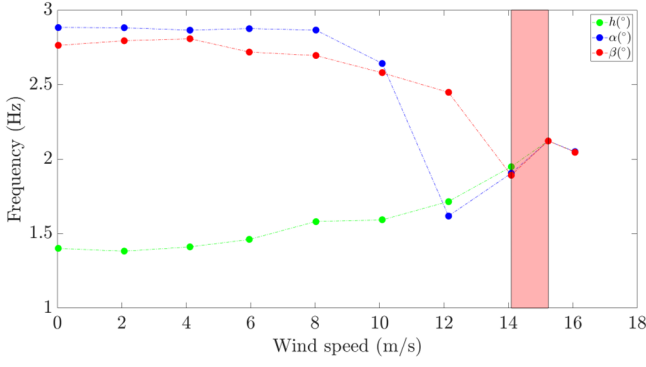


Figure 3.22: Frequency evolution of h , α , β with wind speed in flap-TMD configuration [9]

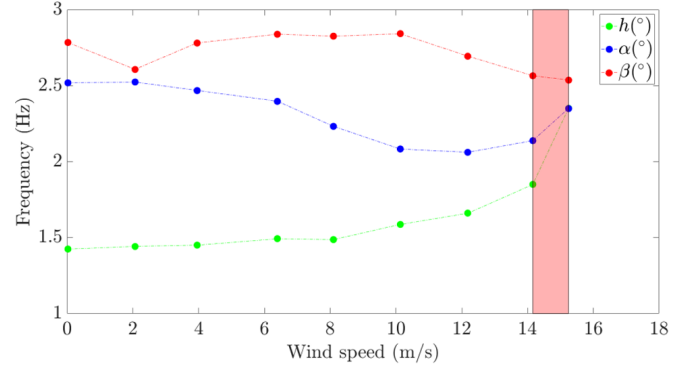


Figure 3.23: Frequency evolution of h , α , β with wind speed in flap-NES configuration [9]

In the post-flutter regime, the 2-DOF baseline stabilised into a bounded, constant-amplitude LCO (Figure 3.24). This behaviour reveals the presence of unaccounted structural or aerodynamic nonlinearities which are not disclosed by the author, as a linear system would undergo unbounded exponential growth past the flutter speed.

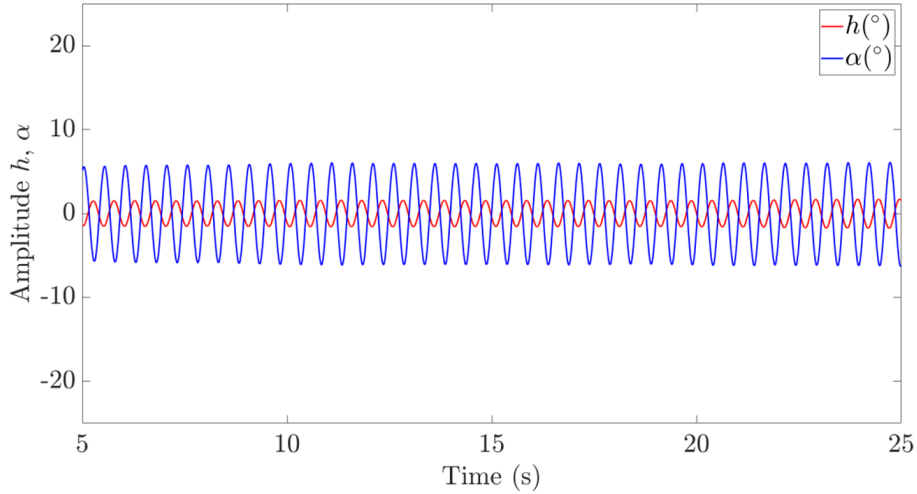


Figure 3.24: Response of 2DOF system at wind speed of 15.35 m/s [9]

The recorded limit cycle oscillation (LCO) signal energy decreased by up to 61% with the Flap-TMD and 79% with the Flap-NES. The Flap-NES exhibited quasi-periodic motion at 16.20 m/s, demonstrating the recurring burst-out and suppression TET mechanism discussed in Section 3.1.

Flap-NES Parametric Study

Garcia Perez [10] aimed to assess the influence of flap-NES parameters in an experimental aeroelastic system. A 2D wing with 3-DOF, which includes the Flap-NES and a cubic stiffness on the pitch DOF (obtained through parallel traction springs, Figure 3.20b) is tested up to 25 m/s with a turbulence rate of $< 0.1\%$. The Flap-NES extends over $1/3$ of the span as illustrated in Figure 3.25 and possesses a small linear stiffness due to spring pre-tensioning. The simulated flap-NES is therefore not a pure NES but rather an NLTVA. The Flap-NES inertia, static moment, nonlinear stiffness, and angle of attack are varied. The baseline comparison is that of the 2-DOF system, where the Flap-NES is locked into position.

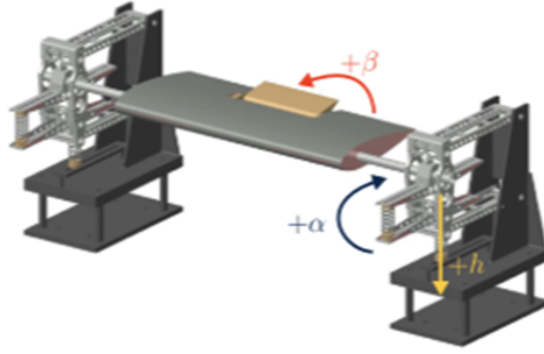


Figure 3.25: Flap-NES parametric study: experimental setup [10]

An initial analysis of the time-response data, processed via Fourier transforms, reveals the evolution of the system's natural frequencies with respect to airspeed (Figure 3.26). The Flap-NES natural frequency is identified at 9.11 Hz, positioning it distinctly above both the heave and pitch modes. This high-frequency characteristic directly mirrors the structural trend previously reported by Fernandez Escudero (Figure 3.23) [9]. Furthermore, unlocking the Flap-NES is shown to alter the baseline structural modes, causing the heave and pitch natural frequencies to decrease from 4.56 Hz and 6.00 Hz to 3.84 Hz and 5.77 Hz, respectively. The linear stiffness of the Flap-NES participates in the linear dynamics of the aeroelastic system.

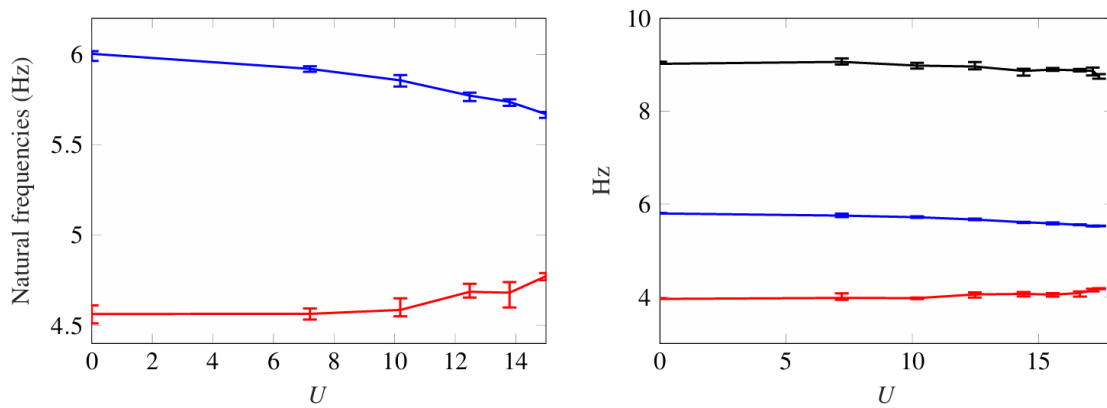


Figure 3.26: Natural frequencies of the main modes for flap OFF (left) and flap ON (right). Black = Flap-NES [10]

Garcia Perez [10] concludes that the Flap-NES is effective in terms of flutter suppression, delaying the onset by 35% and increasing stall flutter by 24% all while reducing LCO amplitudes. This performance is consistent despite changes in AOA. In high amplitude stall cases, the LCOs were reduced by 78% transforming the subcritical bifurcation to supercritical.

However, these results have two primary methodological limitations: the reliance on random, undocumented initial conditions, and the exclusion of low-airspeed data ($U < 14.5$ m/s). As demonstrated by Amar [1], an aeroelastic system equipped with a Flap-NES is highly sensitive to initial conditions due to the co-existing supercritical and subcritical bifurcations. By applying a finite number of initial conditions and limiting the bifurcation sweeps to high airspeeds, only a portion of the post-flutter behaviour was studied.

3.2.2. Numerical

Based on the preceding experimental results, developing a numerical model is necessary to systematically characterise the system dynamics and enable parametric optimisation of the Flap-NES design.

Flap-NES Non-Linear Dynamics

Amar [1] expanded their previous experimental work to compare empirical data with numerical models based on potential-flow theory (Theodorsen and UVLM). The Flap-NES was modeled with a purely cubic nonlinear stiffness. Bifurcation diagrams were generated with time-integration (Ibrahim Time Domain method), applying a single initial condition to the heave degree of freedom across various airspeeds.

Both aerodynamic models accurately predicted the two Hopf bifurcations. Two distinct flutter mechanisms were observed: an initial instability driven by flap-heave modal coalescence, followed by a secondary instability driven by heave-pitch coalescence. Regarding post-flutter dynamics, the UVLM model provided the most accurate quantitative and qualitative agreement with the experimental results.

A primary limitation of this numerical approach is the inherent violation of potential-flow assumptions at the high angles of attack experienced during large-amplitude limit cycle oscillations due to flow separation. Secondly, the use of a fixed initial condition across all airspeeds contradicts the author's primary observation regarding the system's sensitivity to initial conditions. As such, only a portion of the post-flutter behaviour is studied, missing the co-existing bifurcation branches, observed experimentally.

Data-Driven Bifurcation Analysis for Flap-NES Parameter Optimisation

García Pérez [11] addressed three primary research objectives: proposing a precise, reduced-complexity bifurcation forecasting method; analyzing the influence of Flap-NES parameters on the linear flutter speed and bifurcation type; and optimizing these parameters to maximize the flutter boundary, minimize heave LCO amplitude, and transition subcritical bifurcations to supercritical.

The study investigated a 2D 3-DOF nonlinear aeroelastic system featuring cubic pitch stiffness and a purely nonlinear Flap-NES, subjected to unsteady aerodynamics and gust perturbations. The author assumes that the Flap-NES performs optimally with a zero-linear stiffness, as would an ungrounded traditional NES. When developing the nonlinear aeroelastic model, the linear stiffness contribution of the Flap-NES is therefore set to 0.

A data-driven bifurcation forecasting method, based on the critical slowing down mechanism, was developed to extract post-flutter dynamic responses and subsequently validated against MATCONT. The Flap-NES parameters (hinge location, inertia, rotational inertia, linear damping, and cubic stiffness) were varied across both supercritical and subcritical baseline systems. Finally, these parameters were optimized using the NSGA-II algorithm.

The data-driven method accurately predicted Hopf bifurcations as well as supercritical and subcritical LCOs, though accuracy degraded at high amplitudes. While the parametric sweep revealed no distinct global trends, the optimised supercritical configuration yielded a 4.92% increase in the linear flutter speed and a 42.2% reduction in pitch LCO amplitude (evaluated at 105% of the flutter speed). In the subcritical case, the optimised Flap-NES successfully increased the flutter boundary and transitioned the instability from a subcritical to a supercritical bifurcation.

A critical limitation of this study is the complete exclusion of the Flap-NES linear stiffness parameter from the parametric study. The linear stiffness was simply set to zero it was erased from the nonlinear aeroelastic model. Furthermore, the author did not study the linear stability boundary through flutter diagrams, as such the flutter mechanism was not characterized.

3.3. Synthesis and Research Objectives

This section identifies the primary research gaps emerging from the literature review, formulates the research question and presents the foundational hypotheses.

3.3.1. Identification of Research Gaps

Despite the promising experimental and numerical results presented in recent literature, a critical review reveals that the current understanding of the Flap-NES flutter control capabilities is compromised by a combination of fragmented methodological limitations across separate studies. The literature currently lacks a comprehensive framework that simultaneously integrates three core elements:

- **A Representative Baseline:** To assess the Flap-NES' efficacy in controlling flutter, the 3-DOF system with Flap-NES must be evaluated against the 2-DOF system, which represents a locked-flap configuration [9, 10].
- **Bifurcation Diagrams:** Current experimental [9, 10] and numerical [1] studies rely on initial conditions, which map only a portion of the post-flutter dynamics. Bifurcation diagrams track the evolution of the LCO amplitudes across various airspeeds, providing a comprehensive study of the post-flutter regime [10]. With the proper numerical framework, co-existing bifurcation branches and transitions from stable to unstable LCOs can be captured, thereby defining the system's sensitivity to perturbations without relying on arbitrary initial conditions.
- **Flap-NES Linear Stiffness:** Previous studies [1, 9, 10] assert that the Flap-NES achieves optimal performance when operating as a classic ungrounded NES: with a zero linear stiffness (Section 2.4). However, achieving a purely nonlinear stiffness is practically unrealizable in experimental setups [1, 9, 10]. As previously established, the presence of this linear term affects the linear eigenvalue evolution of the system [9, 10]. The Flap-NES with linear stiffness results in elevating the linear flutter boundary. Despite this, the numerical parametric investigation conducted by Garcia-Perez [10] excluded the linear stiffness parameter, setting it to zero. Consequently, a numerical assessment of the Flap-NES linear stiffness parameter is required.

3.3.2. Research Question

Consequently, the central research question is: **How does the integration of the Flap-NES, possessing both linear and cubic stiffness, influence the stability and post-flutter dynamics of a 2D typical section under linear unsteady aerodynamics?**

Working towards this research question, the following sub-questions will guide this research:

- **Linear Stability:** How does the linear stiffness component of the Flap-NES alter the flutter mechanism and impact the linear flutter speed?
- **Nonlinear Regime:** To what extent can the Flap-NES control post-flutter LCOs through TET?
- **Parametric Sensitivity:** How do the parameters of the Flap-NES (linear/nonlinear stiffness, damping, length, inertia) dictate the onset of the TET mechanisms?

Hypothesis 1: Premature Coalescence of Flap-NES

The Flap-NES is subjected to aerodynamic loading while the internal NES is not and relies solely on the wing's energy to activate. With zero linear stiffness, the aerodynamic loads may induce large flap deflections at low airspeeds, which may force a premature mode coalescence with the lower-frequency mode (heave or pitch), resulting in a severe decrease of the linear stability boundary.

Hypothesis 2: Hidden Co-Existing Branches

Because multiple Hopf bifurcations may exist in a 3-DOF system, several LCO branches will emerge. It is hypothesised that mapping the global bifurcation diagram will demonstrate that the Flap-NES is sensitive to initial conditions.

Hypothesis 3: Precise Tuning for Quasi-Periodic Motion Onset

Recurring burst-out and suppression TET manifests as quasi-periodic motion, which requires precise parameter tuning. Consequently, parametric sweeps must be employed to identify the parameter domain which enables TET, which may be narrow.

4

Methodology

This chapter presents the methodology developed to evaluate the flutter suppression capabilities of the Flap-NES. The python accompanying codes are available at: https://github.com/emmabibollet/MSc_Thesis_Emma_Bibollet-Ruche.git

Section 4.1 defines the methodological scope, modeling assumptions, and limitations. Section 4.2 derives the 3-DOF nonlinear aeroelastic system, modeled as a 2D typical section under unsteady aerodynamics and subsequently non-dimensionalized. Section 4.3 establishes the linear stability analysis used to identify the flutter mechanisms and linear stability boundaries.

To analyze the post-flutter regime, Section 4.4 outlines the Harmonic Balance Method (HBM) implemented to predict the system's post-flutter dynamics. Section 4.5 details the subsequent application of Hill's method to assess Limit Cycle Oscillation (LCO) stability and detect Neimark-Sacker bifurcations. Appendix B validates the numerical framework against established literature.

4.1. Methodology Overview and Limitations

To answer the research questions and evaluate the hypotheses (Chapter 3), the methodology must accomplish three primary tasks: model the non-linear aeroelastic system with the Flap-NES, identify the linear stability boundary and predict the post-flutter dynamic behaviour to identify if any TET occurs. Importantly, the Flap-NES will be compared to a 2-DOF baseline where the Flap-NES is locked, to assess if it effectively suppresses aeroelastic instabilities. The methodology overview and primary limitations are presented in Figure 4.1

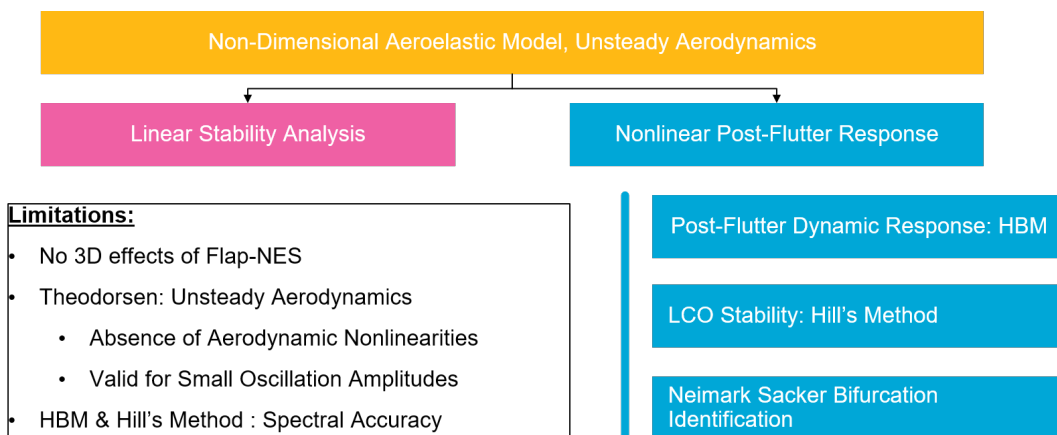


Figure 4.1: Methodology overview and associated limitations

4.1.1. Nonlinear Aeroelastic System Modelling

The aeroelastic system is modelled as a 2D typical section under unsteady aerodynamics, meaning the Flap-NES extends over the entire wingspan. In reality, the Flap-NES is designed to occupy only a portion of the control surface. This formulation provides a baseline to identify key trends and can be expanded to include 3D effects in future work.

The study is limited to unsteady linear aerodynamics described through the Theodorsen formulation which assumes small oscillation amplitudes. High-fidelity nonlinear aerodynamics would introduce a level of complexity which would overtake the effect of the introduced structural nonlinearities of the Flap-NES to control flutter.

Furthermore, unlike a purely internal NES, the Flap-NES is continuously subjected to aerodynamic loading. To accurately assess whether these loads induce premature flutter mechanism, the linear stiffness of the Flap-NES must be retained in the model.

Finally, the governing equations are non-dimensionalised. This ensures that the obtained results apply broadly to a general class of aeroelastic systems, while also preventing numerical instability during matrix inversion.

4.1.2. Numerical Methods for Linear and Post-Flutter Analysis

Firstly, a linear stability analysis of the aeroelastic system with the Flap-NES is performed with through flutter diagrams (eigenvalue analysis) which allow to identify the flutter mechanisms at play and the linear flutter speed.

Evaluating the effectiveness of the Flap-NES to mitigate flutter requires moving beyond linear theory to predict the post-flutter dynamic behaviour through bifurcation diagrams. Three primary numerical methods are available: brute-force time integration, the shooting method, and the harmonic balance method (HBM).

While time integration is exact, it difficulty predicts unstable LCOs and co-existing branches which require testing a high number of arbitrary initial conditions resulting in a high computational cost. [5, 32]. Boundary-value approaches like the shooting method directly calculate the periodic steady-state via a Newton-Raphson scheme. However, the shooting method fundamentally relies on time-marching; integrating an unstable trajectory amplifies numerical errors, rendering the Jacobian ill-conditioned and making automated continuation highly sensitive [32, 8].

HBM approximates the periodic response as a truncated Fourier series. By bypassing time integration entirely, HBM coupled with pseudo-arc-length continuation robustly and efficiently tracks both stable and unstable LCO branches. While limited by its spectral accuracy, the inclusion of a sufficient number of harmonics ensures a highly accurate representation of the physical system [32, 8]. Consequently, HBM is selected as the most computationally viable for obtaining a complete bifurcation diagram and perform a parametric study (Table 4.1).

Table 4.1: Comparison of numerical methods for nonlinear bifurcation tracking

	Time Integration	Shooting	HBM
Computational Cost	High	Medium	Low
Unstable LCO Tracking	Undetectable	Highly Sensitive	Robust
Co-Existing Branches	Difficult to Detect	Detectable	Detectable
Accuracy	Exact	Exact	Spectral Accuracy

While HBM does not inherently determine the stability of the resulting LCO branches. Branch stability is subsequently evaluated using Hill's method, an adaptation of Floquet theory. Finally, the emergence of quasi-periodic behaviour (recurring burstout and suppression TET mechanism) is identified by detecting the Neimark-Sacker bifurcation as discussed in Section 3.1.

4.2. Non Linear Aeroelastic Model

The system is defined as a three-degree-of-freedom (3-DOF) 2D typical section incorporating cubic structural non-linearities. Unsteady aerodynamic forces are calculated using Theodorsen's theory under the assumption of small oscillation amplitudes. To accommodate the numerical continuation analysis, the governing equations of motion are non-dimensionalized and recast into a first-order state-space formulation.

4.2.1. Aeroelastic Model: 2D Typical Section 3-DOF

The aeroelastic system is modelled as a two-dimensional (2D) typical section. The 3-DOF typical section is elastically supported in plunge and pitch by linear springs (if required, hardening cubic non-linearities can be added to model structural geometric stiffening). The flap-NES is attached to the airfoil with a linear and nonlinear spring (Figure 4.2).

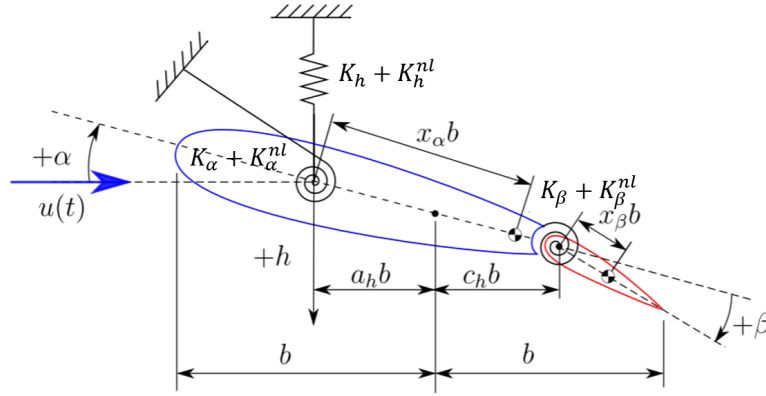


Figure 4.2: Schematic of 3-DOF model of a typical section [11]

Following standard aeroelastic notation, the system's degrees of freedom are defined as follows: h denotes the plunge displacement (positive downward), α represents the pitch angle (positive nose-up), and β defines the flap deflection angle (positive for a downward trailing-edge movement) [6] [13].

Geometrically, the main airfoil is suspended at the elastic axis, located at a distance $a_h b$ from the midchord, where b represents the semi-chord. The wing's centre of mass is positioned at a distance $x_\alpha b$ aft of the elastic axis. The flap is attached to the main airfoil at a hinge located a distance $c_h b$ from the midchord, and the flap's center of mass is positioned at a distance $x_\beta b$ aft of the hinge line.

The linear structural stiffnesses for the heave, pitch, and flap degrees of freedom are denoted by K_h , K_α , and K_β , respectively. The nonlinear cubic stiffnesses are denoted: K_h^{nl} , K_α^{nl} , and K_β^{nl} .

The aeroelastic system, of mass m , is immersed in a horizontal, incompressible airstream with a freestream velocity U and fluid density ρ , flowing from left to right.

The equations of motion for this pitch-plunge-flap two-dimensional wing can be written as:

$$\begin{aligned} m\ddot{h} + S_\alpha\ddot{\alpha} + S_\beta\ddot{\beta} + C_h\dot{h} + K_h h + K_h^{nl} h^3 &= -L \\ S_\alpha\ddot{h} + I_\alpha\ddot{\alpha} + I_{\alpha\beta}\ddot{\beta} + C_\alpha\dot{\alpha} + K_\alpha\alpha + K_\alpha^{nl}\alpha^3 &= M_\alpha \\ S_\beta\ddot{h} + I_{\alpha\beta}\ddot{\alpha} + I_\beta\ddot{\beta} + C_\beta\dot{\beta} + K_\beta\beta + K_\beta^{nl}\beta^3 &= M_\beta \end{aligned} \quad (4.1)$$

where, S_i is the first moment of inertia, and I_i is the second moment of inertia. L denotes the aerodynamic lift, where the negative sign is introduced to ensure consistency with the positive-downward convention of the heave displacement h . The terms M_α and M_β represent the aerodynamic moments evaluated about the main elastic axis and the flap-NES hinge line, respectively. Velocity is defined as $(\dot{})$ and acceleration $(\ddot{})$.

Or, expressed in compact matrix form:

$$\mathbf{A}\ddot{\mathbf{y}} + \mathbf{C}\dot{\mathbf{y}} + \mathbf{E}\mathbf{y} + \mathbf{K}^{nl}\mathbf{y}^3 = \begin{bmatrix} -L \\ M_\alpha \\ M_\beta \end{bmatrix} \quad (4.2)$$

where, the displacement vector is defined as $\mathbf{y} = [h \quad \alpha \quad \beta]^T \in \mathbb{R}^{n \times 1}$ and n the number of DOF. The structural mass matrix ($\mathbf{A} \in \mathbb{R}^{n \times n}$) structural damping matrix ($\mathbf{C} \in \mathbb{R}^{n \times n}$) structural stiffness matrix ($\mathbf{E} \in \mathbb{R}^{n \times n}$) and nonlinear stiffness matrix ($\mathbf{K}^{nl} \in \mathbb{R}^{n \times n}$), are presented in A.1.

4.2.2. Aerodynamic Loads: Unsteady Aerodynamics, Theodorsen

In a 1935 paper, Theodorsen [24] developed the foundational flutter analysis techniques that incorporated the effects of unsteady aerodynamics. This work provided the first analytical solution to the oscillating airfoil problem and will serve as the aerodynamic framework for the present thesis 91 years later! Theodorsen's unsteady aerodynamic theory provides a linearized, small-disturbance solution based on the assumptions of incompressible flow and small oscillation amplitudes.

Aerodynamic Loads Expression: Theodorsen

The aerodynamic loads, L , M_α and M_β , are calculated using Theodorsen's analytical expressions:

$$L(t) = \rho b^2 \left(U\pi\dot{\alpha} + \pi\ddot{h} - \pi b a_h \ddot{\alpha} - U T_4 \dot{\beta} - T_1 b \ddot{\beta} \right) + 2\pi\rho b U C(k)w \quad (4.3)$$

$$\begin{aligned} M_\alpha(t) = & -\rho b^2 \left(-a_h \pi b \ddot{h} + \pi b^2 \left(\frac{1}{8} + a_h^2 \right) \ddot{\alpha} - (T_7 + (c_h - a_h)T_1) b^2 \ddot{\beta} \right) \\ & -\rho b^2 \left(\pi \left(\frac{1}{2} - a_h \right) U b \dot{\alpha} + \left(T_1 - T_8 - (c_h - a_h)T_4 + \frac{T_{11}}{2} \right) U b \dot{\beta} \right) \\ & -\rho b^2 (T_4 + T_{10}) U^2 \beta + 2\rho U b^2 \pi \left(a_h + \frac{1}{2} \right) C(k)w \end{aligned} \quad (4.4)$$

$$\begin{aligned} M_\beta(t) = & -\rho b^2 \left(-T_1 b \ddot{h} + 2T_{13} b^2 \ddot{\alpha} - \frac{1}{\pi} T_3 b^2 \ddot{\beta} \right) \\ & -\rho b^2 \left(\left(-2T_9 - T_1 + T_4 \left(a_h - \frac{1}{2} \right) \right) U b \dot{\alpha} - \frac{1}{2\pi} U b T_4 T_{11} \dot{\beta} \right) \\ & -\frac{\rho b^2 U^2}{\pi} (T_5 - T_4 T_{10}) \beta - \rho b^2 U T_{12} C(k)w \end{aligned} \quad (4.5)$$

where, $C(k)$ is the Theodorsen function, T are the Theodorsen coefficients (presented in Appendix A.2) and w is the aerodynamic downwash.

$$w = U\alpha + \dot{h} + b\left(\frac{1}{2} - a_h\right)\dot{\alpha} + \frac{T_{10}}{\pi}U\beta + b\frac{T_{11}}{2\pi}\dot{\beta} \quad (4.6)$$

Circulatory and Non-Circulatory Components of Aerodynamic Load Equations

The aerodynamic load equations (Equations (4.3) - (4.5)) are separated into circulatory and non-circulatory. Circulatory terms (red) describe the delayed interference caused by the shed wake in unsteady aerodynamics. As the kinematic state of the airfoil changes, the Kutta condition dictates that the flow must remain tangent to, and smoothly leave, the sharp trailing edge. According to Kelvin's circulation theorem, the shed wake vorticity must be equal and opposite to the change in bound circulation over the airfoil, as illustrated in Figure 4.3. As these vortices travel downstream, their induced velocity field interferes with the wing, causing both an amplitude reduction and a phase lag in the resultant aerodynamic forces. Mathematically, this delayed wake interference is captured by the Theodorsen function, $C(k)$. While the circulatory lift in a quasi-static regime is proportional to $2\pi\alpha$, the unsteady interference of the shed wake modifies this relationship, scaling the circulatory lift by $2\pi\alpha C(k)$ [24].

The non-circulatory terms (green) represent all the remaining terms: the instantaneous aerodynamic forces [24].

Theodorsen Theory

Theodorsen's theory is a frequency domain theory because the formulation assumes simple harmonic motion: $C(k)$ depends on the reduced frequency k . The reduced frequency is a dimensionless parameter representing the unsteadiness of the flow and is defined by:

$$k = \frac{\omega c}{2U_\infty} \quad (4.7)$$

As illustrated in Figure 4.4, $C(k)$ is a complex-valued function with a real part, $F(k)$, and an imaginary part, $G(k)$, which dictate the amplitude reduction and phase angle of the aerodynamic response:

$$|C(k)| = \sqrt{F(k)^2 + G(k)^2} \quad (4.8)$$

$$\phi = \arctan\left(\frac{G(k)}{F(k)}\right) \quad (4.9)$$

The physical influence of the wake circulation is characterised by three distinct regimes:

- **Quasi-Steady Limit** ($k \rightarrow 0$): Wake effects are negligible. The imaginary component vanishes, resulting in zero phase angle (no aerodynamic lag) and an unattenuated lift amplitude.
- **Unsteady Region** ($0.05 < k < 0.3$): Wake effects are predominant. The phase lag reaches its maximum magnitude, shifting the aerodynamic forces out of phase with the structural motion and providing the energy-feeding mechanism necessary to drive flutter.
- **Highly Unsteady Limit** ($k \rightarrow \infty$): Wake interference is maximized. The function converges to $C(k) = 0.5 + 0i$, dictating a 50% reduction in the initial circulatory lift amplitude with zero phase lag.

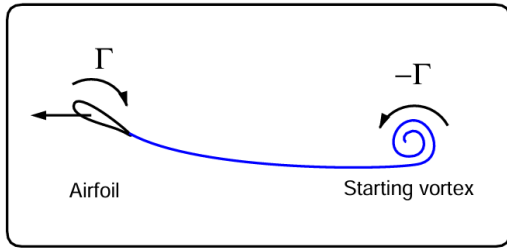


Figure 4.3: Circulation within an initial contour [13]

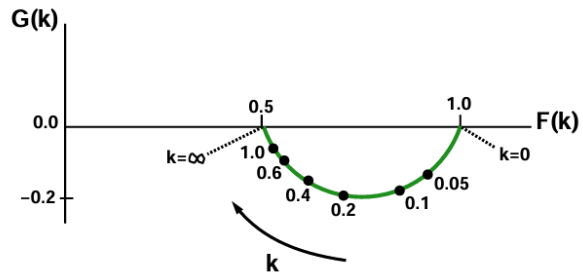


Figure 4.4: Components of the Theodorsen's function $C(k) = F(k) + iG(k)$ [13]

Converting Hybrid Frequency-Time Domain to Time Domain: Wagner Theory

While the non-circulatory terms are formulated in the time domain, the circulatory terms are formulated in a hybrid time-frequency domain. While several methods exist to solve this problem, standard frequency-domain solvers are excluded due to the structural nonlinearities present in the aeroelastic model. Instead, the circulatory aerodynamic loads are converted into a purely time-domain representation using Wagner's theory [31].

Wagner's theory [31] establishes that, under ideal attached flow, the variation in the circulatory lift δl^c , over the chord $2b$ (flat plate) generated by a step in downwash, δw , occurring at time t_0 , is expressed as [6]:

$$\delta l^c(t) = 2\pi\rho b\Phi(t - t_0)\delta w \quad (4.10)$$

With $\phi(t^*)$, the Wagner function, an indicial (step) response, which builds asymptotically from 0.5 to 1 as time increases (Figure 4.5).

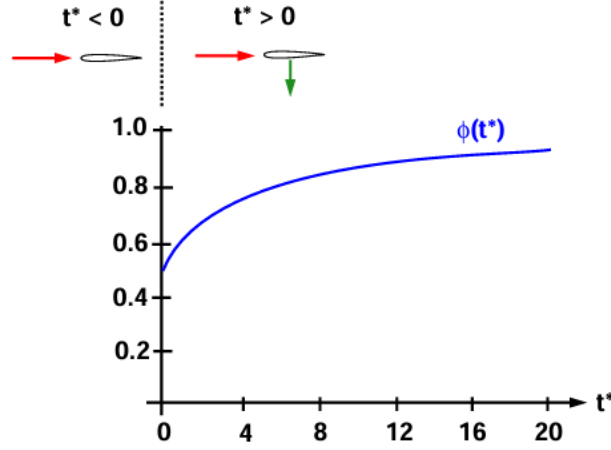


Figure 4.5: Wagner indicial response function to a step change in downwash [13]

The circulatory lift, $L^c(t)$, is computed using the Duhamel integral (Equation 4.11). Comparing Equation 4.11 with Equation (4.3), Wagner's and Theodorsen's circulatory lift expressions can be directly related (Equation 4.12). Finally, applying integration by parts yields Equation 4.13:

$$L^c(t) = 2\pi\rho Ub \left(w(0)\Phi(t) + \int_0^t \Phi(t-t_0) \frac{dw(t_0)}{dt_0} dt_0 \right) \quad (4.11)$$

$$C(k)w(t) = w(0)\Phi(t) + \int_0^t \Phi(t-t_0) \frac{dw(t_0)}{dt_0} dt_0 \quad (4.12)$$

$$C(k)w(t) = w(t)\Phi(0) + \int_0^t \frac{\partial\Phi(t-t_0)}{\partial t_0} w(t_0) dt_0 \quad (4.13)$$

Here, $w_c(t)$ represents the effective circulatory downwash. Wagner's function, $\Phi(t)$, is approximated using Jones' Approximation [15]:

$$\Phi(t) = 1 - \Psi_1 e^{-\epsilon_1 U t/b} - \Psi_2 e^{-\epsilon_2 U t/b} \quad (4.14)$$

where $\Psi_1 = 0.165$, $\Psi_2 = 0.335$, $\epsilon_1 = 0.0455$, and $\epsilon_2 = 0.3$.

Substituting the Wagner approximation and the circulatory downwash term $w_c(t)$ into Equations (4.3)-(4.5) yields the final time-domain aerodynamic loads:

$$L(t) = \rho b^2 \left(U\pi\dot{\alpha} + \pi\ddot{h} - \pi b a_h \ddot{\alpha} - U T_4 \dot{\beta} - T_1 b \ddot{\beta} \right) + 2\pi\rho b U w_c(t) \quad (4.15)$$

$$\begin{aligned} M_\alpha(t) = & -\rho b^2 \left(-a_h \pi b \ddot{h} + \pi b^2 \left(\frac{1}{8} + a_h^2 \right) \ddot{\alpha} - (T_7 + (c_h - a_h) T_1) b^2 \ddot{\beta} \right) \\ & - \rho b^2 \left(\pi \left(\frac{1}{2} - a_h \right) U b \dot{\alpha} + \left(T_1 - T_8 - (c_h - a_h) T_4 + \frac{T_{11}}{2} \right) U b \dot{\beta} \right) \\ & - \rho b^2 (T_4 + T_{10}) U^2 \beta + 2\rho U b^2 \pi \left(a_h + \frac{1}{2} \right) w_c(t) \end{aligned} \quad (4.16)$$

$$\begin{aligned} M_\beta(t) = & -\rho b^2 \left(-T_1 b \ddot{h} + 2T_{13} b^2 \ddot{\alpha} - \frac{1}{\pi} T_3 b^2 \ddot{\beta} \right) \\ & - \rho b^2 \left(\left(-2T_9 - T_1 + T_4 \left(a_h - \frac{1}{2} \right) \right) U b \dot{\alpha} - \frac{1}{2\pi} U b T_4 T_{11} \dot{\beta} \right) \\ & - \frac{\rho b^2 U^2}{\pi} (T_5 - T_4 T_{10}) \beta - \rho b^2 U T_{12} w_c(t) \end{aligned} \quad (4.17)$$

Introducing Lag States

The current, time domain integro-differential equations is modified to a pure system of ODEs through the introduction of 6 lag states. Indeed, applying the integration by parts to the Duhamel integral (Equation 4.13) will render residual convolution integrals due to the exponential. These are defined as six aerodynamic lag states.

$$\begin{aligned}
 w_1(t) &= \int_0^t e^{-\epsilon_1 U(t-t_0)/b} h(t_0) dt_0 & w_2(t) &= \int_0^t e^{-\epsilon_2 U(t-t_0)/b} h(t_0) dt_0 \\
 w_3(t) &= \int_0^t e^{-\epsilon_1 U(t-t_0)/b} \alpha(t_0) dt_0 & w_4(t) &= \int_0^t e^{-\epsilon_2 U(t-t_0)/b} \alpha(t_0) dt_0 \\
 w_5(t) &= \int_0^t e^{-\epsilon_1 U(t-t_0)/b} \beta(t_0) dt_0 & w_6(t) &= \int_0^t e^{-\epsilon_2 U(t-t_0)/b} \beta(t_0) dt_0
 \end{aligned} \tag{4.18}$$

Substituting the downwash (Equation 4.6) into the Duhamel Integral (Equation 4.13), evaluating the integration by parts, and replacing the residual convolution integrals with the newly defined lag states (Equation 4.18) gives:

$$\begin{aligned}
 \Phi(0)w - \int_0^t \frac{\partial \Phi(t-t_0)}{\partial t_0} w(t_0) dt_0 &= \Phi(0)w(t) - \left(h(0) + b \left(\frac{1}{2} - a \right) \alpha(0) + b \frac{T_{11}}{2\pi} \beta(0) \right) \dot{\Phi}(t) \\
 &+ \left(h(t) + b \left(\frac{1}{2} - a \right) \alpha(t) + b \frac{T_{11}}{2\pi} \beta(t) \right) \dot{\Phi}(0) \\
 &- \Psi_1 \left(\frac{\epsilon_1 U}{b} \right)^2 w_1(t) - \Psi_2 \left(\frac{\epsilon_2 U}{b} \right)^2 w_2(t) \\
 &+ \Psi_1 \frac{\epsilon_1 U^2}{b} \left(1 - \epsilon_1 \left(\frac{1}{2} - a \right) \right) w_3(t) \\
 &+ \Psi_2 \frac{\epsilon_2 U^2}{b} \left(1 - \epsilon_2 \left(\frac{1}{2} - a \right) \right) w_4(t) \\
 &+ \Psi_1 \frac{\epsilon_1 U^2}{b\pi} \left(T_{10} - \epsilon_1 \frac{T_{11}}{2} \right) w_5(t) \\
 &+ \Psi_2 \frac{\epsilon_2 U^2}{b\pi} \left(T_{10} - \epsilon_2 \frac{T_{11}}{2} \right) w_6(t)
 \end{aligned} \tag{4.19}$$

Substituting the newly defined aerodynamic lag states into the structural equations of motion introduces six additional unknown variables. Consequently, the aeroelastic system currently consists of three equations governing nine unknowns, rendering it mathematically underdetermined. Six additional equations are required to balance the system. These are derived by applying Leibniz's rule for integrals with variable limits [6] to Equations (4.18).

$$\begin{aligned}
 \dot{w}_1(t) &= h(t) - \frac{\epsilon_1 U}{b} w_1(t) \\
 \dot{w}_2(t) &= h(t) - \frac{\epsilon_2 U}{b} w_2(t) \\
 \dot{w}_3(t) &= \alpha(t) - \frac{\epsilon_1 U}{b} w_3(t) \\
 \dot{w}_4(t) &= \alpha(t) - \frac{\epsilon_2 U}{b} w_4(t) \\
 \dot{w}_5(t) &= \beta(t) - \frac{\epsilon_1 U}{b} w_5(t) \\
 \dot{w}_6(t) &= \beta(t) - \frac{\epsilon_2 U}{b} w_6(t)
 \end{aligned} \tag{4.20}$$

Self-Excited Flutter

Within Equation (4.19), the initial condition excitation vector is defined as:

$$\mathbf{g} = \left(h(0) + b \left(\frac{1}{2} - a \right) \alpha(0) + b \frac{T_{11}}{2\pi} \beta(0) \right) \dot{\Phi}(t) \quad (4.21)$$

For the present analysis of self-excited flutter, the system is assumed to start from rest. Consequently, all initial kinematic displacements ($h(0)$, $\alpha(0)$, $\beta(0)$) are zero, which inherently nullifies the excitation vector $\mathbf{g}$.

Equations of Motion for Pitch-Plunge-Flap

The complete equations of motion for the pitch, plunge airfoil with flap is obtained by assembling equations (4.2), (4.15) - (4.17), (4.19) and (4.20):

$$\begin{aligned} (\mathbf{A} + \rho\mathbf{B})\ddot{\mathbf{y}} + (\mathbf{C} + \rho U\mathbf{D})\dot{\mathbf{y}} + (\mathbf{E} + \rho U^2\mathbf{F})\mathbf{y} + \mathbf{K}^{nl}\mathbf{y}^3 + \rho U^3\mathbf{W}\mathbf{w} &= \mathbf{0} \\ \dot{\mathbf{w}} - \mathbf{W}_1\mathbf{y} - U\mathbf{W}_2\mathbf{w} &= \mathbf{0} \end{aligned} \quad (4.22)$$

where, the displacement vector is described as: $\mathbf{y} = [h \quad \alpha \quad \beta]^T$. The lag vector includes the six lag states: $\mathbf{w} = [w_1 \quad w_2 \quad w_3 \quad w_4 \quad w_5 \quad w_6]^T$

The aerodynamic mass matrix ($\mathbf{B} \in \mathbb{R}^{n \times n}$), aerodynamic damping matrix ($\mathbf{D} \in \mathbb{R}^{n \times n}$), aerodynamic stiffness matrix ($\mathbf{F} \in \mathbb{R}^{n \times n}$) and the aerodynamic states influence matrices ($\mathbf{W} \in \mathbb{R}^{6 \times 1}$, $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{6 \times n}$) are presented in Appendix A.2.

4.2.3. Non-Dimensionalization

The aeroelastic equations of motion are non-dimensionalized with respect to space and time using the semi-chord (b) and freestream velocity (U). Note that the pitch α and flap β DOF do not receive spatial scaling; they are therefore expressed in radians and subsequently post-processed in degrees $^\circ$.

$$\tau = \frac{Ut}{b}, \quad \xi = \frac{h}{b} \quad (4.23)$$

Applying the chain rule, the dimensional time derivatives are transformed into non-dimensional time derivatives. Moving forward, the prime notation ($'$) represents differentiation with respect to non-dimensional time (τ):

$$\begin{aligned} \frac{dh}{dt} &= U\xi' & \frac{d^2h}{dt^2} &= \frac{U^2}{b}\xi'' \\ \frac{d\alpha}{dt} &= \frac{U}{b}\alpha' & \frac{d^2\alpha}{dt^2} &= \frac{U^2}{b^2}\alpha'' \\ \frac{d\beta}{dt} &= \frac{U}{b}\beta' & \frac{d^2\beta}{dt^2} &= \frac{U^2}{b^2}\beta'' \end{aligned} \quad (4.24)$$

Non-Dimensionalized Equations of Motion

Substituting these derivatives into the dimensional equations of motion yields:

$$\begin{aligned} m \frac{U^2}{b} \xi'' + S_\alpha \frac{U^2}{b^2} \alpha'' + S_\beta \frac{U^2}{b^2} \beta'' + C_h U \xi' + K_h b \xi + K_h^{nl} b^3 \xi^3 &= -\pi \rho U^2 b C_L(\tau) \\ S_\alpha \frac{U^2}{b} \xi'' + I_\alpha \frac{U^2}{b^2} \alpha'' + I_{\alpha\beta} \frac{U^2}{b^2} \beta'' + C_\alpha \frac{U}{b} \alpha' + K_\alpha \alpha + K_\alpha^{nl} \alpha^3 &= \pi \rho U^2 b^2 C_{M\alpha}(\tau) \\ S_\beta \frac{U^2}{b} \xi'' + I_{\alpha\beta} \frac{U^2}{b^2} \alpha'' + I_\beta \frac{U^2}{b^2} \beta'' + C_\beta \frac{U}{b} \beta' + K_\beta \beta + K_\beta^{nl} \beta^3 &= \pi \rho U^2 b^2 C_{M\beta}(\tau) \end{aligned} \quad (4.25)$$

Dividing the plunge, pitch, and flap equations by mU^2/b , $I_\alpha U^2/b^2$, and $I_\beta U^2/b^2$ respectively, normalizes the system:

$$\begin{aligned} \xi'' + \frac{S_\alpha}{bm}\alpha'' + \frac{S_\beta}{mb}\beta'' + \frac{C_h b}{mU}\xi' + \frac{K_h b^2}{mU^2}\xi + \frac{K_h^{nl} b^4}{mU^2}\xi^3 &= -C_L(\tau)\frac{\pi\rho b^2}{m} \\ \frac{S_\alpha b}{I_\alpha}\xi'' + \alpha'' + \frac{I_{\alpha\beta}}{I_\alpha}\beta'' + \frac{C_\alpha b}{I_\alpha U}\alpha' + \frac{K_\alpha b^2}{I_\alpha U^2}\alpha + \frac{K_\alpha^{nl} b^2}{I_\alpha U^2}\alpha^3 &= C_{M\alpha}(\tau)\frac{\pi\rho b^4}{I_\alpha} \\ \frac{S_\beta b}{I_\beta}\xi'' + \frac{I_{\alpha\beta}}{I_\beta}\alpha'' + \beta'' + \frac{C_\beta b}{U I_\beta}\beta' + \frac{K_\beta b^2}{U^2 I_\beta}\beta + \frac{K_\beta^{nl} b^2}{U^2 I_\beta}\beta^3 &= C_{M\beta}(\tau)\frac{\pi\rho b^4}{I_\beta} \end{aligned} \quad (4.26)$$

The classical non-dimensional model proposed by Irani et al. [14] non-dimensionalises the flap equation of motion as follows:

$$\frac{x_\beta}{r_\beta^2}\xi'' + \frac{r_\beta^2 + (c_h - a_h)x_\beta}{r_\beta^2}\alpha'' + \beta'' + \frac{2\zeta_\beta\Omega_2}{U^*}\beta' + \frac{\Omega_2^2}{U^{*2}}\beta + \frac{\kappa_\beta\Omega_2^2}{U^{*2}}\beta^3 = \frac{C_{M\beta}(\tau)}{\mu r_\beta^2}$$

In this standard formulation, the frequency ratio (Ω_2), damping ratio (ζ_β), and nonlinear stiffness ratio (κ_β) are all intrinsically coupled to the flap-NES linear stiffness (K_β):

$$\Omega_2 = \frac{1}{\omega_\alpha} \sqrt{\frac{K_\beta}{I_\beta}} \quad \zeta_\beta = \frac{C_\beta}{2\sqrt{I_\beta K_\beta}} \quad \kappa_\beta = \frac{K_\beta^{nl}}{K_\beta}$$

Contrarily, the work by García Pérez et al. [11] sets the linear stiffness to zero ($K_\beta = 0$).

Neither strategy is suitable for the current objective: assessing how Flap-NES parameters influence linear and post-flutter dynamics and whether it enables flutter mitigation via TET. Consequently, the linear flap/pitch stiffness ratio (Ω_2) cannot be omitted ($K_\beta \neq 0$). However, to accurately assess the independent impact of each parameter, they must be decoupled. Therefore, in this formulation, the Flap-NES damping and nonlinear stiffness are explicitly decoupled from the linear stiffness, K_β .

To decouple the flap-NES damping and nonlinear stiffness from the linear stiffness, these parameters are instead normalised against the primary pitch frequency (ω_α) and flap inertia (I_β). This choice is consistent with the non-dimensional strategy as ω_α serves to scale the airspeed (U) to non-dimensional (U^*) and I_β is introduced when normalising the system (Equation 4.26).

The complete set of non-dimensional parameters is defined as follows:

$$\begin{aligned} \mu &= \frac{m}{\rho\pi b^2} & U^* &= \frac{U}{b\omega_\alpha} \\ x_i &= \frac{S_i}{bm} & r_i &= \sqrt{\frac{I_i}{mb^2}} \quad (i = \alpha, \beta) \\ \omega_\xi &= \sqrt{\frac{K_h}{m}} & \omega_i &= \sqrt{\frac{K_i}{I_i}} \quad (i = \alpha, \beta) \\ \zeta_\xi &= \frac{C_h}{2\sqrt{mK_h}} & \zeta_\alpha &= \frac{C_\alpha}{2\sqrt{I_\alpha K_\alpha}} & c_\beta^* &= \frac{C_\beta}{\omega_\alpha I_\beta} \\ \Omega_1 &= \frac{\omega_\xi}{\omega_\alpha} & \Omega_2 &= \frac{\omega_\beta}{\omega_\alpha} \\ \kappa_\xi &= \frac{K_h^{nl} b^2}{K_h} & \kappa_\alpha &= \frac{K_\alpha^{nl}}{K_\alpha} & \Gamma_\beta &= \frac{K_\beta^{nl}}{\omega_\alpha^2 I_\beta} \end{aligned}$$

Consequently, the adapted flap parameters (c_β^* and Γ_β) no longer represent the classical damping and nonlinear stiffness ratios. The physical interpretation of the above non-dimensional parameters is presented in Table 4.2.

Table 4.2: Physical interpretation of non-dimensional aeroelastic parameters

Param.	Definition	Small Value	Large Value
μ	Mass Ratio	Aero forces dominate (heavy fluid)	Inertia dominates (typical aircraft)
U^*	Reduced Velocity	Low airspeed/stiff wing	High airspeed/flexible wing
x_i	Static Unbalance	CG near rotation axis; low inertial coupling.	CG far from rotation axis; high inertial coupling.
r_i	Radius of Gyration	Mass concentrated near rotation axis; low rotational inertia	Mass distributed outward; high rotational inertia.
Ω_1	Heave/Pitch Ratio	Soft Plunge; resistant to twisting.	Soft Pitch; resistant to bending.
Ω_2	Flap/Pitch Ratio	Soft flap (typical for NES).	Stiff flap.
ζ_i	Damping Ratio	Weak structural energy dissipation.	Strong structural energy dissipation.
c_β^*	Flap Damping	Weak energy dissipation by the flap.	Strong energy dissipation by the flap.
κ_i	Nonlinear Ratio	Weak hardening; linear stiffness	Strong hardening; nonlinear stiffness
Γ_β	Flap Nonlinearity	Weak nonlinear restoring force.	Dominant nonlinear restoring force.

Substituting these parameters produces the final non-dimensional equations of motion:

$$\begin{aligned}
 \xi'' + x_\alpha \alpha'' + x_\beta \beta'' + \frac{2\zeta_\xi \Omega_1}{U^*} \xi' + \frac{\Omega_1^2}{U^{*2}} \xi + \frac{\Omega_1^2 \kappa_\xi}{U^{*2}} \xi^3 &= -\frac{C_L(\tau)}{\mu} \\
 \frac{x_\alpha}{r_\alpha^2} \xi'' + \alpha'' + \frac{r_\beta^2 + (c_h - a_h)x_\beta}{r_\alpha^2} \beta'' + \frac{2\zeta_\alpha}{U^*} \alpha' + \frac{1}{U^{*2}} \alpha + \frac{\kappa_\alpha}{U^{*2}} \alpha^3 &= \frac{C_{M\alpha}(\tau)}{\mu r_\alpha^2} \\
 \frac{x_\beta}{r_\beta^2} \xi'' + \frac{r_\beta^2 + (c_h - a_h)x_\beta}{r_\beta^2} \alpha'' + \beta'' + \frac{c_\beta^*}{U^*} \beta' + \frac{\Omega_2^2}{U^{*2}} \beta + \frac{\Gamma_\beta}{U^{*2}} \beta^3 &= \frac{C_{M\beta}(\tau)}{\mu r_\beta^2}
 \end{aligned} \tag{4.27}$$

This system is expressed in compact matrix form as:

$$\begin{aligned}
 \left(\mathbf{A}^* + \frac{1}{\mu} \mathbf{B}^* \right) \mathbf{y}^{*''} + \left(\frac{1}{U^*} \mathbf{C}^* + \frac{1}{\mu} \mathbf{D}^* \right) \mathbf{y}^{*'} + \left(\frac{1}{U^{*2}} \mathbf{E}^* + \frac{1}{\mu} \mathbf{F}^* \right) \mathbf{y}^* + \frac{1}{U^{*2}} \mathbf{K}_{nl}^* \mathbf{y}^{*3} + \frac{1}{\mu} \mathbf{W}^* \mathbf{w}^* &= \mathbf{0} \\
 \mathbf{w}^{*'} - \mathbf{W}_1^* \mathbf{y}^* - \mathbf{W}_2^* \mathbf{w}^* &= \mathbf{0}
 \end{aligned} \tag{4.28}$$

Where the non-dimensional displacement vector is: $\mathbf{y}^* = [\xi \quad \alpha \quad \beta]^T$ and the non-dimensional aerodynamic lag vector is: $\mathbf{w}^* = [w_1^* \quad w_2^* \quad w_3^* \quad w_4^* \quad w_5^* \quad w_6^*]^T$.

The non-dimensional stiffness and aerodynamic matrices $\mathbf{A}^*$, $\mathbf{C}^*$, $\mathbf{E}^*$, $\mathbf{K}_{nl}^*$, $\mathbf{B}^*$, $\mathbf{D}^*$, $\mathbf{F}^*$, $\mathbf{W}^*$, $\mathbf{W}_1^*$, and $\mathbf{W}_2^*$ are presented in Appendix A.3 and have the same size as their dimensional versions.

4.2.4. First-Order State Space Formulation

The governing aeroelastic equations (4.28) in the time domain can be rewritten as a set of first-order ordinary differential equations (ODEs):

$$\mathbf{x}' = \mathbf{Q}\mathbf{x} + \mathbf{f}_{nl}(\mathbf{x}) \quad (4.29)$$

where the state vector is $\mathbf{x} = [\xi' \quad \alpha' \quad \beta' \quad \xi \quad \alpha \quad \beta \quad w_1 \quad \dots \quad w_6]^T \in \mathbb{R}^{(n+6) \times 1}$.

Defining the effective non-dimensional mass matrix as $\mathbf{M} = \mathbf{A}^* + \frac{1}{\mu}\mathbf{B}^* \in \mathbb{R}^{n \times n}$, the first-order linear state-space matrix $\mathbf{Q}$ is constructed by isolating the structural accelerations alongside the kinematic and aerodynamic state derivatives.

The first order matrix $\mathbf{Q} \in \mathbb{R}^{(2n+6) \times (2n+6)}$ is defined as:

$$\mathbf{Q} = \begin{bmatrix} -\mathbf{M}^{-1} \left(\frac{1}{U^*} \mathbf{C}^* + \frac{1}{\mu} \mathbf{D}^* \right) & -\mathbf{M}^{-1} \left(\frac{1}{U^{*2}} \mathbf{E}^* + \frac{1}{\mu} \mathbf{F}^* \right) & -\mathbf{M}^{-1} \left(\frac{1}{\mu} \mathbf{W}^* \right) \\ \mathbf{I}_{3 \times 3} & \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 6} \\ \mathbf{0}_{6 \times 3} & \mathbf{W}_1^* & \mathbf{W}_2^* \end{bmatrix} \quad (4.30)$$

The nonlinear forcing vector $\mathbf{f}_{nl} \in \mathbb{R}^{(2n+6) \times n}$ isolates the cubic stiffness strictly to the non-dimensional displacements ($\mathbf{y}^*$):

$$\mathbf{f}_{nl} = \begin{bmatrix} -\mathbf{M}^{-1} \mathbf{K}_{nl}^* \\ \mathbf{0}_{9 \times 3} \end{bmatrix} \mathbf{y}^{*3} \quad (4.31)$$

This state-space assembly logic applies directly to the dimensional model by substituting the corresponding dimensional matrices from Equation (4.22).

4.3. Linear Stability Analysis

Flutter diagrams are used to characterise the flutter mechanism and identify the Hopf bifurcation airspeeds. This initial step determines the linear stability boundary of the nonlinear aeroelastic system.

Before flutter onset, structural displacements remain infinitesimally small. In this regime, the purely cubic structural nonlinearities are mathematically negligible, and the system's stability is governed entirely by the linear state matrix, $\mathbf{Q}$. Consequently, the flap-NES operates solely through its linear properties: size (c_h), mass distribution (x_β, r_β), damping (c_β^*), and its linear stiffness (Ω_2).

4.3.1. Flutter Diagrams: Eigenvalue Analysis

Flutter diagrams identify the onset of aeroelastic instability and characterise the underlying flutter mechanism by plotting the damping and frequency of the system's natural modes against airspeed.

Before flutter occurs, structural displacements are small enough that cubic stiffness terms are negligible. The system is therefore governed by the linear, first-order state equation:

$$\mathbf{x}'(t) = \mathbf{Q}\mathbf{x}(t) \quad (4.32)$$

where $\mathbf{Q}$ is the aeroelastic state matrix at a given non-dimensional airspeed U^* .

To extract the natural modes, a solution is assumed in the form:

$$\mathbf{x}(t) = \phi e^{\lambda t} \quad (4.33)$$

where $\lambda = \sigma \pm i\omega$ is the complex eigenvalue and ϕ is the complex eigenvector (mode shape).

Substituting this assumed solution into the state equation and factoring out the exponential term yields the standard eigenvalue problem:

$$(\mathbf{Q} - \lambda\mathbf{I})\phi = \mathbf{0} \quad (4.34)$$

For a non-trivial solution ($\phi \neq \mathbf{0}$), the determinant must be zero:

$$\det(\mathbf{Q} - \lambda\mathbf{I}) = 0 \quad (4.35)$$

Solving this characteristic equation at incremental airspeeds yields the complex eigenvalues used to construct the flutter diagrams:

Damping Ratio: The real part of the complex eigenvalue, σ . A negative value represents a stable focus while a positive value represents an unstable focus (Table 4.3). Following standard aeroelastic conventions, the real part of the eigenvalue is normalized and plotted as the dimensionless damping ratio, defined as $\zeta = -\frac{\sigma}{\omega_n}$ [4].

Natural Frequency: The imaginary part of the complex eigenvalue and defined as $\omega_n = |\lambda|$.

Table 4.3: Classification of fixed points based on eigenvalues

Fixed point	Eigenvalues	Response
Stable focus	Complex with negative real parts	Decaying oscillation
Centre	Imaginary	Constant oscillation
Unstable focus	Complex with positive real parts	Diverging oscillation

4.3.2. Hopf Bifurcation

In nonlinear dynamics, the transition from a stable static equilibrium to a bounded, periodic Limit Cycle Oscillation (LCO) is defined as a Hopf bifurcation. Accurately detecting this transition is the fundamental first step required to anchor subsequent nonlinear continuation algorithms.

A Hopf bifurcation occurs when a complex conjugate pair of eigenvalues from the linear state matrix ($\mathbf{Q}$) becomes purely imaginary ($\sigma = 0, \omega \neq 0$). The Hopf bifurcation occurs at the airspeed where the static equilibrium solution changes stability.

Numerical Detection

To pinpoint the Hopf bifurcation numerically, the Hopf Test Function (τ_H) is defined to monitor the stability of the complex conjugate eigenvalue pairs extracted from $\mathbf{Q}$:

$$\tau_H(U) = \prod_{j=1}^{n_c/2} (\lambda_j(U) + \lambda_j^*(U)) \quad (4.36)$$

where λ_j and λ_j^* represent the j -th complex conjugate pair, and n_c is the total number of complex eigenvalues.

The sum of a conjugate pair equates to twice its real part (2σ). When a pair crosses the imaginary axis ($\sigma = 0$), the sum equals zero, rendering the test function $\tau_H = 0$ [7].

The detection algorithm evaluates $\mathbf{Q}$ over incrementally increasing airspeeds using a step size of ΔU . A potential bifurcation is bracketed when the test function changes sign between consecutive velocity steps:

$$\tau_{H,i+1} \cdot \tau_{H,i} < 0 \quad (4.37)$$

Once a sign change is detected, the exact Hopf bifurcation airspeed (U_f) and its associated flutter frequency (ω_f) are isolated using a secant root-finding method within a specified numerical tolerance [7].

4.4. Nonlinear Post-Flutter Response

This section establishes the numerical framework required to compute the post-flutter global dynamic response of the nonlinear aeroelastic system. Following the Hopf bifurcation, the system may converge to an LCO or diverge along an unstable equilibrium branch. By approximating the periodic response as a truncated Fourier series, the Harmonic Balance Method (HBM) generates the complete bifurcation diagram, including both stable and unstable LCO branches, at a low computational cost, therefore facilitating the parametric study of the flap-NES.

4.4.1. Principle of HBM [32]

In an effort of progressive complexity, the HBM method will be explained through a forced n degrees of freedom duffing system governed by the following equation of motion:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) + \mathbf{f}_{nl}(\mathbf{x}, \dot{\mathbf{x}}) = \mathbf{P}(\omega t) \quad (4.38)$$

with, $\mathbf{x}(t) \in \mathbb{R}^{n \times 1}$ the displacement vector containing the n DOFs, $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{n \times n}$, the generalized mass, damping and stiffness matrices, $\mathbf{f}_{nl} \in \mathbb{R}^{n \times 1}$ the vector of nonlinear forces, $\mathbf{P}(\omega t) \in \mathbb{R}^{n \times 1}$ the periodic excitation forces and ω the excitation frequency.

HBM assumes a periodic steady-state response (excludes the transient regimes). Consequently, displacements, excitation, and nonlinear forces are approximated as truncated Fourier series of order H , the number of included harmonics.

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{X}^0 + \sum_{k=0}^H \mathbf{X}_k^c \cos(k\omega t) + \mathbf{X}_k^s \sin(k\omega t) \\ \mathbf{f}_{nl} &= \mathbf{F}^0 + \sum_{k=0}^H \mathbf{F}_k^c \cos(k\omega t) + \mathbf{F}_k^s \sin(k\omega t) \end{aligned} \quad (4.39)$$

$$\mathbf{P}(\omega t) = \mathbf{P}^0 + \sum_{k=0}^H \mathbf{P}_k^c \cos(k\omega t) + \mathbf{P}_k^s \sin(k\omega t) \quad (4.40)$$

The Fourier coefficients are assembled into global vectors of unknowns $\mathbf{X}, \mathbf{F}_{nl}, \mathbf{P} \in \mathbb{R}^{n(2H+1) \times 1}$, defined as:

$$\begin{aligned} \mathbf{X} &= [\mathbf{X}_0^T, \mathbf{X}_c^{1T}, \mathbf{X}_s^{1T}, \dots, \mathbf{X}_c^{HT}, \mathbf{X}_s^{HT}]^T \\ \mathbf{F}_{nl} &= [\mathbf{F}_0^T, \mathbf{F}_c^{1T}, \mathbf{F}_s^{1T}, \dots, \mathbf{F}_c^{HT}, \mathbf{F}_s^{HT}]^T \\ \mathbf{P} &= [\mathbf{P}_0^T, \mathbf{P}_c^{1T}, \mathbf{P}_s^{1T}, \dots, \mathbf{P}_c^{HT}, \mathbf{P}_s^{HT}]^T \end{aligned} \quad (4.41)$$

To formulate the system in a compact matrix notation, the vector of trigonometric basis functions, $\mathbf{T}(\omega t) \in \mathbb{R}^{1 \times (2H+1)}$, is first defined as:

$$\mathbf{T}(\omega t) = [1, \cos(\omega t), \sin(\omega t), \dots, \cos(H\omega t), \sin(H\omega t)] \quad (4.42)$$

By utilizing the Kronecker tensor product ($\otimes$) and the identity matrix, denoted as $\mathbf{I}_n \in \mathbb{R}^{n \times n}$, the time-domain displacement vector $\mathbf{x}(t)$, the nonlinear force vector $\mathbf{f}_{nl}(\mathbf{x}, \dot{\mathbf{x}})$, and the external excitation $\mathbf{P}(t)$ are expressed directly in terms of their spectral coefficients:

$$\begin{aligned} \mathbf{x}(t) &= (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) \mathbf{X} \\ \mathbf{f}_{nl}(\mathbf{x}, \dot{\mathbf{x}}) &= (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) \mathbf{F}_{nl} \\ \mathbf{P}(t) &= (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) \mathbf{P} \end{aligned} \quad (4.43)$$

To compute the time derivatives required for the equations of motion, a block-diagonal derivative operator $\nabla \in \mathbb{R}^{(2H+1) \times (2H+1)}$ is introduced. This matrix accounts for the differentiation of the harmonic terms and is structured as:

$$\nabla = \text{diag}(0, \nabla_1, \dots, \nabla_j, \dots, \nabla_H) \quad \text{with} \quad \nabla_j = j \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad (4.44)$$

Consequently, the velocity and acceleration vectors are derived by applying this operator to the harmonic coefficients, scaled by the angular frequency ω :

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \omega (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) (\nabla \otimes \mathbf{I}_n) \mathbf{X} = \omega ((\mathbf{T}(\omega t) \nabla) \otimes \mathbf{I}_n) \mathbf{X} \\ \ddot{\mathbf{x}}(t) &= \omega^2 (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) (\nabla^2 \otimes \mathbf{I}_n) \mathbf{X} = \omega^2 ((\mathbf{T}(\omega t) \nabla^2) \otimes \mathbf{I}_n) \mathbf{X} \end{aligned} \quad (4.45)$$

Galerkin Procedure

Substituting the truncated Fourier approximations (Eqs(4.43), (4.45)) into the governing equation of motion (Eq. (4.38)) introduces a residual error vector, $\mathbf{R}(\mathbf{X}, \omega, t)$, as the finite series cannot perfectly satisfy the equality at every instant.

$$\begin{aligned} \mathbf{R}(\mathbf{X}, \omega, t) &= \omega^2 \mathbf{M} [(\mathbf{T}(\omega t) \nabla^2) \otimes \mathbf{I}_n] \mathbf{X} + \omega \mathbf{C} [(\mathbf{T}(\omega t) \nabla) \otimes \mathbf{I}_n] \mathbf{X} \\ &\quad + \mathbf{K} (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) \mathbf{X} + (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) \mathbf{F}_{nl} - (\mathbf{T}(\omega t) \otimes \mathbf{I}_n) \mathbf{P} \end{aligned} \quad (4.46)$$

To transform the continuous time-domain differential equations into a discrete frequency domain algebraic system, a Galerkin projection is employed. The residual, Eq (4.46), is projected on the trigonometric basis $\mathbf{T}(\omega t)$.

$$\frac{2}{\tau} \int_0^\tau (\mathbf{T}(\omega t))^T \otimes \mathbf{I}_n \mathbf{R}(\mathbf{X}, \omega, t) dt \quad (4.47)$$

The evaluation of this integral relies on the fundamental orthogonality properties:

$$\begin{aligned} \langle \sin(m\tau), \cos(n\tau) \rangle &= 0 \quad \forall m, n \\ \langle \cos(m\tau), \cos(n\tau) \rangle &= \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \neq 0 \\ 2\pi & \text{if } m = n = 0 \end{cases} \\ \langle \sin(m\tau), \sin(n\tau) \rangle &= \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \neq 0 \\ 0 & \text{if } m = n = 0 \end{cases} \end{aligned} \quad (4.48)$$

As defined in Eqs.(4.48), all cross-terms between differing frequencies or differing phases integrate to zero. Consequently, the projection operation acts as a filter, isolating the kinetics of each harmonic component. This decouples the continuous time-domain problem into a set of $n(2H + 1)$ independent algebraic equations in the frequency domain:

$$\mathbf{R}(\mathbf{X}, \omega) = \mathbf{Z}(\omega) \mathbf{X} + \mathbf{F}_{nl} - \mathbf{P} = 0 \quad (4.49)$$

where,

$$\mathbf{Z}(\omega) = \omega^2 \nabla^2 \otimes \mathbf{M} + \omega \nabla \otimes \mathbf{C} + \mathbf{I}_{2H+1} \otimes \mathbf{K} \quad (4.50)$$

The solutions of Equation (4.49) are obtained through an iterative Newton-Raphson procedure described below.

From an initial guess, $\mathbf{X}^0$, a linear correction vector $\delta\mathbf{X}^k$ is computed at every iteration k , by solving:

$$\frac{dR^k}{d\mathbf{X}}\delta\mathbf{X}^k = -\mathbf{R}^k \quad (4.51)$$

The solution is then updated for subsequent iterations:

$$\mathbf{X}^{k+1} = \mathbf{X}^k + \delta\mathbf{X} \quad (4.52)$$

This iterative process is repeated until the norm of the residual vector, $\|\mathbf{R}^k\|$, falls below a predefined convergence tolerance, ϵ_{NR} .

4.4.2. Treatment of Nonlinearities (AFT Method)

Evaluating the nonlinear restoring forces, modelled as a cubic hardening stiffness, directly in the frequency domain requires convolution operations. While it is possible to obtain these analytically, the nonlinearities will be evaluated in the time domain using the Alternating Frequency/Time (AFT) method (Figure 4.6) [32, 8]. Note, however, that in the case of systems with many DOFs may result in higher numerical cost.

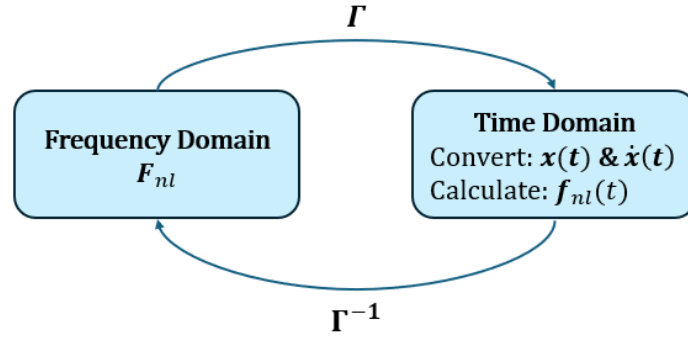


Figure 4.6: Alternating frequency/time (AFT) method

The fundamental period of oscillation is discretised into N uniformly spaced time samples. The state variables, displacement $\mathbf{x}(t)$ and velocity $\dot{\mathbf{x}}(t)$, along with the resulting nonlinear force vector $\mathbf{f}_{nl}(t)$, are evaluated at these discrete time steps. This transformation from the frequency domain to the time domain uses the Discrete Fourier Transform (DFT) matrix, $\mathbf{\Gamma} \in \mathbb{R}^{N \times (2n+1)}$, defined as:

$$\mathbf{\Gamma} = \begin{bmatrix} \mathbf{T}(\omega t_1) \\ \vdots \\ \mathbf{T}(\omega t_N) \end{bmatrix} = \begin{bmatrix} 1 & \cos(\theta_1) & \sin(\theta_1) & \dots & \cos(H\theta_1) & \sin(H\theta_1) \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 1 & \cos(\theta_N) & \sin(\theta_N) & \dots & \cos(H\theta_N) & \sin(H\theta_N) \end{bmatrix} \quad (4.53)$$

$$\mathbf{x}(t) = (\mathbf{\Gamma} \otimes \mathbf{I}_n) \mathbf{X} \quad (4.54)$$

$$\dot{\mathbf{x}}(t) = \omega[(\mathbf{\Gamma} \nabla) \otimes \mathbf{I}_n] \mathbf{X} \quad (4.55)$$

After computing the nonlinear force in the time domain, it is transformed back into the frequency domain to yield the Fourier coefficients, $\mathbf{F}_{nl}$. This projection is achieved using the inverse DFT matrix, $\mathbf{\Gamma}^{-1} \in \mathbb{R}^{(2n+1) \times N}$:

$$\mathbf{F}_{nl} = (\mathbf{\Gamma}^{-1} \otimes \mathbf{I}_n) \mathbf{f}_{nl}(t) \quad (4.56)$$

where,

$$\mathbf{\Gamma}^{-1} = \frac{1}{N} \begin{bmatrix} 1 & \dots & 1 \\ 2\cos(\theta_1) & \dots & 2\cos(\theta_N) \\ 2\sin(\theta_1) & \dots & 2\sin(\theta_N) \\ \dots & \dots & \dots \\ 2\cos(H\theta_1) & \dots & 2\cos(H\theta_N) \\ 2\sin(H\theta_1) & \dots & 2\sin(H\theta_N) \end{bmatrix} \quad (4.57)$$

Within the previously established Newton-Raphson scheme, calculating the Jacobian matrix $\frac{d\mathbf{R}}{d\mathbf{X}}$ necessitates the derivative of this nonlinear force vector:

$$\frac{d\mathbf{R}}{d\mathbf{X}} = \mathbf{Z}(\omega) + \frac{d\mathbf{F}_{nl}}{d\mathbf{X}} \quad (4.58)$$

The nonlinear force Jacobian block is computed numerically using a forward finite difference method.

4.4.3. Pseudo-Arclength Continuation [32]

A linear system has exactly one output for a given input. In contrast, nonlinear responses can fold back on themselves or two periodic branches may co-exist, producing multi-valued solutions for the same input parameter. At these limit points (folds), standard Newton-Raphson schemes fail due to singular Jacobian matrices. Coupling a pseudo-arc length continuation scheme with the HBM enables the continuous tracking of both stable and unstable solution branches.

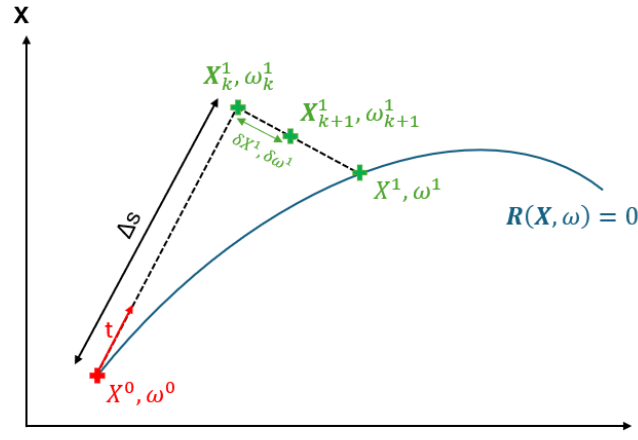


Figure 4.7: Pseudo arc length continuation scheme, k describes the iteration

The system is treated as having two unknowns, $\mathbf{X}$ and ω . After obtaining the first starting point $\mathbf{X}^0$ and ω^0 , the tangent $\mathbf{t} = (\Delta\mathbf{X}^T, \Delta\omega)$ is computed by solving:

$$\begin{bmatrix} \frac{d\mathbf{R}}{d\mathbf{X}} & \frac{d\mathbf{R}}{d\omega} \end{bmatrix} \begin{bmatrix} \Delta\mathbf{X} \\ \Delta\omega \end{bmatrix} = \mathbf{R}(\mathbf{X}, \omega) = \mathbf{0} \quad (4.59)$$

where,

$$\frac{d\mathbf{R}}{d\omega} = \frac{d\mathbf{X}}{d\omega} \mathbf{X} = (2\omega \nabla^2 \otimes \mathbf{M} + \nabla \otimes \mathbf{C}) \mathbf{X} \quad (4.60)$$

A normalization condition, $||\mathbf{t}|| = \Delta \mathbf{X}^T \Delta \mathbf{X} + \Delta \omega^2 = 1$, makes the tangent vector unit length. The following solution, $\mathbf{X}_k^1$ and ω_k^1 is estimated at a specific step size Δs along the direction of $\mathbf{t}$.

$$\begin{bmatrix} \Delta \mathbf{X}_k^1 \\ \Delta \omega_k^1 \end{bmatrix} = \begin{bmatrix} \Delta \mathbf{X}^0 \\ \Delta \omega^0 \end{bmatrix} + \Delta s \mathbf{t} \quad (4.61)$$

This initial prediction is subsequently corrected along the orthogonal direction of $\mathbf{t}$ (Figure 4.7), resulting in an augmented Newton-Raphson scheme. By coupling the primary residual equations with this geometric constraint, the augmented Jacobian matrix remains strictly non-singular. The corrections $\delta \mathbf{X}$ and $\delta \omega$ calculated iteratively by:

$$\begin{bmatrix} \frac{dR^k}{d\mathbf{X}} & \frac{dR^k}{d\omega} \\ \Delta \mathbf{X} & \Delta \omega \end{bmatrix} \begin{bmatrix} \delta \mathbf{X}^{k+1} \\ \delta \omega^{k+1} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}^k \\ 0 \end{bmatrix} \quad (4.62)$$

The solution is updated for each subsequent iteration k :

$$\begin{aligned} \mathbf{X}^{k+1} &= \mathbf{X}^k + \delta \mathbf{X}^{k+1} \\ \omega^{k+1} &= \omega^k + \delta \omega^{k+1} \end{aligned} \quad (4.63)$$

This iterative process is repeated until the norm of the residual vector, $||\mathbf{R}^k||$, falls below the predefined convergence tolerance, ϵ_{NR} .

Finally, the step size, Δs , is adapted relative to the number of Newton-Raphson iterations required for convergence and ensures a robust continuation.

4.4.4. HBM for Nonlinear Aeroelastic System

To capture the post-flutter response of the 3-DOF nonlinear aeroelastic system, the previously established HBM and pseudo-arc length continuation methods are adapted.

As derived in Section 4.2.4, the governing equations of the nonlinear aeroelastic system are expressed in state-space form as:

$$\mathbf{x}' = \mathbf{Q}\mathbf{x} + \mathbf{f}_{nl} \quad (4.64)$$

where the state vector is $\mathbf{x} = [\xi' \quad \alpha' \quad \beta' \quad \xi \quad \alpha \quad \beta \quad w_1 \quad \dots \quad w_6]^T$ of size $n = 12$

The nonlinear forcing vector $\mathbf{f}_{nl}$ isolates the cubic stiffness strictly to the structural displacements, defined as $\mathbf{y}^* = [\xi \quad \alpha \quad \beta]^T$:

$$\mathbf{f}_{nl} = \begin{bmatrix} -\mathbf{M}^{-1} \mathbf{K}_{nl}^* \\ \mathbf{0}_{9 \times 3} \end{bmatrix} \mathbf{y}^{*3} \quad (4.65)$$

Critically, the linear state matrix $\mathbf{Q}$ is a function of the non-dimensional airspeed, U^* . Note that time is non-dimensional and represented by τ . Following the same procedure applied to the duffing system, the displacement, velocity and nonlinear force vectors are expanded into their Fourier spectral coefficients:

$$\begin{aligned} \mathbf{x}(\tau) &= (\mathbf{T}(\omega\tau) \otimes \mathbf{I}_n) \mathbf{X} \\ \mathbf{f}_{nl}(\tau) &= (\mathbf{T}(\omega\tau) \otimes \mathbf{I}_n) \mathbf{F}_{nl} \end{aligned}$$

Applying the Galerkin procedure projects the system into the frequency domain, and the residual error vector, $\mathbf{R}(\mathbf{X}, \omega, U^*)$ is now composed of $n(2H + 1)$ independent algebraic equations:

$$\mathbf{R}(\mathbf{X}, \omega, U^*) = \mathbf{Z}(\omega, U^*)\mathbf{X} - \mathbf{F}_{nl}(\mathbf{X}) \quad (4.66)$$

where, $\mathbf{Z}(\omega, U^*) = \omega \nabla \otimes \mathbf{I}_n - \mathbf{I}_{2H+1} \mathbf{Q}$

Pseudo Arc-Length Continuation

The fundamental shift in this formulation is the expansion of the continuation space to include three independent unknowns: $\mathbf{X}$ (Fourier coefficients), ω (LCO frequency) and U^* (non-dimensional airspeed).

From a known initial solution, $\mathbf{X}^0$, ω^0 and U^{*0} (the prediction of which is detailed in the subsequent section), the tangent predictor vector $\mathbf{t} = (\Delta \mathbf{X}, \Delta \omega, \Delta U^*)$ is determined by solving:

$$\begin{bmatrix} \frac{d\mathbf{R}}{d\mathbf{X}} & \frac{d\mathbf{R}}{d\omega} & \frac{d\mathbf{R}}{dU^*} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{X} \\ \Delta \omega \\ \Delta U^* \end{bmatrix} = \mathbf{0} \quad (4.67)$$

where,

$$\begin{aligned} \frac{d\mathbf{R}}{d\mathbf{X}} &= \mathbf{Z}(\omega, U^*) - \frac{d\mathbf{F}_{nl}}{d\mathbf{X}} \\ \frac{d\mathbf{R}}{d\omega} &= \frac{d\mathbf{X}(\omega, U^*)}{d\omega} \mathbf{X} = \nabla \otimes \mathbf{I}_n \mathbf{X} \\ \frac{d\mathbf{R}}{dU^*} &= \frac{d\mathbf{Z}(\omega, U^*)}{dU^*} \mathbf{X} \end{aligned} \quad (4.68)$$

The nonlinear force Jacobian, $\frac{d\mathbf{F}_{nl}}{d\mathbf{X}}$, and the parameter derivative, $\frac{d\mathbf{Z}(\omega, U^*)}{dU^*}$, are approximated numerically via a forward finite-difference scheme.

By normalising the tangent vector to unit length, the following solution, $\mathbf{X}_k^1$, ω_k^1 and U_k^{*1} , is estimated at a specific step size Δs along the direction of $\mathbf{t}$.

$$\begin{bmatrix} \Delta \mathbf{X}_k^1 \\ \Delta \omega_k^1 \\ \Delta U_k^{*1} \end{bmatrix} = \begin{bmatrix} \Delta \mathbf{X}^0 \\ \Delta \omega^0 \\ \Delta U^{*0} \end{bmatrix} + \Delta s \mathbf{t} \quad (4.69)$$

This prediction must be iteratively corrected onto the exact solution branch. Because the continuation space is now three-dimensional, the temporal phase of the oscillation is arbitrary. To ensure a mathematically unique solution during the augmented Newton-Raphson correction, an additional phase constraint, $\mathbf{E}_j^T$, is enforced within the augmented Jacobian matrix to anchor the periodic orbit [8].

$$\begin{bmatrix} \frac{d\mathbf{R}^k}{d\mathbf{X}} & \frac{d\mathbf{R}^k}{d\omega} & \frac{d\mathbf{R}^k}{dU^*} \\ \Delta \mathbf{X} & \Delta \omega & \Delta U^* \\ \mathbf{E}_j^T & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta \mathbf{X}^{k+1} \\ \delta \omega^{k+1} \\ \delta U^{*k+1} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}^k \\ 0 \\ 0 \end{bmatrix} \quad (4.70)$$

The phase condition is achieved by anchoring one specific Fourier coefficient. $\mathbf{E}_j^T \in \mathbb{R}^{n \times (2n+1)}$ has all its elements set to zero, except for a value of 1 at the specific index, j , corresponding to the constrained coefficient.

The selection of this phase-locking coordinate depends on the dominant physical dynamics. As a standard practice for this aeroelastic system, the constraint is applied to the fundamental sine harmonic of the specific degree of freedom driving the critical flutter eigenmode [8].

Finally, the step size, Δs is adapted relative to the number of Newton-Raphson iterations required for convergence and ensures a robust continuation. In the numerical implementation, the step size is bounded by the user by defining Δs_{max} and Δs_{min} .

Initial Condition for Pseudo-Arc Length Continuation

The continuation algorithm requires a first point on the LCO bifurcation branch to begin tracking the remaining branch. This first point is therefore the initial condition for the pseudo-arc-length continuation scheme.

The known Hopf bifurcation point, of airspeed U_f^* and frequency ω_f , is perturbed. The goal is to step off the trivial equilibrium solution and onto the bifurcation branch. To achieve this, the airspeed is either slightly increased or decreased: $U^* = U_f^* \pm \Delta U^*$. The sign of this perturbation depends on whether the solution is supercritical or subcritical; this is determined through trial and error. The next step is to find the LCO amplitude and frequency at this fixed airspeed.

The initial guess for frequency is the Hopf bifurcation frequency ($\omega \approx \omega_f$). A small, non-zero amplitude perturbation is applied to the DOF for which the phase-locking is applied: the critical flutter eigenmode. Note that the Fourier spectral coefficients, $\mathbf{X}$, are separated into sine and cosine terms: the amplitude is applied to the first-order sine Fourier coefficient [8]. This initial guess is then corrected through an iterative Newton-Raphson scheme.

$$\begin{bmatrix} \frac{dR^k}{d\mathbf{X}} & \frac{dR^k}{d\omega} \\ \mathbf{E}_j^T & 0 \end{bmatrix} \begin{bmatrix} \delta\mathbf{X}^{k+1} \\ \delta\omega^{k+1} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}^k \\ 0 \end{bmatrix} \quad (4.71)$$

The solution is updated for each subsequent iteration k :

$$\begin{aligned} \mathbf{X}^{k+1} &= \mathbf{X}^k + \delta\mathbf{X}^{k+1} \\ \omega^{k+1} &= \omega^k + \delta\omega^{k+1} \end{aligned} \quad (4.72)$$

This iterative process is repeated until the norm of the residual vector, $\|\mathbf{R}^k\|$, falls below a predefined convergence tolerance, ϵ_{NR} . The continuation scheme starts from the initial conditions: $\mathbf{X}$, ω and U .

4.4.5. Architecture and Implementation

The overall algorithm architecture of the HBM and pseudo arc-length continuation scheme for the studied nonlinear aeroelastic system is illustrated in Figure 4.8. The accompanying codes are available at:

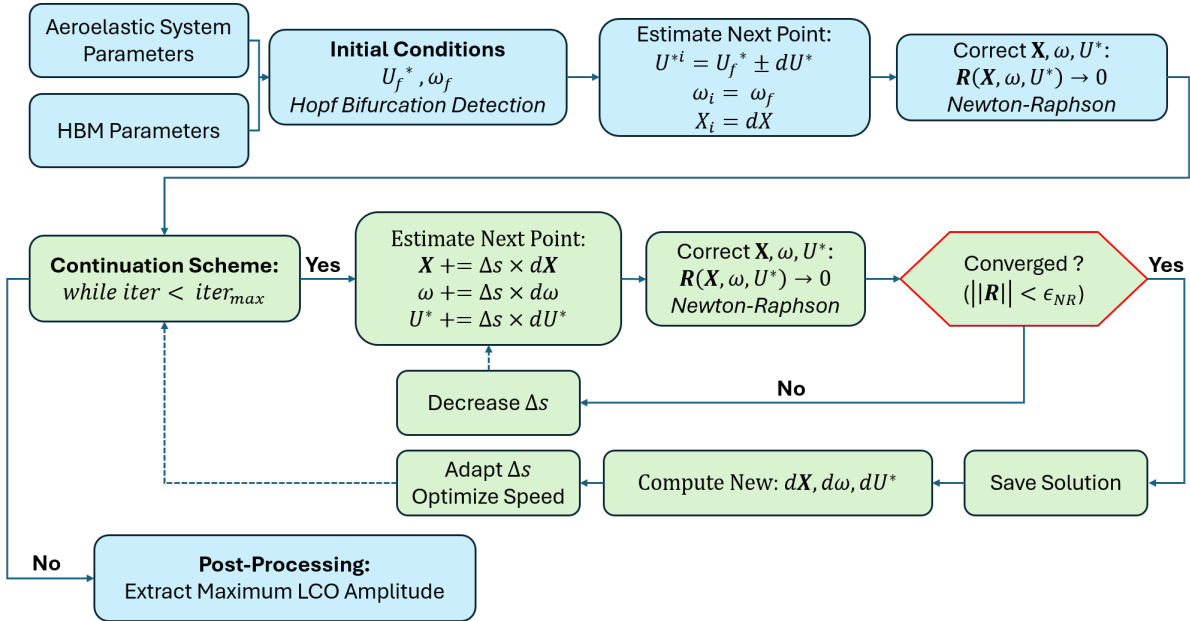


Figure 4.8: Aeroelastic HBM algorithm flowchart

4.5. Stability and Bifurcation Points

HBM, coupled with an arc-length continuation algorithm, resolves the post-flutter nonlinear dynamics of the aeroelastic system. However, this continuation procedure does not inherently determine the stability of the resulting periodic branches. Therefore, stability is evaluated using Hill's method, a frequency-domain formulation of Floquet theory. [7] [30] [20]. Finally, bifurcation boundaries are identified [20, 32]. Specifically, isolating the Neimark-Sacker bifurcation which marks the emergence of stable quasi-periodic branch.

4.5.1. Stability Analysis

Floquet Theory

Floquet theory evaluates the stability of periodic solutions, such as those determined via the implemented HBM. A periodic orbit is considered stable if an infinitesimally small perturbation decays over time. [7] [20]

Consider a general nonlinear system with n states:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \quad (4.73)$$

Let $\mathbf{x}^*(t)$ be a known periodic solution of period T , such that $\mathbf{x}^*(t) = \mathbf{x}^*(t + T)$. To assess its stability, an infinitesimal disturbance $\Psi(t)$ is imposed onto the periodic solution:

$$\mathbf{x}(t) = \mathbf{x}^*(t) + \Psi(t) \quad (4.74)$$

Substituting this perturbed state into the governing system yields:

$$\dot{\mathbf{x}}^* + \dot{\Psi} = \mathbf{f}(\mathbf{x}^* + \Psi) \quad (4.75)$$

Assuming the disturbance $\Psi(t)$ is small, the nonlinear function is expanded using a first-order Taylor series around $\mathbf{x}^*$:

$$\dot{\mathbf{x}}^* + \dot{\Psi} = \mathbf{f}(\mathbf{x}^*) + \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]_{\mathbf{x}^*} \Psi + \mathcal{O}(|\Psi|^2) \quad (4.76)$$

Because the unperturbed solution satisfies the original system ($\dot{\mathbf{x}}^* = \mathbf{f}(\mathbf{x}^*)$), these terms cancel. Neglecting the higher-order terms ($\mathcal{O}(|\Psi|^2)$) yields the first variation equation:

$$\dot{\Psi} = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]_{\mathbf{x}^*} \Psi \quad (4.77)$$

The Jacobian matrix is evaluated along the moving periodic trajectory $\mathbf{x}^*(t)$. Consequently, it is a time-periodic matrix with the same period T . Defining this matrix as $\mathbf{A}(t)$, the equation governing the perturbation reduces to a first-order linear ordinary differential equation with time-varying, periodic coefficients:

$$\dot{\Psi} = \mathbf{A}(t)\Psi \quad \text{where} \quad \mathbf{A}(t) = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]_{\mathbf{x}^*(t)} \quad (4.78)$$

Floquet's Theorem states that:

$$\Psi(t + T) = \Psi(t)\Psi(0)^{-1}\Psi(T) = \Psi(t)\mathcal{M} \quad (4.79)$$

where $\mathcal{M} = \Psi(0)^{-1}\Psi(T)$ is defined as the Monodromy matrix.

The Monodromy matrix acts as a discrete transition operator, mapping the perturbed state of the system over exactly one full period. Computationally, it is obtained by integrating Equation (4.78) over a single period T [7] [20].

The stability of the periodic solution is governed by the eigenvalues of the Monodromy matrix, defined as the Floquet multipliers (μ_i). The system is asymptotically stable if all Floquet multipliers lie strictly within the unit circle in the complex plane ($|\mu_i| < 1$), as illustrated in Figure 4.9 [20].

Throughout the remainder of this thesis, stability is assessed via the Floquet exponents, λ_i . These exponents arise from the assumption that the perturbation takes the form $\Psi(t) = \mathbf{p}(t)e^{\lambda t}$ (where $\mathbf{p}(t)$ is T -periodic), leading to the relationship $\mu_i = e^{\lambda_i T}$. Consequently, the stability criterion shifts from the unit circle to the imaginary axis: a system is stable if the real parts of all Floquet exponents are strictly negative ($\text{Re}(\lambda_i) < 0$) as illustrated in Figure 4.10. [20]

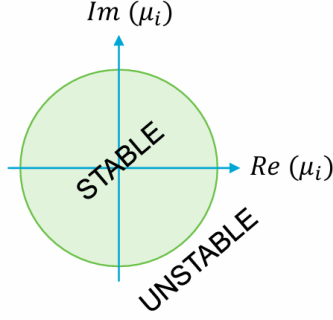


Figure 4.9: Stability criterion using Floquet multipliers

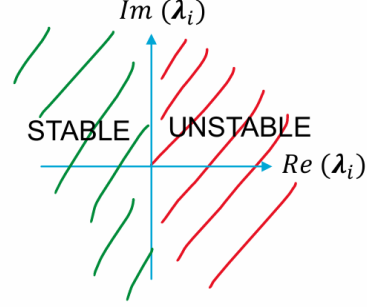


Figure 4.10: Stability criterion using Floquet exponents

Hill's Method

The post-flutter dynamic response of the nonlinear aeroelastic system is resolved in the frequency domain through HBM. To assess the stability of these resulting periodic branches, classical Floquet theory is adapted into a time-invariant eigenvalue problem via Hill's method [30] [20]. Hill's method consists of perturbing the equations of motion around a periodic solution ($\mathbf{x}_p(t)$) and applying the same Galerkin procedure as in section 4.4.1 for HBM [32].

The nonlinear aeroelastic system is described in state-space form as:

$$\dot{\mathbf{x}}(t) = \mathbf{Q}\mathbf{x}(t) + \mathbf{f}_{nl}(\mathbf{x}(t)) \quad (4.80)$$

The system is perturbed by a small variation, $\Delta\mathbf{x}(t)$, around the known periodic solution, $\mathbf{x}_p(t)$:

$$\dot{\mathbf{x}}_p(t) + \Delta\dot{\mathbf{x}}(t) = \mathbf{Q}(\mathbf{x}_p(t) + \Delta\mathbf{x}(t)) + \mathbf{f}_{nl}(\mathbf{x}_p(t) + \Delta\mathbf{x}(t)) \quad (4.81)$$

Assuming the perturbation $\Delta\mathbf{x}(t)$ is sufficiently small, the nonlinear force vector is approximated using a first-order Taylor expansion evaluated along the periodic trajectory:

$$\mathbf{f}_{nl}(\mathbf{x}_p(t) + \Delta\mathbf{x}(t)) \approx \mathbf{f}_{nl}(\mathbf{x}_p(t)) + \left[\frac{\partial \mathbf{f}_{nl}}{\partial \mathbf{x}} \right]_{\mathbf{x}=\mathbf{x}_p(t)} \Delta\mathbf{x}(t) \quad (4.82)$$

Substituting the Taylor expansion back into the perturbed state equation yields:

$$\dot{\mathbf{x}}_p(t) + \Delta\dot{\mathbf{x}}(t) = \mathbf{Q}\mathbf{x}_p(t) + \mathbf{Q}\Delta\mathbf{x}(t) + \mathbf{f}_{nl}(\mathbf{x}_p(t)) + \left[\frac{\partial \mathbf{f}_{nl}}{\partial \mathbf{x}} \right]_{\mathbf{x}=\mathbf{x}_p(t)} \Delta\mathbf{x}(t) \quad (4.83)$$

The original state equation around the known periodic solution, $\mathbf{x}_p(t)$ cancels, thereby obtaining:

$$\Delta\dot{\mathbf{x}}(t) = \mathbf{Q}\Delta\mathbf{x}(t) + \left[\frac{\partial \mathbf{f}_{nl}}{\partial \mathbf{x}} \right]_{\mathbf{x}=\mathbf{x}_p(t)} \Delta\mathbf{x}(t) \quad (4.84)$$

Applying Floquet theory, the perturbation is assumed to be of the form $\Delta \mathbf{x}(t) = e^{\lambda t} \phi(t)$, where $\phi(t)$ is the Floquet mode, a periodic function which oscillates at the base frequency ω and λ is the complex Floquet exponent. Substituting and simplifying by $e^{\lambda t}$ yields:

$$\lambda \phi(t) + \dot{\phi}(t) - \mathbf{Q} \phi(t) - \left[\frac{\partial \mathbf{f}_{nl}}{\partial \mathbf{x}} \right]_{\mathbf{x}=\mathbf{x}_p(t)} \phi(t) = 0 \quad (4.85)$$

To transform the continuous time-domain differential equations into a discrete frequency domain algebraic system, a Galerkin projection is employed (as described in Section 4.4.1). As such, $\phi(t) \rightarrow \Phi$, $\dot{\phi}(t) \rightarrow \Phi \omega [\nabla \otimes \mathbf{I}_n]$ and $\lambda \rightarrow \Lambda$. The eigenvalue problem is described as:

$$\Phi \Lambda + \Phi \frac{d\mathbf{R}}{d\mathbf{X}} = 0 \quad (4.86)$$

where, $\Phi = \Phi_1 + i\Phi_2$ are complex eigenvectors and $\frac{d\mathbf{R}}{d\mathbf{X}}$ is the residual derivative defined in Section 4.4.1.

The Hill's matrix, defined in Equation 4.87, and corresponding complex eigenvalues, Λ , are calculated at each step of the continuation algorithm described in Section 4.4.3

$$\mathbf{H} = -\frac{d\mathbf{R}}{d\mathbf{X}} \quad (4.87)$$

The size of the Hill matrix is $n(2H+1)$, with $n = 12$ the total number of DOF in the state variables in the first-order system, which comprises 6 structural states (velocities and displacements) and 6 states of aerodynamic lag states. Consequently, the Hill matrix yields $n(2H+1)$ complex eigenvalues of which only n with the smallest imaginary part are expected to have a physical meaning and correspond to Floquet exponents, denoted as λ_i , $i = 1 \dots n$ [32] [20].

The stability of each periodic solution is determined by evaluating the sign of the real part of its corresponding Floquet exponents (Figure 4.10). The stability threshold is ϵ_{stab} .

4.5.2. Bifurcation Point Identification

Following the stability analysis using the Hill's method presented in Section 4.5.1, the bifurcation which marks the transition from a stable to unstable LCO branch is identified through the imaginary part of the Floquet exponent λ_i as it crosses the imaginary axis. As presented in Figure 4.12 there are four notable bifurcation points which can be identified:

- **Limit Point (LP):** Also known as a fold bifurcation represents a change in direction. At this bifurcation, the stability changes; however, the solution remains periodic (Figure 4.11 (a)).
- **Branch Point (BP):** An intersection where the primary solution branch changes stability and a new stable periodic branch emerges (Figure 4.11 (b)).
- **Period Doubling (PD):** A bifurcation where an existing limit cycle destabilises and a new periodic branch with exactly twice the original temporal period appears.
- **Neimark-Sacker (NS):** A change of stability from stable to unstable and the appearance of a new quasi-periodic branch (Figure 4.11 (c)) [20].

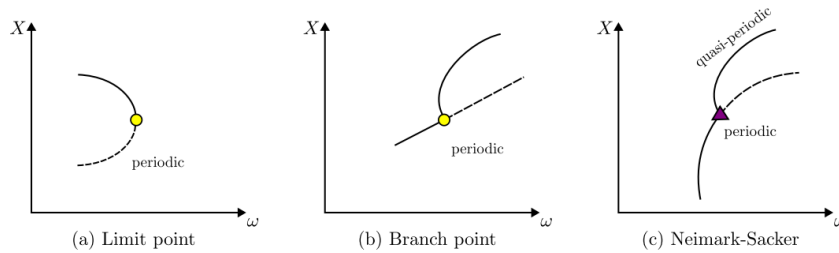


Figure 4.11: Bifurcations of LCO branches [20]

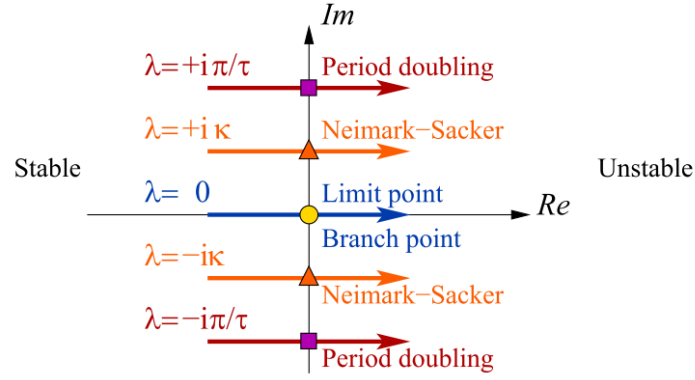


Figure 4.12: Bifurcation point identification through Floquet exponents [32]

Bifurcation points are identified within the continuation loop by monitoring stability changes between consecutive equilibrium steps. The specific bifurcation type is classified by isolating the critical Floquet exponent, λ_c and evaluating its imaginary component. The critical exponent is defined by its proximity to the imaginary axis: during a stable-to-unstable transition, λ_c is the exponent with the strictly minimum positive real part; during an unstable-to-stable transition, λ_c is the exponent with the strictly maximum negative real part. The bifurcation identification procedure is presented below in Figure 4.13.

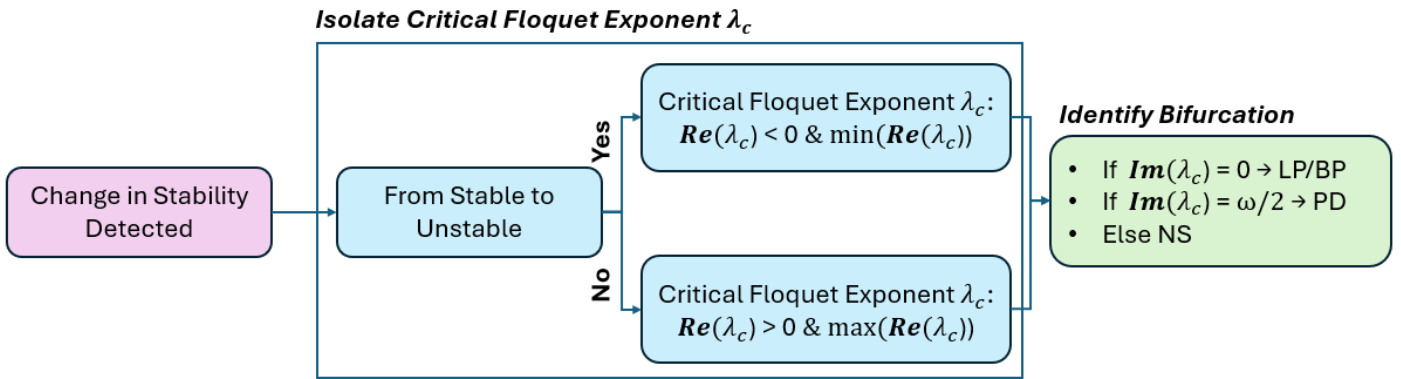


Figure 4.13: Bifurcation identification within continuation loop

The precision of the bifurcation localisation algorithm is governed by the defined HBM parameters, including the number of harmonics, H , the number of time samples for AFT, N , the minimum and maximum continuation step sizes, Δs_{min} and Δs_{max} , the Newton-Raphson convergence threshold, ϵ_{NR} and the stability criterion ϵ_{stab} . The selected HBM parameters are summarised in Table 4.4.

Table 4.4: Harmonic balance method numerical parameters

H	N	Δs_{min}	Δs_{max}	ϵ_{NR}	ϵ_{stab}
5	2^8	10^{-4}	0.01	10^{-9}	10^{-6}

5

Results

This chapter evaluates the efficacy of the Flap-NES in controlling flutter, benchmarking its performance against a locked-Flap, 2-DOF baseline configuration. Section 5.1 defines the comparative baseline and the criteria the flap-NES must follow to claim successful flutter control. Section 5.2 presents the parametric study of the Flap-NES parameters. Based on these results, two regimes are identified: Flap-NES with zero linear stiffness (Section 5.3) and Flap-NES with a tuned linear stiffness (Section 5.4). For both configurations, the linear stability boundaries are quantified and qualified via flutter diagrams, while the nonlinear post-flutter dynamics are mapped using bifurcation diagrams obtained with the harmonic balance method (HBM). Finally, the findings are discussed alongside experimental results from preceding Flap-NES literature [1, 9, 10].

5.1. Comparative Baseline and Flap-NES Success Criteria

In this section, the baseline representing the system with a locked flap-NES is presented alongside the criteria which the Flap-NES must follow to claim successful flutter control.

5.1.1. 2-DOF Baseline: Flap-NES Locked

The baseline 2-DOF system represents the airfoil with the Flap-NES in a locked position. In line with the previous study by [11], this linear-wing configuration utilises the parameters outlined in Table 5.1 to model a representative wing under standard flight conditions.

The structural damping is set to a minimal 0.2%. This ensures that when the flap is subsequently unlocked, any observed aerodynamic energy dissipation can be strictly attributed to the Flap-NES rather than the baseline structure. A non-zero structural damping was chosen to avoid numerical instability of the HBM solver.

Table 5.1: Baseline 2-DOF wing parameters

μ	a_h	x_α	r_α	ζ_h	ζ_α	Ω_1
100	-0.5	0.25	0.5	0.002	0.002	0.5

As shown in Figure 5.1, the 2-DOF baseline system exhibits classic pitch-heave frequency coalescence. The heave mode destabilises when its damping ratio (ζ) becomes negative ¹ at the critical flutter airspeed, $U_f^* = 4.76$.

Because this 2-DOF baseline model is purely linear, U_f^* defines a strict stability boundary; exceeding this airspeed induces divergent oscillations that lead to catastrophic failure.

¹Recall from Linear Stability analysis (Section 4.3): the damping ratio is defined as $\zeta = -\frac{\sigma}{\omega_n}$ with the eigenvalue $\lambda = \sigma \pm i\omega$

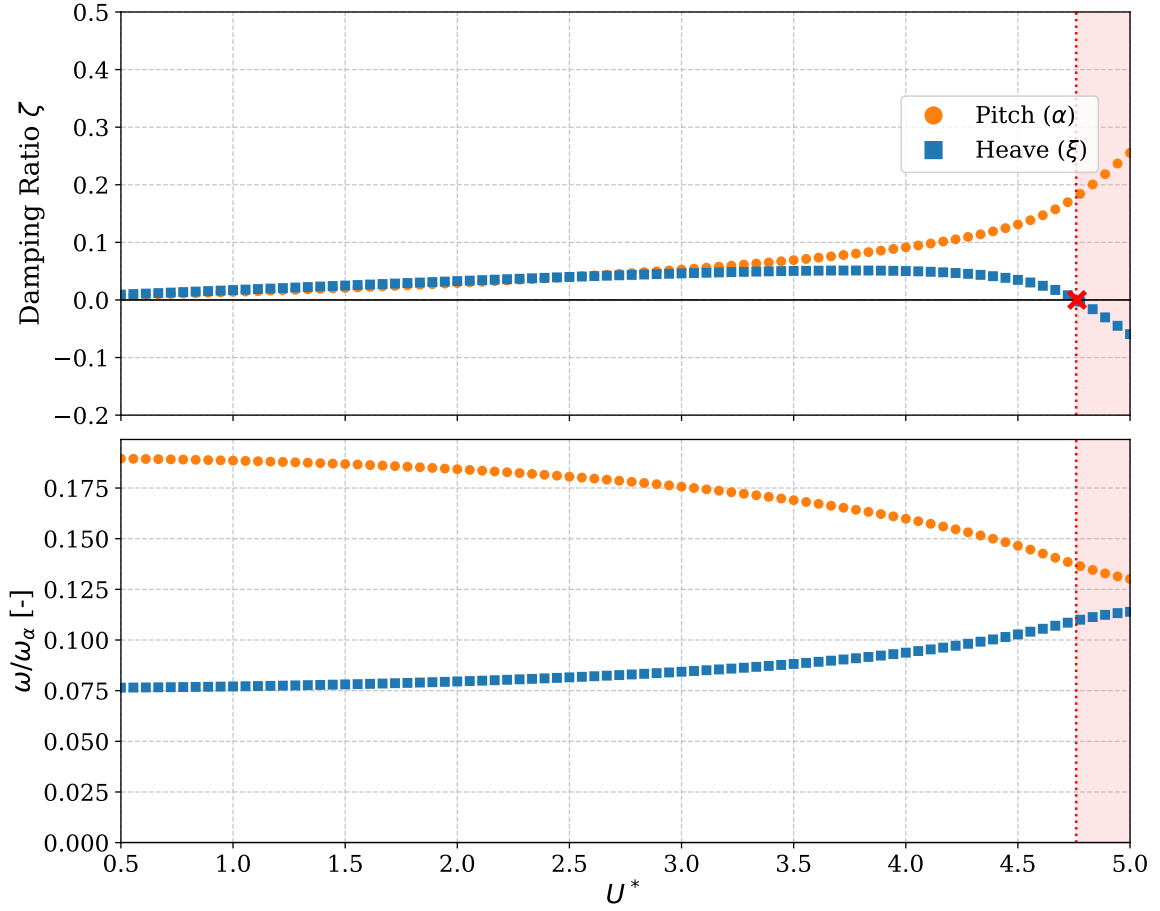


Figure 5.1: Flutter diagram: 2-DOF baseline. Baseline flutter speed: $U_f^* = 4.764$

Parameters: $\mu = 100$, $a_h = -0.5$, $x_\alpha = 0.25$, $r_\alpha = 0.5$, $\Omega_1 = 0.5$, $\zeta_h = \zeta_\alpha = 0.002$.

5.1.2. Criteria for Flap-NES Success

To successfully control flutter, the Flap-NES must expand the 2-DOF baseline stability envelope. This is achieved if the device successfully engages in any or all of the three Targeted Energy Transfer (TET) mechanisms detailed in Section 3.1: complete elimination (increasing the linear flutter speed beyond $U_f^* = 4.76$), intermediate suppression (bounding post-flutter LCOs to low amplitudes), or recurring burst-out and suppression (inducing stable quasi-periodic motion).

The acceptable maximum amplitudes in heave² and pitch³ are set as: $\xi \ll 1$ and $\alpha < 10^\circ$.

These limits ensure the flow remains attached to the airfoil, which is strictly required to preserve the validity of Theodorsen's potential flow theory. At low subsonic Mach numbers, boundary layer separation typically occurs between 12° and 15° angle of attack, depending on the airfoil section and Reynolds number [27]. The true aerodynamic angle of attack depends on both the pitch (α) and plunge velocity ($\dot{\xi}$) [2]. However, because restricting the heave amplitude ($\xi \ll 1$) inherently keeps the plunge velocity small, capping the structural pitch at 10° provides a conservative boundary.

Finally, to accurately evaluate the Flap-NES against the 2-DOF baseline, the total mass distribution of the system must remain constant. Consequently, any chosen flap size (c_h) strictly dictates the corresponding inertia parameters: the centre of gravity (x_β) and the radius of gyration (r_β). These three coupled parameters are derived via the inertial equivalence method (Appendix C).

²The heave displacement is non-dimensionalised such that $\xi = \frac{h}{b}$; the acceptable limit here is $h \ll b$ with b the semi-chord

³The pitch, α and flap β , did not receive any spatial scaling during the non-dimensionalization of the aeroelastic system equations of motion (Section 4.2.3). They are solved in radians homogeneity but are post-processed in degrees for clarity.

5.2. Parametric Study

To understand how the Flap-NES affects linear stability and post-flutter dynamics, a parametric sweep is performed. This sweep leverages the main advantage of the implemented HBM (Section 4.4.1): its low computational cost allows for the rapid (ranging between 2 and 10 min per combination) extraction of bifurcation diagrams across a wide range of parameter combinations. From these results, the Flap-NES linear stiffness is the critical parameter, and three regimes are identified: low, intermediate and high linear stiffness.

5.2.1. Tested Parameter Ranges

The parametric study of the Flap-NES evaluates four primary design variables, summarised in Table 5.2. Recall, the Flap-NES size (c_h) is coupled to its inertial parameters x_β and r_β . Furthermore, the non-dimensional aeroelastic model, the linear stiffness of the Flap-NES is characterised by the frequency ratio $\Omega_2 = \frac{\omega_\alpha}{\omega_\beta}$, where the uncoupled flap natural frequency is defined as $\omega_\beta = \sqrt{\frac{K_\beta}{I_\beta}}$. Finally, the linear structural damping (c_β^*) is intentionally prescribed a low value. This design choice prevents the system from becoming overdamped by accounting for the inherent aerodynamic damping the flap already receives from the freestream flow.

Table 5.2: Flap-NES parametric sweep: Evaluated ranges for the four primary design variables

Design Variable	Symbol	Range
Flap Size	(c_h, x_β, r_β)	$(0.2, 0.070, 0.171) - (0.5, 0.018, 0.069)$
Linear Stiffness	Ω_2	$0 - 1.5$
Linear Damping	c_β^*	$0.02 - 0.08$
Nonlinear Stiffness	Γ_β	$10 - 25$

5.2.2. Critical Parameter: Flap-NES Linear Stiffness, Ω_2

The complete results of the parametric study are detailed in Appendix D. This section extracts the primary trends and conclusions obtained from that data.

The linear stability analysis reveals that the system's stability boundary is fundamentally dictated by the Flap-NES linear stiffness (Ω_2). Specifically, the placement of the flap's natural frequency relative to the heave and pitch modes governs the linear flutter speed. Importantly, no parameter combination increased the linear flutter speed; all configurations reduced the baseline stability boundary by at least 35%. The parametric sweep categorised the system into three distinct regimes:

- **Low Stiffness** ($0 \leq \Omega_2 < \Omega_1$): The Flap-NES frequency lies strictly below the heave and pitch modes. This regime reduced the linear stability boundary by at least 67%. For every parameter combination tested, it introduced a subcritical bifurcation and failed to mitigate flutter. The case of zero linear stiffness ($\Omega_2 = 0$) is presented in Section 5.3.
- **Intermediate Stiffness** ($\Omega_1 < \Omega_2 < 1$): The Flap-NES frequency is sandwiched between the heave and pitch modes. This regime reduced the stability boundary by at least 61%. While it successfully induced quasi-periodic motion (recurring burstout and suppression TET) hypothetically driven by interacting limit cycles (Section 3.1.1), the resulting LCO amplitudes vastly exceeded the acceptable criteria for effective mitigation. Because placing the flap frequency between the two main structural modes inherently forces premature frequency coalescence and results in high LCO amplitudes, this regime will not be studied further.
- **High Stiffness** ($\Omega_2 > 1$): The Flap-NES frequency is placed above both the heave and pitch modes. This regime was the least detrimental to the linear stability boundary (a minimum reduction of 35%). However, two parameter combinations successfully engaged in intermediate suppression and recurring burstout and suppression TET post-flutter. Achieving this behaviour therefore requires careful tuning of the Flap-NES' linear parameters, namely the Flap-NES size, linear stiffness and linear damping. The case with $\Omega_2 = 1.2$ will be presented in Section 5.4.

5.3. Flap-NES with Zero Linear Stiffness: $\Omega_2 = 0$

This section analyses the linear stability and post-flutter dynamics of the Flap-NES operating with zero linear stiffness ($\Omega_2 = 0$). With a pure nonlinear stiffness, this configuration aligns with the foundational design principles of the Flap-NES discussed in Section 3.2. Theoretically, the absence of a linear stiffness, and consequently, the absence of a fixed natural frequency, allows the device to perform Targeted Energy Transfer (TET) across a broadband frequency range once a specific activation energy threshold is reached. These results, however, will challenge this design principle in the final discussion.

5.3.1. Linear Stability

Figure 5.2 presents the flutter diagram for the Flap-NES with zero linear stiffness ($\Omega_2 = 0$). The goal is to evaluate the Flap-NES' impact on the flutter mechanisms and linear flutter speed when unlocked.

Compared to the 2-DOF baseline (Figure 5.1), the Flap-NES introduces a new flutter mechanism: a premature coalescence of the heave and Flap-NES frequencies. This triggers an initial Hopf bifurcation at $U_{f,1}^* = 1.557$ as the heave mode destabilises, reducing the linear flutter speed by 67% relative to the baseline. A second Hopf bifurcation occurs at $U_{f,2}^* = 4.079$, where the pitch mode destabilises via a single-degree-of-freedom (SDOF) flutter mechanism.

At higher airspeeds ($U^* > 5$), the pitch and heave modes approach frequency coalescence. However, this interaction is purely mathematical. Because two of the system's three degrees of freedom are already unstable at these speeds, the wing is operating within the post-flutter dynamic regime.

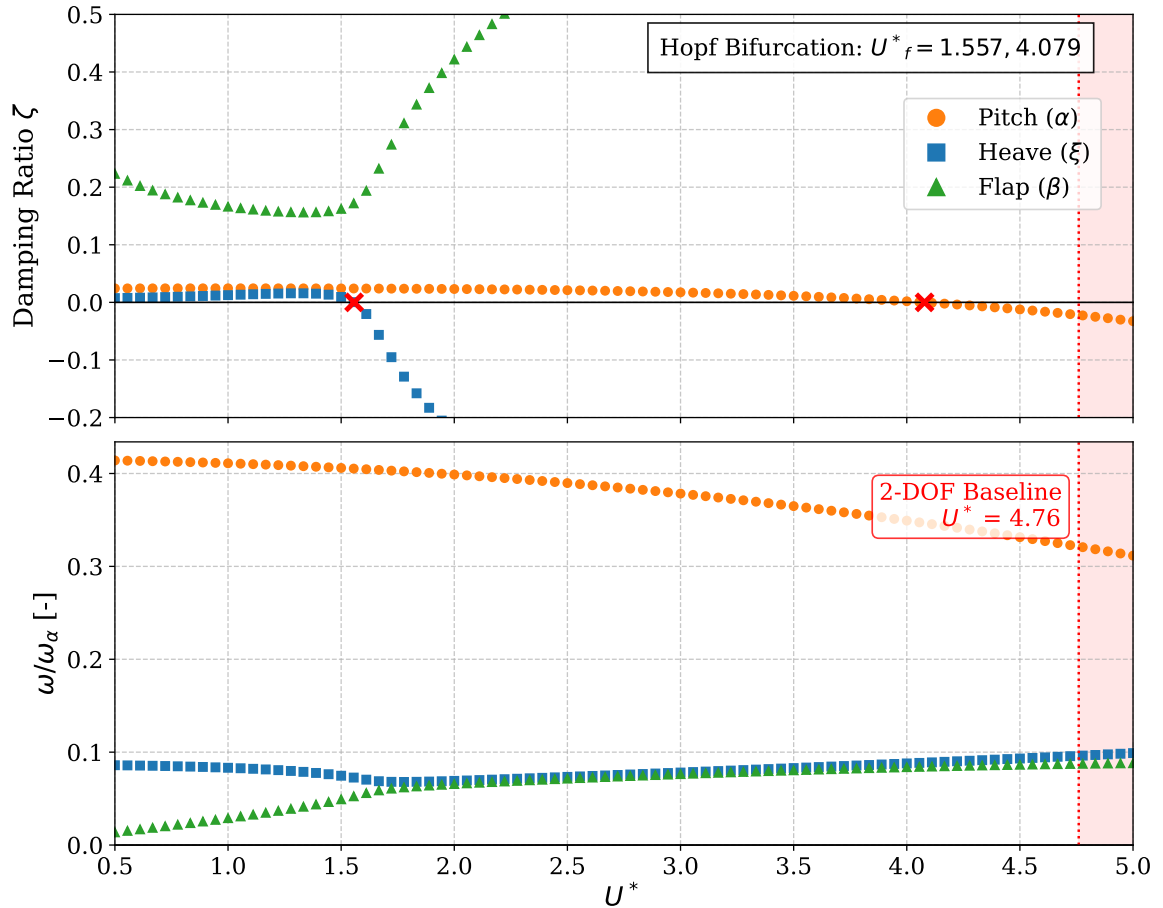


Figure 5.2: Flutter diagram: Zero linear stiffness ($\Omega_2 = 0$)

Parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.2$, $x_\alpha = 0.25$, $x_\beta = 0.070$, $r_\alpha = 0.5$, $r_\beta = 0.171$, $\Omega_1 = 0.5$, $\Omega_2 = 0$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$.

5.3.2. Post-Flutter Behaviour

Figure 5.3 (zone (a)) shows that the initial Hopf bifurcation ($U_{f,1}^*$) is subcritical, characterised by an unstable LCO branch that folds backwards. This subcritical branch introduces a structural risk. At airspeeds strictly below the linear flutter boundary, a sufficient external perturbation can push the system across this unstable threshold, where the system abruptly jumps to the nearest stable branch.

The subcritical behaviour can be explained by its associated flutter diagram (Figure 5.2). As the flap and heave modes coalesce, the cubic nonlinearity in the flap activates, causing the instantaneous eigenvalue to shift upwards and engage in a deep resonance with the heave mode, driving the subcritical bifurcation.

As airspeed increases, the unstable branch stabilises via a Neimark-Sacker (NS) bifurcation at $U_{NS}^* = 1.4$. Beyond this point, LCO amplitudes for the heave, pitch, and flap modes grow continuously. Although the cubic stiffness of the Flap-NES successfully bounds the response, the overall energy dissipation is insufficient to maintain safe operation. Consequently, the pitch amplitude exceeds the 10° acceptable limit at $U^* = 2.75$ (zone (b)).

These trends were observed not only for a Flap-NES with zero linear stiffness but generally low linear stiffness: $0 \leq \Omega_2 < \Omega_1$ as described in Appendix D.1. Additionally, no tested parameter combination of the Flap-NES parameters under this regime allowed to modify the subcritical bifurcation to supercritical.

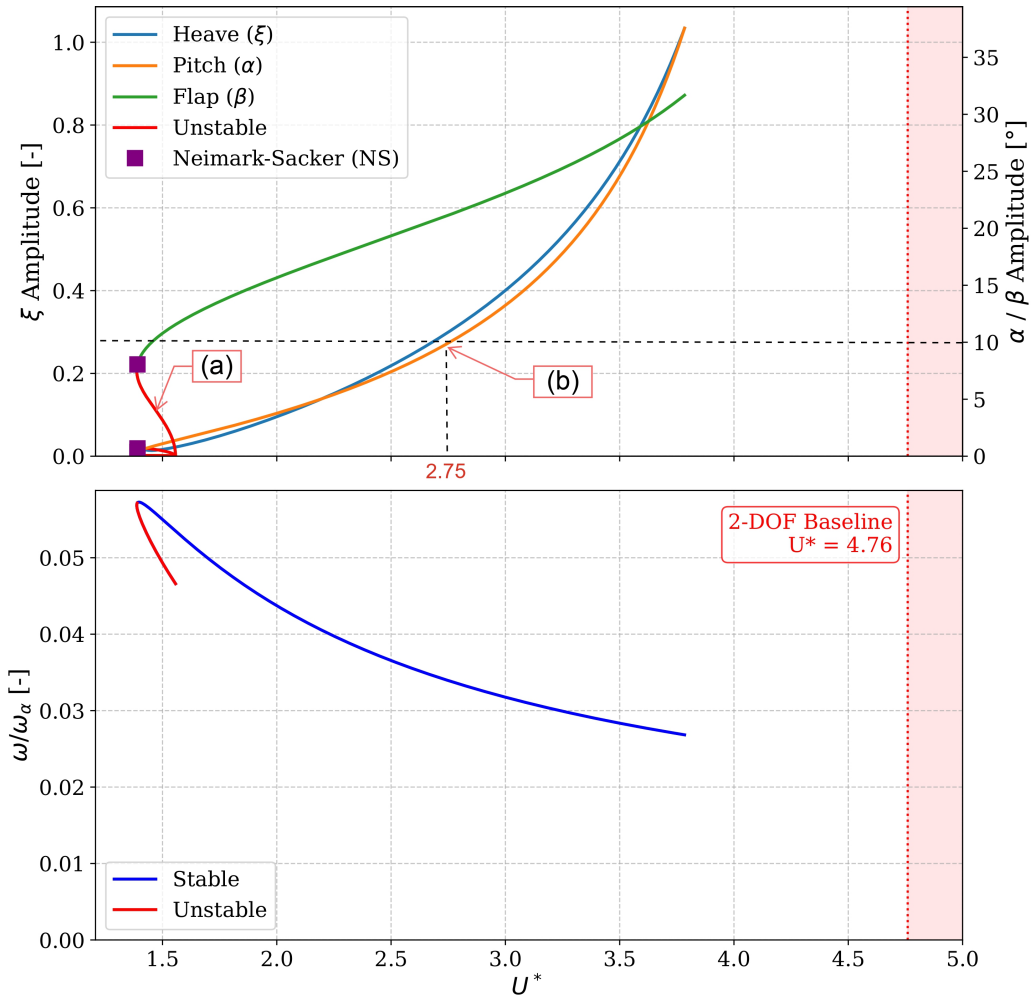


Figure 5.3: Bifurcation diagram: Zero linear stiffness ($\Omega_2 = 0$)

Parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.2$, $x_\alpha = 0.25$, $x_\beta = 0.070$, $r_\alpha = 0.5$, $r_\beta = 0.171$, $\Omega_1 = 0.5$, $\Omega_2 = 0.01$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$, $\kappa_h = \kappa_\alpha = 0$, $\Gamma_\beta = 10$.

HBM parameters: $H = 5$, $N = 2^8$, $\Delta s_{min} = 10^{-4}$, $\Delta s_{max} = 0.01$, $\epsilon_{NR} = 10^{-9}$, $\epsilon_{stab} = 10^{-6}$, $dU^* = 0.0001$, $dX = 0.01$.

5.3.3. Discussion: Flap-NES $\neq$ NES

The results demonstrate that a Flap-NES with zero linear stiffness fails to mitigate flutter; instead, it decreases the linear stability boundary and induces high-amplitude subcritical bifurcations. Consequently, this specific design strictly deteriorates the baseline system's aeroelastic performance.

This behaviour fundamentally contrasts with the traditional ungrounded NES tuning strategy, which achieves optimal energy pumping when its stiffness is purely nonlinear [29]. As presented in the literature review (Section 3.1), a NES, with pure nonlinear stiffness, placed inside the airfoil, controls flutter by engaging in three TET mechanisms [19].

The difference in performance between the internal NES and the Flap-NES is strictly governed by threshold activation energy. Because an internal NES is shielded from external aerodynamic loads, it remains dormant and only activates when the primary structure exceeds a specific internal energy threshold. Conversely, the Flap-NES acts as an aerodynamic control surface and is continuously excited by the freestream flow. The concept of an internal activation threshold is therefore eliminated, forcing the device to operate under continuous external forcing.

Consequently, the Flap-NES cannot operate successfully with zero linear stiffness. This conclusion contradicts a prevailing assumption in previous literature [1, 9, 10], which posited that the Flap-NES requires strictly zero linear stiffness to mimic classical NES behaviour.

While Garcia-Perez [10] numerically demonstrated that a theoretical zero-stiffness Flap-NES could increase flutter speeds and transform subcritical bifurcations to supercritical, the nonlinear aeroelastic model entirely omitted the linear stiffness term from the equations of motion. By mathematically erasing this term, the simulated device possessed no distinct natural frequency, artificially preventing the premature modal coalescence identified in the present study. Furthermore, the underlying linear flutter mechanisms were left unverified, as flutter diagrams were not plotted.

Finally, all successfully tested Flap-NES prototypes in experimental campaigns [1, 9, 10] inherently possessed a linear stiffness. Currently, there is no experimental evidence supporting the assumption that a Flap-NES with strictly zero linear stiffness can effectively mitigate aeroelastic flutter.

The parametric study (Appendix D) demonstrates that the Flap-NES is ineffective not only at strictly zero linear stiffness but across the entire low-stiffness regime. Successful flutter mitigation is achieved only when the Flap-NES natural frequency is positioned above the heave and pitch modes, necessitating a significantly higher linear stiffness. This case will be developed in the following section.

5.4. Flap-NES with Tuned Linear Stiffness: $\Omega_2 = 1.2$

The previous section established that a Flap-NES with zero linear stiffness cannot control flutter. Instead, the parametric study (Section 5.2) shows that an effective Flap-NES requires a linear stiffness high enough to place its natural frequency above the heave and pitch modes. This frequency separation prevents premature frequency coalescence.

The linear stiffness transforms the system into a Nonlinear Tuned Vibration Absorber (NLTVA). Operating as an NLTVA, the Flap-NES no longer possesses an activation energy threshold, but it retains the ability to perform TET. It can also present isolated solutions in its bifurcation diagram [20].

5.4.1. Linear Stability

As illustrated in Figure 5.4, the non-dimensional natural frequency of the Flap-NES is significantly higher ($\omega_n^* = 0.32$) than those of the pitch ($\omega_n^* = 0.15$) and heave ($\omega_n^* = 0.075$) modes. Two successive Hopf bifurcations emerge at $U_{f,1}^* = 3.076$ and $U_{f,2}^* = 3.104$, destabilizing the heave and flap modes, respectively. These critical airspeeds are separated by a mere $\Delta U^* = 0.03$.

The frequency evolution with respect to airspeed demonstrates that the onset of flutter is not driven by classical frequency coalescence; rather, the system exhibits two independent SDOF flutter mechanisms. The Flap-NES therefore successfully decouples the classical heave-pitch flutter mechanism, though the linear flutter speed is reduced by 35%.

This behaviour requires careful tuning of the Flap-NES linear parameters. As detailed in the parametric sweep (Appendix D.3), only two specific parameter combinations produced these results.

At higher airspeeds ($U^* > 5$), the heave and pitch modes approach frequency coalescence. However, this interaction is purely mathematical, as two of the three degrees of freedom are already unstable.

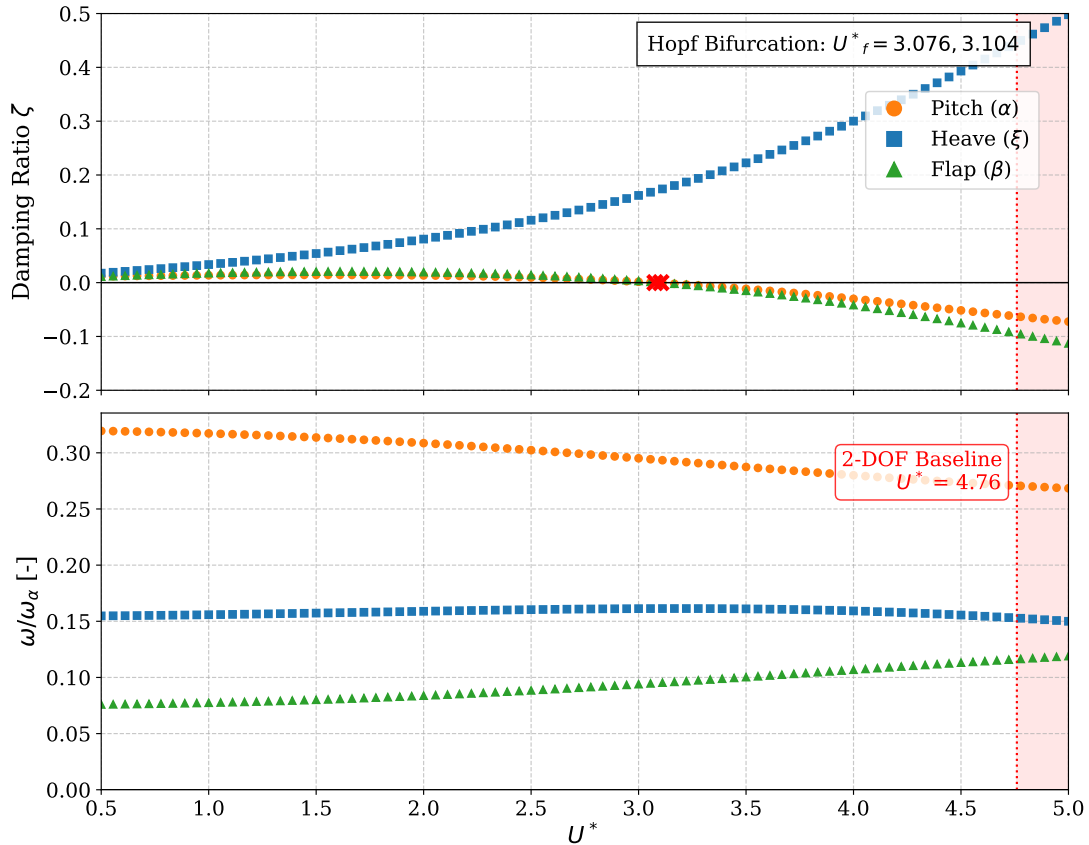


Figure 5.4: Flutter diagram: Tuned linear stiffness ($\Omega_2 = 1.2$)

Parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.5$, $x_\alpha = 0.25$, $x_\beta = 0.018$, $r_\alpha = 0.5$, $r_\beta = 0.069$, $\Omega_1 = 0.5$, $\Omega_2 = 1.2$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$.

5.4.2. Post-Flutter Behaviour

Following the initial Hopf bifurcation driven by heave instability ($U_{f,1}^*$), the emerging branch is supercritical (Figure 5.5).

The heave and pitch responses are bounded to low LCO amplitudes, staying well below the defined acceptable amplitude limits (Section 5.1.2), reaching 0.005 and 1.56° , respectively, at an airspeed of $U^* = 3.99$ with a Flap-NES amplitude of 10° . This bounded response indicates that the system engages in intermediate suppression TET. Although unlocking the Flap-NES lowers the linear stability boundary, the device dissipates sufficient energy to maintain a steady, low-amplitude oscillation.

At $U_{NS}^* = 3.99$, a Neimark-Sacker (NS) bifurcation is detected. This marks the onset of a stable quasi-periodic branch and the initiation of the recurring burstout and suppression TET mechanism. Recall that the HBM numerical framework cannot plot the stable quasi-periodic branch as it does not represent an LCO. Instead, it can be obtained through time-integration (RK45) as illustrated in Figure 5.6. The distinct non-repeating, drifting pattern is clearly observed in the heave and pitch DOF for which the maximum amplitude is capped at 0.0027 and 1.4° respectively.

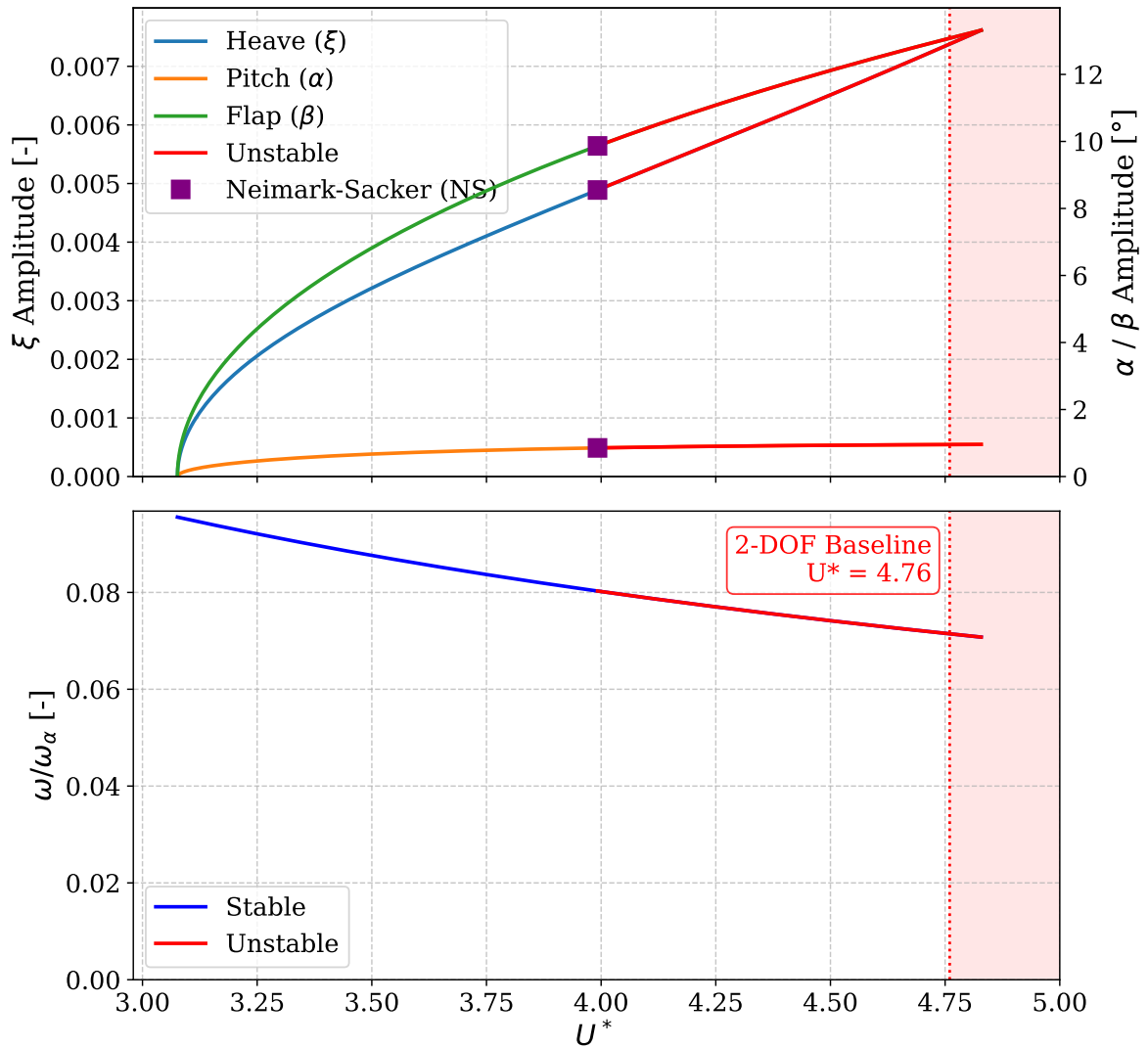


Figure 5.5: Bifurcation diagram: Tuned linear Stiffness ($\Omega_2 = 1.2$)

Parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.5$, $x_\alpha = 0.25$, $x_\beta = 0.018$, $r_\alpha = 0.5$, $r_\beta = 0.069$, $\Omega_1 = 0.5$, $\Omega_2 = 1.2$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$, $\kappa_h = \kappa_\alpha = 0$, $\Gamma_\beta = 25$.

HBM parameters: $H = 5$, $N = 2^8$, $\Delta s_{min} = 10^{-4}$, $\Delta s_{max} = 0.01$, $\epsilon_{NR} = 10^{-9}$, $\epsilon_{stab} = 10^{-6}$, $dU^* = 0.0001$, $dX = 0.01$.

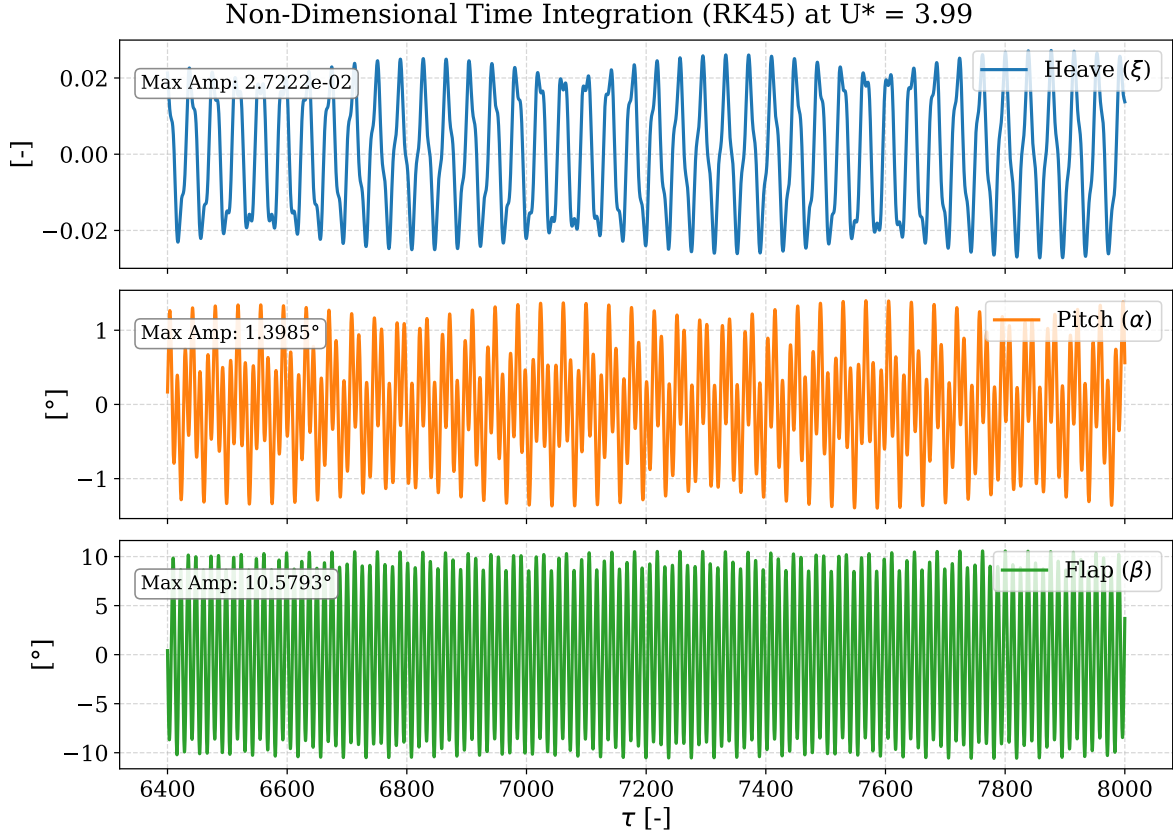


Figure 5.6: Time integration: Quasi-periodic motion. Initial conditions $\beta = 5^\circ$ at $U^* = 3.99$

The recurring burstout and suppression TET mitigation zone is bounded by an entrance and an exit NS bifurcation (Section 3.1). In Figure 5.5, no exit NS bifurcation is observed at higher airspeeds, leaving the upper boundary of the quasi-periodic branch undetermined. Further time-integration analysis is recommended to identify the transition from stable quasi-periodic motion to chaotic behaviour and to evaluate the mitigation capacity of the Flap-NES beyond the baseline 2-DOF flutter airspeed of $U^* = 4.76$.

As detailed in Appendix D.3, achieving these dynamics demands precise tuning of the linear parameters (Ω_2 , c_h , r_β , x_β , c_β^*). Conversely, the cubic nonlinear stiffness functions purely as an amplitude scalar: higher values directly compress the resulting LCO bounds.

Second Bifurcation Branch ($U_{f,2}^* = 3.104$)

As established in the linear stability analysis (Figure 5.4), the aeroelastic system exhibits two Hopf bifurcations and therefore possesses two bifurcation branches. This subsection will first focus on presenting the second bifurcation branch and then discuss the consequences of the resulting coexisting branches.

The second LCO branch, which emanates from $U_{f,2}^*$, presents a supercritical but initially unstable branch (Figure 5.7 and Figure 5.8 for a zoomed view of zone (a)), stabilising only after reaching a NS bifurcation at $U_{NS}^* = 3.11$. It is hypothesised that this localised region of instability may indicate the presence of an undetected isolated solution (isola). As established by [20], integrating significant linear stiffness transforms the Flap-NES into a NLTVA, prone to exhibiting isolas: closed solution branches completely detached from the primary equilibrium path (Figure 5.9).

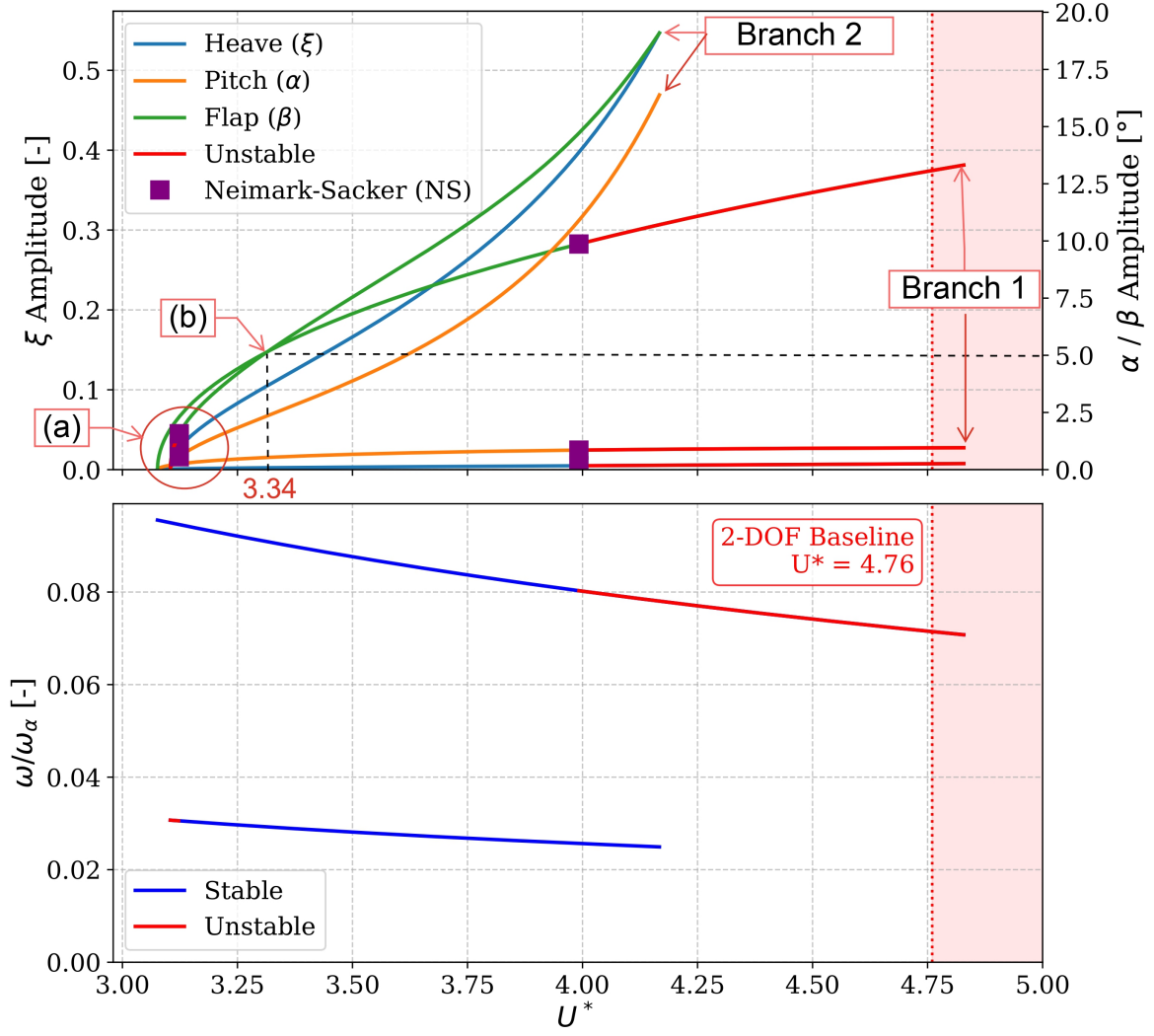


Figure 5.7: Bifurcation diagram with 2 branches: Tuned linear Stiffness ($\Omega_2 = 1.2$)

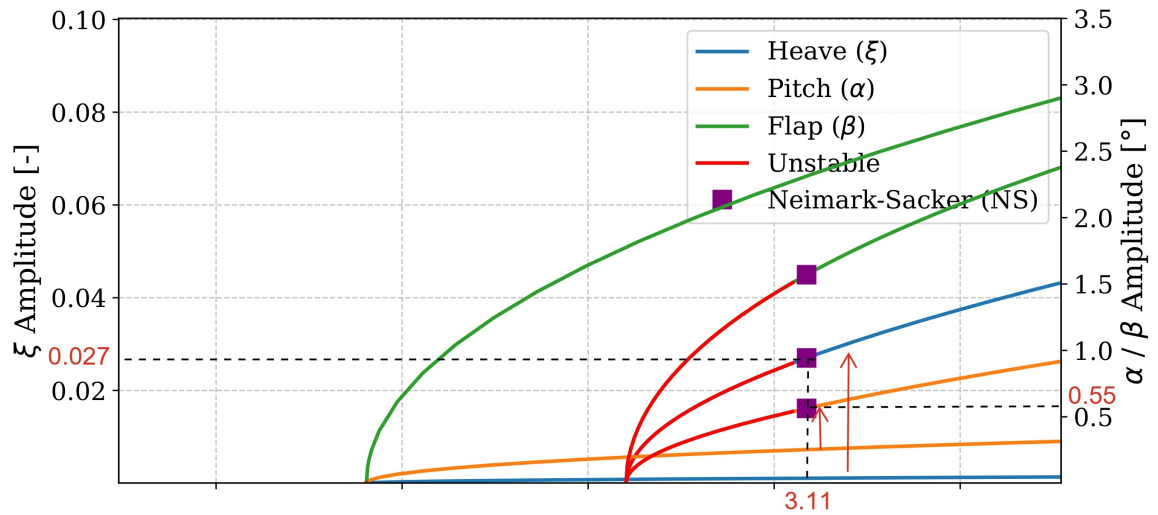


Figure 5.8: Bifurcation diagram: Zone (a) zoomed view

Parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.5$, $x_\alpha = 0.25$, $x_\beta = 0.018$, $r_\alpha = 0.5$, $r_\beta = 0.069$, $\Omega_1 = 0.5$, $\Omega_2 = 1.2$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$, $\kappa_h = \kappa_\alpha = 0$, $\Gamma_\beta = 25$.

HBM parameters: $H = 5$, $N = 2^8$, $\Delta s_{min} = 10^{-4}$, $\Delta s_{max} = 0.01$, $\epsilon_{NR} = 10^{-9}$, $\epsilon_{stab} = 10^{-6}$, $dU^* = 0.0001$, $dX = 0.01$.

Note that HBM is capable of capturing such isolated solutions as exhibited by the predicted response for a 2-DOF oscillating system in Figure 5.9 [20]. However, the current implementation tracks a continuous bifurcation branch emanating from the system's primary Hopf bifurcations. The isola is detached from the main equilibrium path; therefore, a new continuation scheme must start from an explicit initial guess that lies directly on the isola itself.

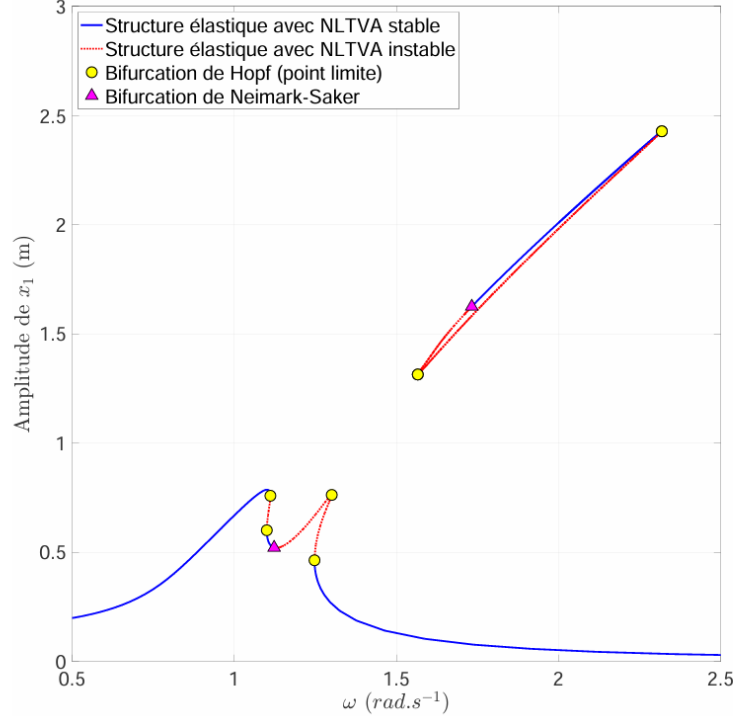


Figure 5.9: NLTVA frequency response for an external force amplitude of 0.11 [20]. Blue = stable; red = unstable

Regardless of whether this specific region is defined as a continuous unstable branch or a detached isola, the system possesses two coexisting bifurcation branches. Given that the two Hopf bifurcations emerge at near equal airspeeds, that they possess similar LCO amplitudes and incommensurate frequencies in the vicinity of the bifurcation point it can be defined as a Hopf-Hopf bifurcation. It is therefore the Hopf-Hopf bifurcation which drives the quasi-periodic motion observed on branch 1 (Section 3.1) [4]. The coexistence of the two branches is therefore required to achieve recurring burst-out and suppression TET.

However, the coexistence of these branches introduces a sensitivity to initial conditions. In zone (a) of Figure 5.7, the stable and unstable branches share similar LCO amplitudes near the Hopf bifurcation airspeeds. Consequently, if a disturbance pushes the system onto the unstable branch, the trajectory will jump to the adjacent stable LCO. The structural risk associated with this jump is limited: the unstable branch stabilizes at $U^* = 3.11$, where the resulting amplitudes remain low (0.027 in heave and 0.55° in pitch) rendering a small jump ($\Delta\xi = 0.025$ and $\Delta\alpha = 0.25^\circ$) as observed in Figure 5.8.

Focusing now, on zone (b) in Figure 5.7: At airspeeds above $U^* = 3.32$, the flap amplitude of the second branch surpasses the first (the flap crossover amplitude is 5.23°). The system now has two bifurcation branch: a desired low-amplitude LCO (Branch 1) and a dangerous high-amplitude LCO (Branch 2). If forced onto Branch 2, the system reaches its maximum amplitude limits at $U^* = 3.85$.

The system is therefore sensitive to its initial conditions and its final state is governed its basins of attraction. In nonlinear dynamics, a stable periodic orbit acts as an attractor, pulling neighboring trajectories into its cycle, while an unstable orbit repels them. The basin of attraction defines the specific range of initial conditions that pull the system into a given stable orbit [22]. While a bifurcation diagram proves these states exist, it does not measure the size or strength of their respective basins.

With the concept of basin of attraction in hand, the stable quasi-periodic motion emanating from the NS bifurcation of branch 1, marker of recurring burstout and suppression TET, is revisited. While Figure 5.6 demonstrated a seemingly stable recurring burstout and suppression TET, extending the time-integration (Figure 5.10) reveals that the system escapes the quasi-periodic regime at $\tau = 13000$ and is pulled onto the dominant LCO of Branch 2 (reaching stable amplitudes of 0.39 in heave, 10.7° in pitch, and 14.52° in flap at $U^* = 3.99$). The quasi-periodic motion is therefore not sustained and affected by the second branch's attractivity. Mapping the system's basin of attraction is therefore the next logical step to assess the efficacy of the Flap-NES in controlling flutter.

In all, the coexistence of bifurcation branches is essential for initiating recurring burst-out and suppression TET. However, the dominance of the second branch ultimately neutralizes quasi-periodic motion, forcing the system into a high-amplitude LCO.

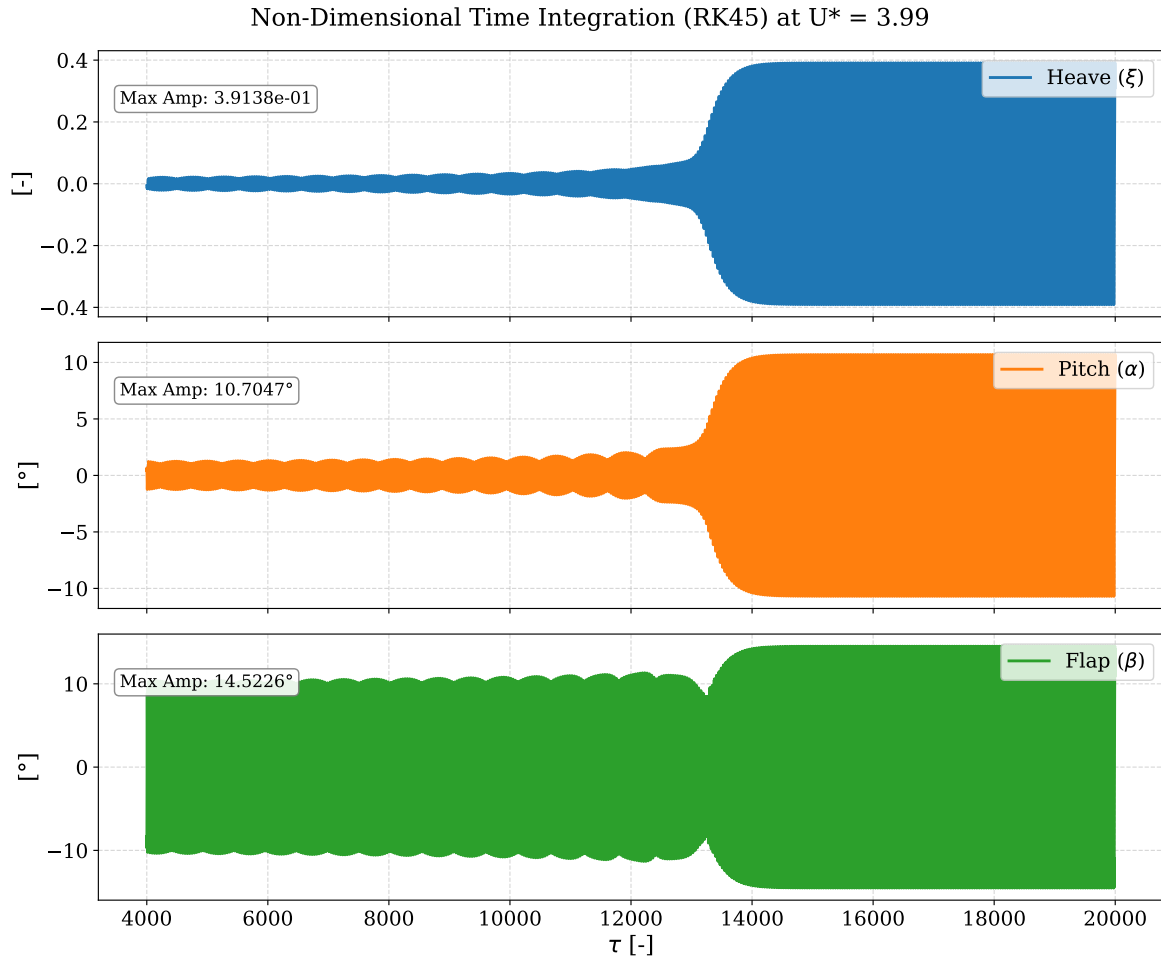


Figure 5.10: Extended time integration: Quasi-periodic motion. Initial conditions $\beta = 5^\circ$ at $U^* = 3.99$

5.4.3. Discussion

Through careful tuning of the Flap-NES linear parameters ($\Omega_2, c_h, r_\beta, x_\beta, c_\beta^*$), the first bifurcation branch demonstrates intermediate suppression (bounding the heave and pitch responses to low amplitudes) and recurring burstout and suppression (quasi-periodic motion) TET. Conversely, the second branch emerges as initially unstable before transitioning to a stable, high-amplitude LCOs. While the coexistence of these bifurcation branches is necessary to trigger the recurring burst-out and suppression TET, the dominant second branch ultimately neutralizes this mitigation over time, forcing the system to settle into the high-amplitude state.

The unsustained quasi-periodic motion is attributed to the highly conservative structural damping assumed in the heave and pitch modes ($\zeta_h = \zeta_\alpha = 0.002$). This near-zero damping was intentionally selected to represent a strict worst-case scenario, forcing the Flap-NES to provide all system dissipation. In physical applications, inherent structural damping typically ranges between 2% and 3%. It is hypothesized that this baseline dissipation would prevent the transition to the second branch, thereby sustaining the quasi-periodic state and achieving long-term flutter suppression.

At this stage, the current numerical model does not demonstrate successful, long-term flutter control. The Flap-NES fails to increase the linear flutter boundary, restricts its mitigation to intermediate suppression below the baseline 2-DOF flutter speed, and cannot permanently sustain recurring burstout and suppression TET. Despite these functional limitations, the underlying dynamics establish a critical structural criterion: to control flutter, the Flap-NES natural frequency must be tuned strictly above the heave and pitch modes.

This frequency placement hypothesis is explicitly corroborated by previous experimental campaigns (Section 3.2, Figures 3.24 and 3.26). These physical studies successfully achieved increased linear stability boundaries, bounded post-flutter LCOs, and sustained quasi-periodic motion. Crucially, these experimental successes relied on the exact frequency separation strategy identified in the current work. Therefore, while the numerical implementation requires further work to match experimental results, the fundamental Flap-NES design rule is supported.

The discrepancies between the numerical model and the experimental literature highlight key areas for future research. First, the parametric study should be re-evaluated with realistic structural damping values. Second, the current model strictly couples the flap's inertial properties (x_β and r_β) to its geometric size (c_h) to maintain a uniform mass distribution relative to the 2-DOF baseline; decoupling these inertial parameters could offer additional tuning possibilities. Third, the experimental setups possessed structural nonlinearities in heave and pitch not included in the current framework, which likely assisted in bounding the quasi-periodic motion. Finally, a formal basin of attraction study is required to fully quantify the system's sensitivity to initial conditions and map the robustness of the coexisting bifurcation branches.

6

Conclusion

This Master's thesis evaluated the capacity of a Flap-NES (Nonlinear Energy Sink), possessing both linear and cubic stiffness, to control flutter of a 2D linear typical section under linear unsteady aerodynamics.

When an NES is placed inside of an airfoil, it suppresses flutter through three Targeted Energy Transfer (TET) mechanisms: recurring burstout and suppression, intermediate suppression and complete elimination [19]. Previous literature on the Flap-NES provided an extensive experimental and numerical foundation demonstrating a capacity to control flutter [1, 9, 10]. However, it lacks a comprehensive framework that simultaneously integrates three core elements: a representative baseline, bifurcation diagrams, and the Flap-NES linear stiffness.

This work mapped the linear stability boundaries and post-flutter bifurcation diagrams to provide a general understanding of the Flap-NES' effect on the aeroelastic system. The system is modelled as a 2D 3-DOF aeroelastic system featuring cubic stiffness only in the flap and subjected to unsteady aerodynamics (Theodorsen). Linear stability is studied through flutter diagrams to characterise the flutter mode and detect the Hopf Bifurcations. The post-flutter dynamic response is obtained through the Harmonic Balance Method coupled with pseudo arc length continuation. The stability of the branches is subsequently assessed via the Hill's method. A parametric study is conducted to evaluate the Flap-NES against a 2-DOF reference system, where the Flap-NES is locked.

Unlike a traditional internal NES, which activates strictly upon reaching a specific aeroelastic energy threshold, the Flap-NES is continuously subjected to external aerodynamic loads. Consequently, its linear stiffness cannot be zero; an unanchored flap induces a premature frequency coalescence mechanism that drives the system into a dangerous subcritical Hopf bifurcation.

While the current numerical implementation does not increase the linear flutter boundary, indicating that complete suppression TET was not achieved, careful tuning of the flap-NES' linear parameters (size, inertia, damping, and linear stiffness) establishes a critical aeroelastic design rule. By positioning the Flap-NES natural frequency strictly above those of the heave and pitch modes, the system can achieve a Hopf-Hopf bifurcation. In the post-flutter regime, the first bifurcation branch engages in intermediate suppression TET, followed by an initiated quasi-periodic motion. However, the coexistence of a dominant secondary branch eventually erases the quasi-periodic motion, attracting the system onto it's a high-amplitude stable LCO branch.

It is recommended that future research follow the Flap-NES tuning criterion established in this study: the natural frequency of the Flap-NES must be tuned to lie above the heave and pitch modes. It can directly expand current work by incorporating a more realistic structural damping, integrate heave and pitch structural hardening nonlinearities characteristic of high-aspect-ratio wings. Finally, formal basin of attraction mapping must accompany standard bifurcation analyses to fully quantify the system's sensitivity to initial conditions.

The Flap-NES cannot yet control flutter, but showed viability by engaging in TET with proper tuning.

7

Perspectives

This research evaluated the efficacy of the Flap-NES, a passive device that converts a control surface into a Non-Linear Energy Sink, for wing flutter suppression. Modelled as a 2D 3-DOF typical section with cubic stiffness in the flap degree of freedom (DOF) under linear unsteady aerodynamics, the system's response is characterised through linear stability and nonlinear post-flutter bifurcation analyses.

It is recommended to follow the established design rules for the Flap-NES: the linear parameters (size, inertia, damping and linear stiffness) must be tuned such that the natural frequency of the flap lies above the heave and pitch modes. Below are direct areas of interest which can be developed working towards flutter control of the Flap-NES.

Post-Flutter and Chaotic Dynamics While an initial Neimark-Sacker (NS) bifurcation successfully identified the onset of quasi-periodic motion, an exit NS bifurcation was not detected. Consequently, the maximum airspeed for sustained stable quasi-periodic motion remains undefined, leaving it unclear whether the TET mechanism persists beyond the 2-DOF baseline flutter speed. Using time-integration bifurcation analysis, the transition from quasi-periodic beating to chaotic motion can be defined.

Higher Structural Damping The current numerical framework used a highly conservative structural damping ratio ($\zeta_h = \zeta_\alpha = 0.002$) to ensure that dissipation was governed by the Flap-NES. In physical aerospace structures, damping typically ranges between 2% and 3%. Future work should re-evaluate the parametric study under these realistic structural damping conditions for the heave and pitch modes. The increased damping could assist the system to permanently sustain quasi-periodic motion.

Basin of Attraction The system exhibits high sensitivity to initial conditions due to the presence of a co-existing bifurcation branch. Furthermore, the second branch pushes the system out of quasi-periodic motion attracting it onto its stable high amplitude LCO. Reading the bifurcation diagram alongside a basin of attraction study would quantify the system's sensitivity to initial conditions.

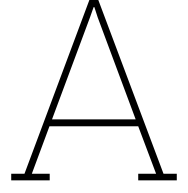
Three-Dimensional Flap-NES Effects The current 2D framework assumes a full-span Flap-NES. Because practical applications will occupy a partial span, future work must quantify the effect of span-wise sizing on mitigation efficacy. Additionally, the model should be expanded to a 4-DOF system (heave, pitch, primary control surface, and Flap-NES) and evaluated against a 3-DOF baseline.

Integration with High-Aspect-Ratio Flexible Wings The baseline main airfoil in this study was modelled as purely linear. However, the Flap-NES is ultimately intended for high-aspect-ratio wings, which are inherently subject to structural nonlinearities driven by increased flexibility and large geometric deflections. Future analyses must introduce cubic stiffness into the main heave and pitch degrees of freedom to evaluate how intrinsic structural nonlinearities interact with the Flap-NES during post-flutter dynamics.

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Appendix A: Aeroelastic Model

A.1. Structural Mass Matrix: Inertial Coupling

The structural mass matrix $\mathbf{A}$, which describes the inertial coupling of the system is defined as:

$$\mathbf{A} = \begin{bmatrix} m & S_\alpha & S_\beta \\ S_\alpha & I_\alpha & I_{\alpha\beta} \\ S_\beta & I_{\alpha\beta} & I_\beta \end{bmatrix} \quad (\text{A.1})$$

The structural damping matrix $\mathbf{C}$ and the linear structural stiffness matrix $\mathbf{E}$ are given by:

$$\mathbf{C} = \begin{bmatrix} C_h & 0 & 0 \\ 0 & C_\alpha & 0 \\ 0 & 0 & C_\beta \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} K_h & 0 & 0 \\ 0 & K_\alpha & 0 \\ 0 & 0 & K_\beta \end{bmatrix} \quad (\text{A.2})$$

Finally, the nonlinear structural stiffness matrix, $\mathbf{K}^{nl}$ is given by:

$$\mathbf{K}^{nl} = \begin{bmatrix} K_h^{nl} & 0 & 0 \\ 0 & K_\alpha^{nl} & 0 \\ 0 & 0 & K_\beta^{nl} \end{bmatrix} \quad (\text{A.3})$$

A.2. Aerodynamic Matrices

Aerodynamic Mass Matrix

$$\mathbf{B} = b^2 \begin{bmatrix} \pi & -\pi a_h b & -T_1 b \\ -\pi a_h b & \pi b^2 \left(\frac{1}{8} + a_h^2 \right) & -(T_7 + (c_h - a_h)T_1) b^2 \\ -T_1 b & 2T_{13} b^2 & -\frac{T_3 b^2}{\pi} \end{bmatrix} \quad (\text{A.4})$$

Aerodynamic Damping Matrix

$$\mathbf{D}_1 = b^2 \begin{bmatrix} 0 & \pi & -T_4 \\ 0 & \pi \left(\frac{1}{2} - a_h \right) b & (T_1 - T_8 - (c_h - a_h)T_4 + \frac{T_{11}}{2}) b \\ 0 & (-2T_9 - T_1 + T_4 (a_h - \frac{1}{2})) b & -\frac{T_4 T_{11} b}{2\pi} \end{bmatrix} \quad (\text{A.5})$$

$$\mathbf{D}_2 = \begin{bmatrix} 2\pi b & 2\pi b^2 \left(\frac{1}{2} - a_h \right) & b^2 T_{11} \\ -2\pi b^2 \left(a_h + \frac{1}{2} \right) & -2\pi b^3 \left(a_h + \frac{1}{2} \right) \left(\frac{1}{2} - a_h \right) & -b^3 \left(a_h + \frac{1}{2} \right) T_{11} \\ b^2 T_{12} & b^3 T_{12} \left(\frac{1}{2} - a_h \right) & \frac{b^3 T_{12} T_{11}}{2\pi} \end{bmatrix} \quad (\text{A.6})$$

$$\mathbf{D} = \mathbf{D}_1 + \Phi(0)\mathbf{D}_2 \quad (\text{A.7})$$

Where, $\Phi(0) = 1 - \Psi_1 + \Psi_2$.

Aerodynamic Stiffness Matrix

$$\mathbf{F}_1 = b^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_4 + T_{10} \\ 0 & 0 & \frac{T_5 - T_4 T_{10}}{\pi} \end{bmatrix} \quad (\text{A.8})$$

$$\mathbf{F}_2 = \begin{bmatrix} 0 & 2\pi b & 2bT_{10} \\ 0 & -2\pi b^2 \left(a_h + \frac{1}{2}\right) & -2b^2 \left(a_h + \frac{1}{2}\right) T_{10} \\ 0 & b^2 T_{12} & \frac{b^2 T_{12} T_{10}}{\pi} \end{bmatrix} \quad (\text{A.9})$$

$$\mathbf{F}_3 = \begin{bmatrix} 2\pi b & 2\pi b^2 \left(\frac{1}{2} - a_h\right) & b^2 T_{11} \\ -2\pi b^2 \left(a_h + \frac{1}{2}\right) & -2\pi b^3 \left(a_h + \frac{1}{2}\right) \left(\frac{1}{2} - a_h\right) & -b^3 \left(a_h + \frac{1}{2}\right) T_{11} \\ b^2 T_{12} & b^3 T_{12} \left(\frac{1}{2} - a_h\right) & \frac{b^3 T_{12} T_{11}}{2\pi} \end{bmatrix} \quad (\text{A.10})$$

$$\mathbf{F} = \mathbf{F}_1 + \Phi(0)\mathbf{F}_2 + \dot{\Phi}(0)\mathbf{F}_3 \quad (\text{A.11})$$

Where, $\Phi(0) = 1 - \Psi_1 + \Psi_2$ and $\dot{\Phi}(0) = \frac{\Psi_1 \epsilon_1}{b} + \frac{\Psi_2 \epsilon_2}{b}$.

Aerodynamic State Influence Matrices

$$\mathbf{W}_0 = \begin{bmatrix} -\Psi_1 \left(\frac{\epsilon_1}{b}\right)^2 \\ -\Psi_2 \left(\frac{\epsilon_2}{b}\right)^2 \\ \frac{\Psi_1 \epsilon_1}{b} \left(1 - \epsilon_1 \left(\frac{1}{2} - a_h\right)\right) \\ \frac{\Psi_2 \epsilon_2}{b} \left(1 - \epsilon_2 \left(\frac{1}{2} - a_h\right)\right) \\ \frac{\Psi_1 \epsilon_1}{\pi b} \left(T_{10} - \frac{\epsilon_1 T_{11}}{2}\right) \\ \frac{\Psi_2 \epsilon_2}{\pi b} \left(T_{10} - \frac{\epsilon_2 T_{11}}{2}\right) \end{bmatrix} \quad (\text{A.12})$$

$$\mathbf{W} = \begin{bmatrix} 2\pi b \mathbf{W}_0^T \\ -2\pi b^2 \left(a_h + \frac{1}{2}\right) \mathbf{W}_0^T \\ b^2 T_{12} \mathbf{W}_0^T \end{bmatrix} \quad (\text{A.13})$$

$$\mathbf{W}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{A.14})$$

$$\mathbf{W}_2 = \begin{bmatrix} -\frac{\epsilon_1}{b} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\epsilon_2}{b} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\epsilon_1}{b} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\epsilon_2}{b} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\epsilon_1}{b} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\epsilon_2}{b} \end{bmatrix} \quad (\text{A.15})$$

Theodorsen Coefficients

$$\begin{aligned}
T_1 &= -\frac{1}{3}\sqrt{1-c_h^2}(2+c_h^2) + c_h\nu \\
T_2 &= c_h(1-c_h^2) - \sqrt{1-c_h^2}(1+c_h^2)\nu + c_h\nu^2 \\
T_3 &= -\left(\frac{1}{8} + c_h^2\right)\nu^2 + \frac{1}{4}c_h\sqrt{1-c_h^2}\nu(7+2c_h^2) - \frac{1}{8}(1-c_h^2)(5c_h^2+4) \\
T_4 &= -\nu + c_h\sqrt{1-c_h^2} \\
T_5 &= -(1-c_h^2) - \nu^2 + 2c_h\sqrt{1-c_h^2}\nu \\
T_6 &= T_2 \\
T_7 &= -\left(\frac{1}{8} + c_h^2\right)\nu + \frac{1}{8}c_h\sqrt{1-c_h^2}(7+2c_h^2) \\
T_8 &= -\frac{1}{3}\sqrt{1-c_h^2}(2c_h^2+1) + c_h\nu \\
T_9 &= \frac{1}{2}\left[\frac{1}{3}(1-c_h^2)^{3/2} + aT_4\right] \\
T_{10} &= \sqrt{1-c_h^2} + \nu \\
T_{11} &= \nu(1-2c_h) + \sqrt{1-c_h^2}(2-c_h) \\
T_{12} &= \sqrt{1-c_h^2}(2+c_h) - \nu(2c_h+1) \\
T_{13} &= \frac{1}{2}[-T_7 - (c_h-a)T_1] \\
T_{14} &= \frac{1}{16} + \frac{1}{2}ac_h
\end{aligned} \tag{A.16}$$

Where, $\nu = \arccos c_h$.

A.3. Non-Dimensional: Aeroelastic Matrices

Non-Dimensional: Mass Matrix

$$\mathbf{A}^* = \begin{bmatrix} 1 & x_\alpha & x_\beta \\ \frac{x_\alpha}{r_\alpha^2} & 1 & \frac{r_\beta^2 + (c_h - a_h)x_\beta}{r_\alpha^2} \\ \frac{x_\beta}{r_\beta^2} & \frac{r_\beta^2 + (c_h - a_h)x_\beta}{r_\beta^2} & 1 \end{bmatrix} \tag{A.17}$$

Non-Dimensional: Damping Matrix

$$\mathbf{C}^* = \begin{bmatrix} 2\zeta_z\omega_1 & 0 & 0 \\ 0 & 2\zeta_\alpha & 0 \\ 0 & 0 & c_\beta^* \end{bmatrix} \tag{A.18}$$

Non-Dimensional: Stiffness Matrix

$$\mathbf{E}^* = \begin{bmatrix} \Omega_1^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \Omega_2^2 \end{bmatrix} \tag{A.19}$$

Non-Dimensional: Nonlinear Stiffness Matrix

$$\mathbf{K}_{nl}^* = \begin{bmatrix} \kappa_\zeta & 0 & 0 \\ 0 & \kappa_\alpha & 0 \\ 0 & 0 & \Gamma_\beta \end{bmatrix} \tag{A.20}$$

Non-Dimensional: Aerodynamic Mass Matrix

$$\mathbf{B}^* = \begin{bmatrix} 1 & -a_h & -\frac{T_1}{\pi} \\ -\frac{a_h}{r_\alpha^2} & \frac{\frac{1}{8}+a_h^2}{r_\alpha^2} & -\frac{T_7+(c_h-a_h)T_1}{\pi r_\alpha^2} \\ -\frac{T_1}{\pi r_\beta^2} & \frac{2T_{13}}{\pi r_\beta^2} & -\frac{T_3}{(\pi r_\beta)^2} \end{bmatrix} \quad (\text{A.21})$$

Non-Dimensional: Aerodynamic Damping Matrix

$$\mathbf{D}^*_1 = \begin{bmatrix} 0 & 1 & -\frac{T_4}{\pi} \\ 0 & \frac{\frac{1}{2}-a_h}{r_\alpha^2} & \frac{T_1-T_8-(c_h-a_h)T_4+\frac{T_{11}}{2}}{\pi r_\alpha^2} \\ 0 & \frac{-2T_9-T_1-T_4(0.5-a_h)}{\pi r_\beta^2} & -\frac{T_4T_{11}}{2\pi^2 r_\beta^2} \end{bmatrix} \quad (\text{A.22})$$

$$\mathbf{D}^*_2 = \begin{bmatrix} 2 & 2\left(\frac{1}{2}-a_h\right) & \frac{T_{11}}{\pi} \\ -\frac{2(a_h+\frac{1}{2})}{r_\alpha^2} & -\frac{2(a_h+\frac{1}{2})\left(\frac{1}{2}-a_h\right)}{r_\alpha^2} & -\frac{(a_h+\frac{1}{2})T_{11}}{\pi r_\alpha^2} \\ \frac{T_{12}}{\pi r_\beta^2} & \frac{T_{12}\left(\frac{1}{2}-a_h\right)}{\pi r_\beta^2} & \frac{T_{12}T_{11}}{2\pi^2 r_\beta^2} \end{bmatrix} \quad (\text{A.23})$$

$$\mathbf{D}^* = \mathbf{D}^*_1 + \Phi(0)\mathbf{D}^*_2 \quad (\text{A.24})$$

Where, $\Phi(0) = 1 - \Psi_1 + \Psi_2$.

Non-Dimensional: Aerodynamic Stiffness Matrix

$$\mathbf{F}^*_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{T_4+T_{10}}{\pi r_\alpha^2} \\ 0 & 0 & \frac{T_5-T_4T_{10}}{\pi^2 r_\beta^2} \end{bmatrix} \quad (\text{A.25})$$

$$\mathbf{F}^*_2 = \begin{bmatrix} 0 & 2 & \frac{2T_{10}}{\pi} \\ 0 & -\frac{2(a_h+\frac{1}{2})}{r_\alpha^2} & -\frac{2(a_h+\frac{1}{2})T_{10}}{\pi r_\alpha^2} \\ 0 & \frac{T_{12}}{\pi r_\beta^2} & \frac{T_{12}T_{10}}{\pi^2 r_\beta^2} \end{bmatrix} \quad (\text{A.26})$$

$$\mathbf{F}^*_3 = \begin{bmatrix} 2 & 2\left(\frac{1}{2}-a_h\right) & \frac{T_{11}}{\pi} \\ -\frac{2(a_h+\frac{1}{2})}{r_\alpha^2} & -\frac{2(a_h+\frac{1}{2})\left(\frac{1}{2}-a_h\right)}{r_\alpha^2} & -\frac{(a_h+\frac{1}{2})T_{11}}{\pi r_\alpha^2} \\ \frac{T_{12}}{\pi r_\beta^2} & \frac{T_{12}\left(\frac{1}{2}-a_h\right)}{\pi r_\beta^2} & \frac{T_{12}T_{11}}{2\pi^2 r_\beta^2} \end{bmatrix} \quad (\text{A.27})$$

$$\mathbf{F}^* = \mathbf{F}^*_1 + \Phi(0)\mathbf{F}^*_2 + \dot{\Phi}(0)\mathbf{F}^*_3 \quad (\text{A.28})$$

Where, $\Phi(0) = 1 - \Psi_1 + \Psi_2$ and $\dot{\Phi}(0) = \Psi_1\epsilon_1 + \Psi_2\epsilon_2$.

Non-Dimensional: Aerodynamic State Influence Matrices

$$\mathbf{W}^*_0 = \begin{bmatrix} -\Psi_1\epsilon_1^2 \\ -\Psi_2\epsilon_2^2 \\ \Psi_1\epsilon_1\left(1-\epsilon_1\left(\frac{1}{2}-a_h\right)\right) \\ \Psi_2\epsilon_2\left(1-\epsilon_2\left(\frac{1}{2}-a_h\right)\right) \\ \frac{\Psi_1\epsilon_1}{\pi}\left(T_{10}-\frac{\epsilon_1T_{11}}{2}\right) \\ \frac{\Psi_2\epsilon_2}{\pi}\left(T_{10}-\frac{\epsilon_2T_{11}}{2}\right) \end{bmatrix} \quad (\text{A.29})$$

$$\mathbf{W}^* = \begin{bmatrix} 2\mathbf{W}_0^T \\ -\frac{2(a_h+\frac{1}{2})}{r_\alpha^2}\mathbf{W}_0^T \\ \frac{T_{12}}{\pi r_\beta^2}\mathbf{W}_0^T \end{bmatrix} \quad (\text{A.30})$$

$$\mathbf{W}^*_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{A.31})$$

$$\mathbf{W}^*_2 = \begin{bmatrix} -\varepsilon_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\varepsilon_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\varepsilon_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\varepsilon_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\varepsilon_1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\varepsilon_2 \end{bmatrix} \quad (\text{A.32})$$

B

Appendix B: Validation

This section details the validation of the developed numerical framework (Chapter 4). The dimensional flutter diagrams and HBM coupled with pseudo arc-length continuation algorithm are first verified against literature benchmarks [4] across linear stability, supercritical and subcritical bifurcations. Next, the non-dimensional aeroelastic model is validated against the corresponding dimensional results. Finally, the analytical stability and bifurcation detection algorithms are validated against the time-integration results for the torus case extracted from [4].

B.1. Dimensional Aeroelastic Model

Although the remainder of this thesis relies exclusively on the proposed non-dimensional aeroelastic model, the dimensional aeroelastic model is retained initially for validation purposes. This approach intentionally leverages the extensive availability of dimensional benchmark data in the existing literature, specifically the test cases detailed in [4].

B.1.1. Linear Stability Analysis: Flutter Diagrams

The linear stability analysis presented in Section 4.3.1 is validated against example 5.1 (p.162) from [4]. The corresponding dimensional parameters are:

Table B.1: Example 5.1 3-DOF Aeroelastic System Parameters [4]

<i>Fluid & Geometric Properties</i>				<i>Inertial Properties</i>				
ρ	c	a_h	c_h	m	S_α	S_β	I_α	I_β
1.225	0.25	-0.2	0.5	13.5	0.3375	0.1055	0.0787	0.0044
<i>Stiffness & Damping</i>								
K_h	K_α	K_β	C_h	C_α	C_β	K_h^{nl}	K_α^{nl}	K_β^{nl}
2131.835	198.9712	17.3489	$K_h/1000$	$K_\alpha/1000$	$K_\beta/1000$	0	$100K_\alpha$	0

Figure B.1 demonstrates that the dimensional aeroelastic model correctly predicts the evolution of the natural frequencies and damping ratios as the freestream velocity increases according to [4] under unsteady aerodynamics.

B.1.2. Supercritical

Following the detection of the Hopf bifurcation (Section 4.3.2), the post-flutter dynamics are captured via HBM coupled with an arc-length continuation scheme (Section 4.4). This section verifies these numerical implementations against the supercritical case with hardening stiffness (Table B.1) presented in Section 5.4 entitled "Hardening Stiffness" (p.170) of [4]. Because the reference relies on equivalent linearization, only one harmonic is included in the HBM model.

Figure B.2 illustrates the validation of the predicted LCO amplitudes and frequencies against [4]. The Hopf bifurcation is precisely detected at $U = 40.95$ m/s. While the heave, pitch, and frequency variations exhibit excellent correlation, the flap amplitude is slightly underestimated at airspeeds exceeding 43 m/s (maximum relative error of 9.72% at $U = 44.2$ m/s). This error falls within acceptable margins and is attributed to a combination of methodological differences (equivalent linearization vs HBM coupled with arc-length continuation) and minor manual data digitisation errors.

B.1.3. Subcritical

To validate the continuation scheme's capacity to resolve subcritical bifurcations and folded LCO branches, Example 5.6 from [4] is reproduced. The system parameters are identical to those presented in Table B.1, however, the nonlinear stiffness is now localised to the heave DOF and is equal to $K_h^{nl} = 1000K_h$. Again, the reference results rely on equivalent linearization therefore the numerical HBM scheme is correspondingly configured with a single harmonic.

As depicted in Figure B.3, the predicted Hopf bifurcation, alongside the heave LCO amplitude and frequency variations, aligns precisely with the reference data from [4]. The HBM arc-length continuation scheme successfully predicts folded unstable branches.

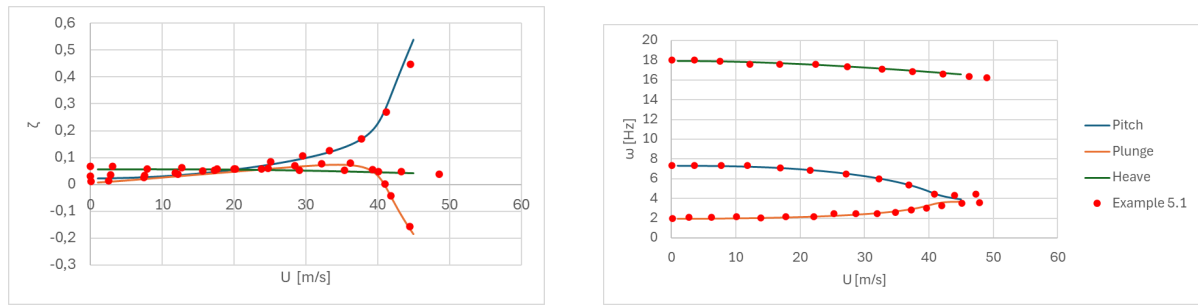


Figure B.1: Linear stability dimensional: Flutter diagrams

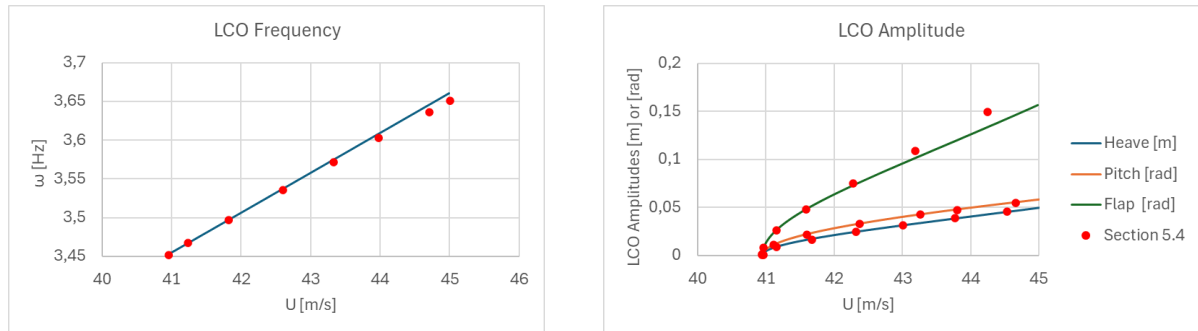


Figure B.2: Supercritical dimensional: LCO amplitude and frequency variation with airspeed

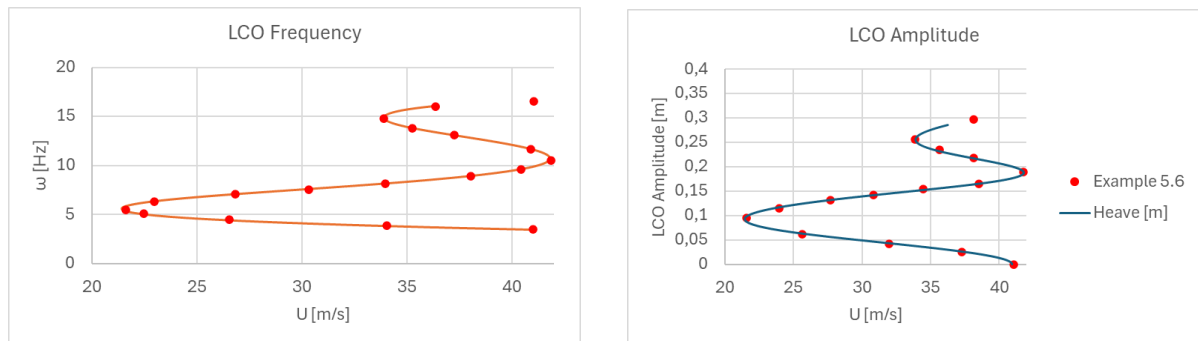


Figure B.3: Subcritical dimensional: LCO amplitude and frequency variation with airspeed

B.2. Non-Dimensional Aeroelastic Model

As established in Section 4.2.3, the governing aeroelastic equations are non-dimensionalized in the remainder of this Thesis. To validate its implementation, the dimensional validation results from the previous section will act as a benchmark after being converted into their dimensionless equivalents for the outputs of the non-dimensional HBM solver. The non-dimensional system parameters for the supercritical and subcritical cases (corresponding to the dimensional parameters of Table B.1) are detailed in Table B.2. A number of 3 harmonics are included in the HBM scheme.

As illustrated in Figures B.4 and B.5, the non-dimensional aeroelastic model precisely captures the Hopf bifurcation at $U^* = 6.51$, as well as the post-flutter amplitudes across all degrees of freedom and the LCO frequency ratio ($\omega^* = \frac{\omega U^*}{\omega_\alpha}$). By accurately reproducing both the supercritical and subcritical behaviours, the non-dimensional model is considered validated.

Table B.2: Non-dimensional aeroelastic system parameters

Fluid & Geometric Properties				Inertial Properties			
μ	c	a_h	c_h	x_α	x_β	r_α	r_β
224.51	0.25	-0.2	0.5	0.2	0.0625	0.611	0.144
Stiffness & Damping							
ζ_ζ	ζ_α	c_β^*	Ω_1	Ω_2	κ_ζ	κ_α	Γ_β
0.006	0.025	0.078	0.25	1.25	15.625 (Subcritical)	100 (Supercritical)	0

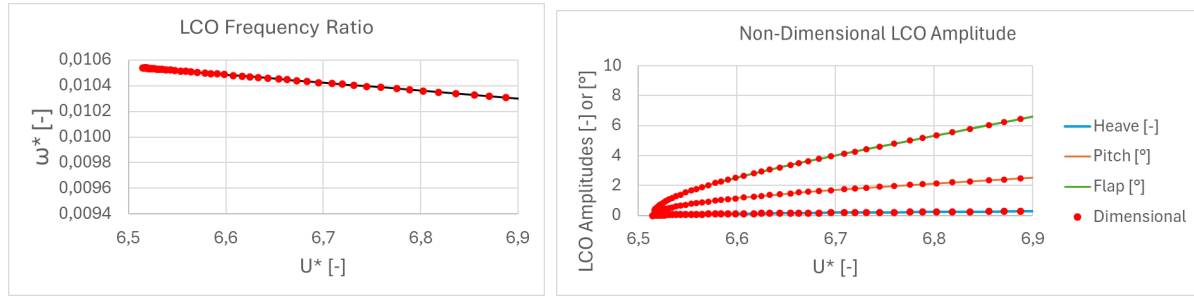


Figure B.4: Supercritical non-dimensional: LCO amplitude and frequency variation with airspeed

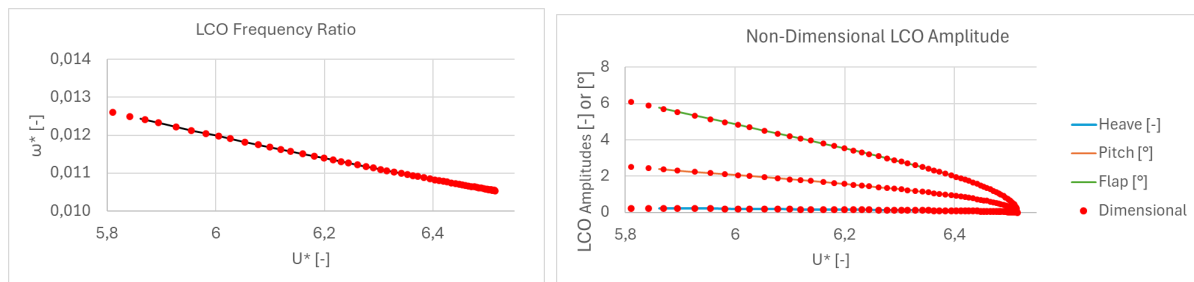


Figure B.5: Subcritical non-dimensional: LCO amplitude and frequency variation with airspeed

B.3. Stability and Bifurcation Identification

Following the prediction of the post-flutter LCO states via HBM, the stability and bifurcation identification algorithms (Section 4.5) must be validated. This is achieved by reproducing the Torus bifurcation case from [4], for which the corresponding non-dimensional parameters provided in Table B.3.

Table B.3: Non-dimensional aeroelastic system parameters: torus (Example 5.9 p 211) [4]

Fluid & Geometric Properties				Inertial Properties			
μ	a_h	c_h	x_α	x_β	r_α	r_β	
224.51	-0.2	0.6	0.2	0.04	0.611	0.102	
Stiffness & Damping							
ζ_ζ	ζ_α	c_β^*	Ω_1	Ω_2	κ_ζ	κ_α	Γ_β
0.003	0.038	0.026	0.083	0.590	0	0	34.779

The validation methodology is outlined in Figure B.6. Following the HBM continuation, the state vector corresponding to each resolved LCO point is extracted to act as the initial condition for the time-integration solver. Standard time-integration inherently converges to whichever attractor's basin encompasses the arbitrary initial conditions, often masking co-existing dynamic responses. With precise initial conditions, these alternative branches can be isolated and tracked. Once the time-domain simulation reaches a steady state at the target airspeed, the peaks are recorded to construct the bifurcation diagram. A stable Period-1 LCO yields a single, consistent peak amplitude, manifesting as a discrete point. Conversely, a quasi-periodic response generates a densely modulated amplitude, appearing as a continuous vertical band of points, whereas chaotic motion results in scatter. Finally, the sampled time-integration peaks are superimposed onto the HBM continuation branches to validate stability and bifurcation point localisation.

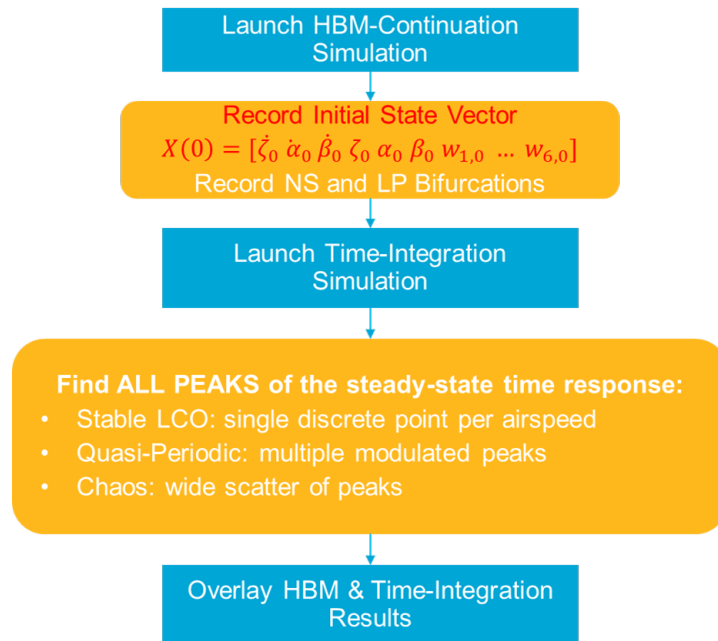


Figure B.6: Stability and bifurcation identification validation

A convergence study determined that retaining five harmonics is necessary to accurately model the post-flutter dynamics, stability and bifurcation points. The system possesses two co-existing responses: a primary supercritical branch originating at a Hopf bifurcation of $U^* = 4.98$, and a secondary subcritical branch emerging at $U^* = 6.92$.

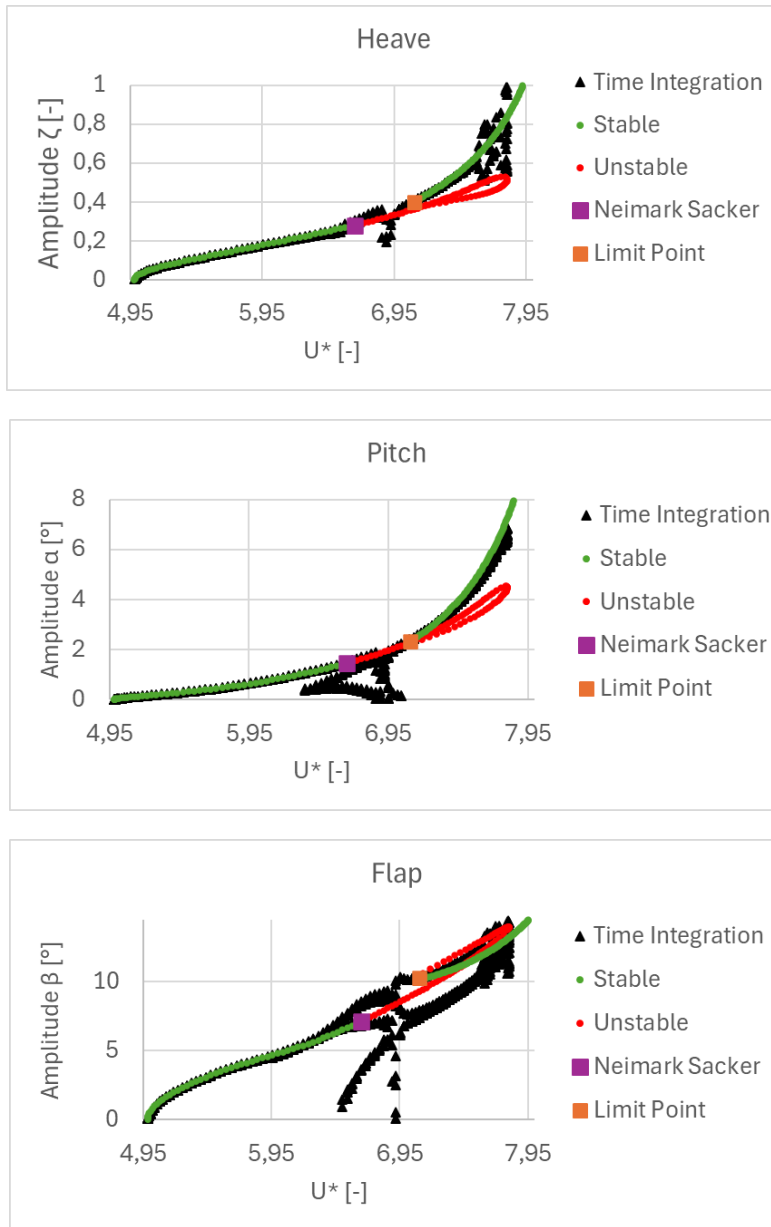


Figure B.7: Branch 1: Post-flutter HBM vs time-integration

Figure B.7 illustrates the post-flutter dynamics for Branch 1.

- **Stable** ($U^* = 6.98$ to ≈ 6.45): The HBM accurately predicts the stable LCO amplitudes across all three degrees of freedom originating from the Hopf bifurcation.
- **Quasi-Periodic Onset** ($U^* \approx 6.45$): Time-integration reveals modulated peaks between $U^* \approx 6.45$ and 6.50 , indicating quasi-periodic behaviour. HBM detects the corresponding Neimark-Sacker bifurcation at $U^* = 6.65$. This minor 2–3% error is acceptable and likely stems from residual transients during time-domain sampling.
- **Re-stabilization and Chaos** ($U^* = 6.3$ to 7.55): The branch remains unstable until a limit point (fold) at $U^* = 6.3$. After this fold, HBM matches the stable LCO amplitudes obtained through time integration up to $U^* = 7.55$. Beyond this airspeed, the physical response becomes chaotic. HBM does not capture this final transition from stable to unstable.

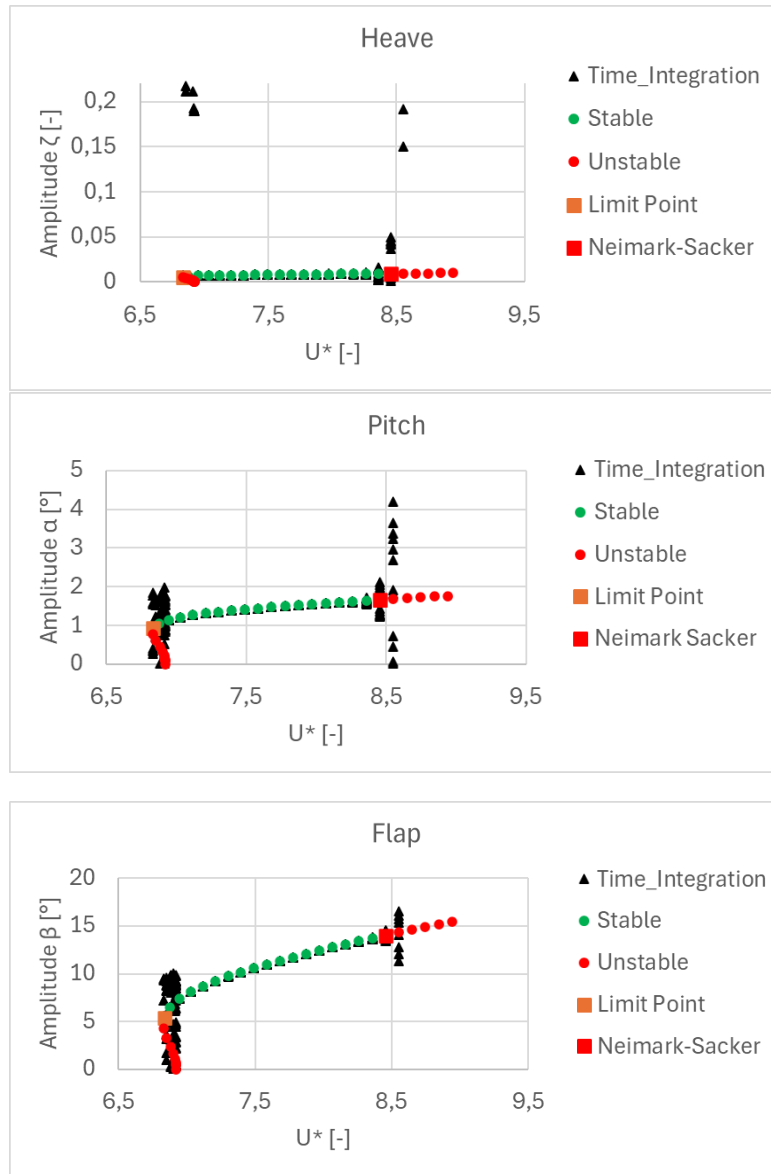


Figure B.8: Branch 2: Post-flutter HBM vs time-integration

Figure B.8 details the post-flutter dynamics for Branch 2 (B2).

- **Unstable Fold ($U^* = 6.92$ to 6.83):** An unstable branch emerges from the Hopf bifurcation until reaching a limit point at $U^* = 6.83$, predicted by HBM stability analysis. The dense vertical band observed in the time-integration data confirms the instability in this specific airspeed range.
- **Stable LCO ($U^* = 6.83$ to 8.46):** Following the limit point, the branch regains stability. The HBM accurately tracks the stable LCO amplitudes across all three degrees of freedom, aligning perfectly with the time-integration results.
- **Quasi-Periodic Onset ($U^* = 8.46$):** The HBM stability analysis correctly detects a Neimark-Sacker bifurcation at $U^* = 8.46$, which is confirmed by the time-integration results showing the physical emergence of modulated quasi-periodic peaks at the same freestream velocity.

Appendix C: Inertial Equivalence

When unlocking the Flap-NES, the global mass, center of gravity and rotational inertia of the wing assembly must remain identical to the 2-DOF baseline (Figure C.1) in order to assess the Flap-NES' capacity to mitigate flutter. Consequently, this section establishes the methodology required to calculate the equivalent inertial parameters for the 3-DOF system, the non-dimensional parameters x_β and r_β (Table 4.2).

2-DOF Baseline (Locked Flap)

- Heave & Pitch
- Hinge Fixed

3-DOF with Flap-NES (Unlocked Flap)

- Static Unbalanced: x_β
- Radius of Gyration: r_β

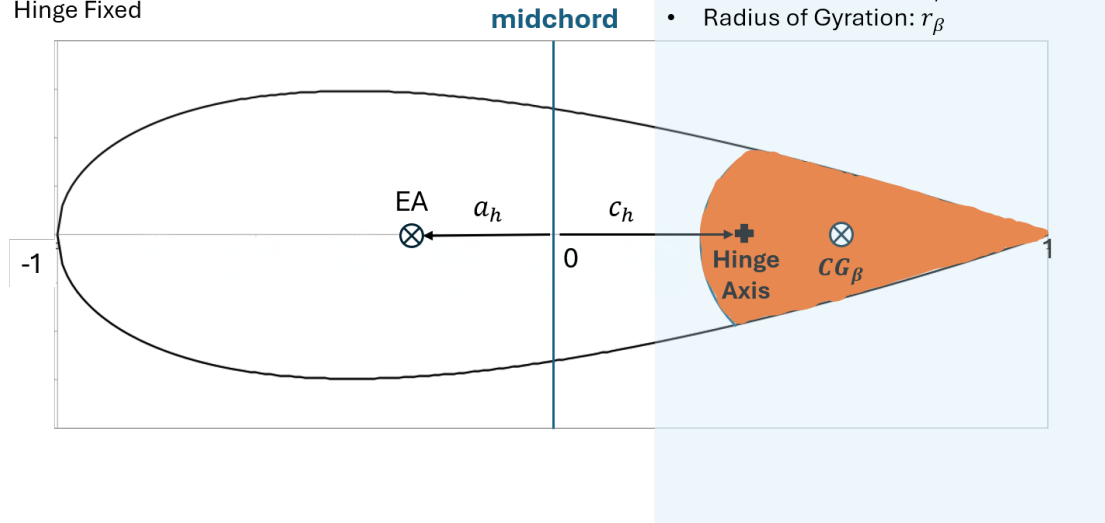


Figure C.1: Schematic representation of the inertial parameter equivalence between the locked 2-DOF baseline and the unlocked 3-DOF Flap-NES configuration

The symmetric NACA 0012 airfoil is employed throughout the remainder of this study. Its maximum thickness is located at 30% of the chord, with a magnitude of 12% of the total chord length [26]. According to the NACA four-digit series convention [23], the half-thickness distribution is defined as:

$$\pm y(x) = \frac{t_{max}}{0.2} (0.2959\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \quad (C.1)$$

where $x \in [-1, 1]$ is the normalized coordinate along the chord, and $t_{max} = 0.12$ is the maximum thickness ratio. The total non-dimensional thickness function is given by $t(x) = 2y(x)$ [23].

First Moment of Inertia (S_β) : The first moment of inertia is given as the integral of every infinitesimal mass element, dm , multiplied by its perpendicular distance from the pivot axis.

$$S = \int r dm \quad (C.2)$$

For the flap, the pivot axis is the hinge at c_h . The flap stretches from $c_h - 1$, and the distance r of any mass from the hinge is $\xi - c_h$ ¹. Therefore, the flap's first moment of inertia is given as:

$$\begin{aligned} S_\beta &= \int_{c_h}^1 b(\xi - c_h) [\rho t(\xi) b] d\xi \\ S_\beta &= \rho b^2 \int_{c_h}^1 (\xi - c_h) t(\xi) d\xi \end{aligned} \quad (C.3)$$

To obtain the resulting x_β , the first moment of inertia must be normalised by the total airfoil mass and half chord b . The total mass of the wing is given as: $m = \rho b \int_{c_h}^1 t(\xi) d\xi$ therefore:

$$x_\beta = \frac{S_\beta}{mb} = \frac{\rho b^2 \int_{c_h}^1 (\xi - c_h) t(\xi) d\xi}{\rho b^2 \int_{c_h}^1 t(\xi) d\xi} = \frac{\int_{c_h}^1 (\xi - c_h) t(\xi) d\xi}{\int_{c_h}^1 t(\xi) d\xi} \quad (C.4)$$

Second Moment of Inertia (I_β) : The second moment of inertia of the flap about the hinge line is defined as the integral of the mass element multiplied by the square of its perpendicular distance from the pivot axis:

$$I = \int r^2 dm \quad (C.5)$$

Following the same procedure as for the first moment of inertia, the corresponding non-dimensional parameter, the flap radius of gyration, r_β is given as:

$$r_\beta = \sqrt{\frac{I_\beta}{mb^2}} = \sqrt{\frac{\int_{c_h}^1 (\xi - c_h)^2 t(\xi) d\xi}{\int_{c_h}^1 t(\xi) d\xi}} \quad (C.6)$$

¹The coordinates of the airfoil are non-dimensionalised such that $x = \xi b$; therefore $dx = b d\xi$

D

Appendix D: Parametric Study

The following parametric study identified the linear stiffness as the critical criteria for the Flap-NES. Three regions emerge: low linear stiffness $0 \leq \Omega_2 < \Omega_1$, intermediate linear stiffness $\Omega_1 \leq \Omega_2 < 1$ and higher linear stiffness $\Omega_2 > 1$. For clarity the results will be presented as a summary matrix for each region for both the linear and post-flutter analysis.

D.1. Low Linear Stiffness $0 \leq \Omega_2 < \Omega_1$

The Flap-NES with low linear stiffness fails to suppress flutter, instead reducing the stability margin by at least 67.32% via premature flap/heave coalescence. This instability induces a subcritical Hopf bifurcation, where the cubic stiffness triggers a severe deep resonance with the heave mode. Consequently, the system experiences extreme limit cycle oscillations well below the baseline flutter speed ($U^* < 4.76$), reaching pitch and heave amplitudes of 84.29° and 1.98, respectively

D.1.1. Linear Stability Analysis

The summarized trends observed from Figure D.1 are presented in Table D.1. For all tested parameter combinations flutter is driven by a premature flap-heave frequency coalescence. The first Hopf bifurcation is detected as the heave mode becomes unstable. The second Hopf bifurcation is detected when the pitch becomes unstable.

The larger linear stiffness (Figure D.1a), increases the Flap-NES natural frequency (ω_n^*) rendering it closer to the heave natural frequency and therefore decreasing the first Hopf bifurcation airspeed U_f^* . While Flap-NES size (c_h) and damping (c_β^*) do not alter the fundamental nature of these instabilities, increasing c_h (Figure D.1b) significantly lowers the pitch natural frequency. This occurs because a smaller flap (larger c_h) translates to a proportionately larger main airfoil, which inherently reduces the natural pitch frequency.

Ultimately, for the tested parameters, releasing the Flap-NES with low stiffness introduces a premature flutter mode, reducing the linear stability margin by at least 67.32% compared to the 2-DOF baseline.

Table D.1: Flutter diagrams: Summary of parametric sensitivities lower linear stiffness

Parameter Increased	Flap-NES ω_n^*	Flutter Mechanism	Unstable mode	$U_{f,1}^*$
Linear Stiffness (Ω_2)	Increases	Flap/Heave	Heave	Decreases
Flap Size (c_h)	No change	Flap/Heave	Heave	Increases
Damping (c_β^*)	No change	Flap/Heave	Heave	No change

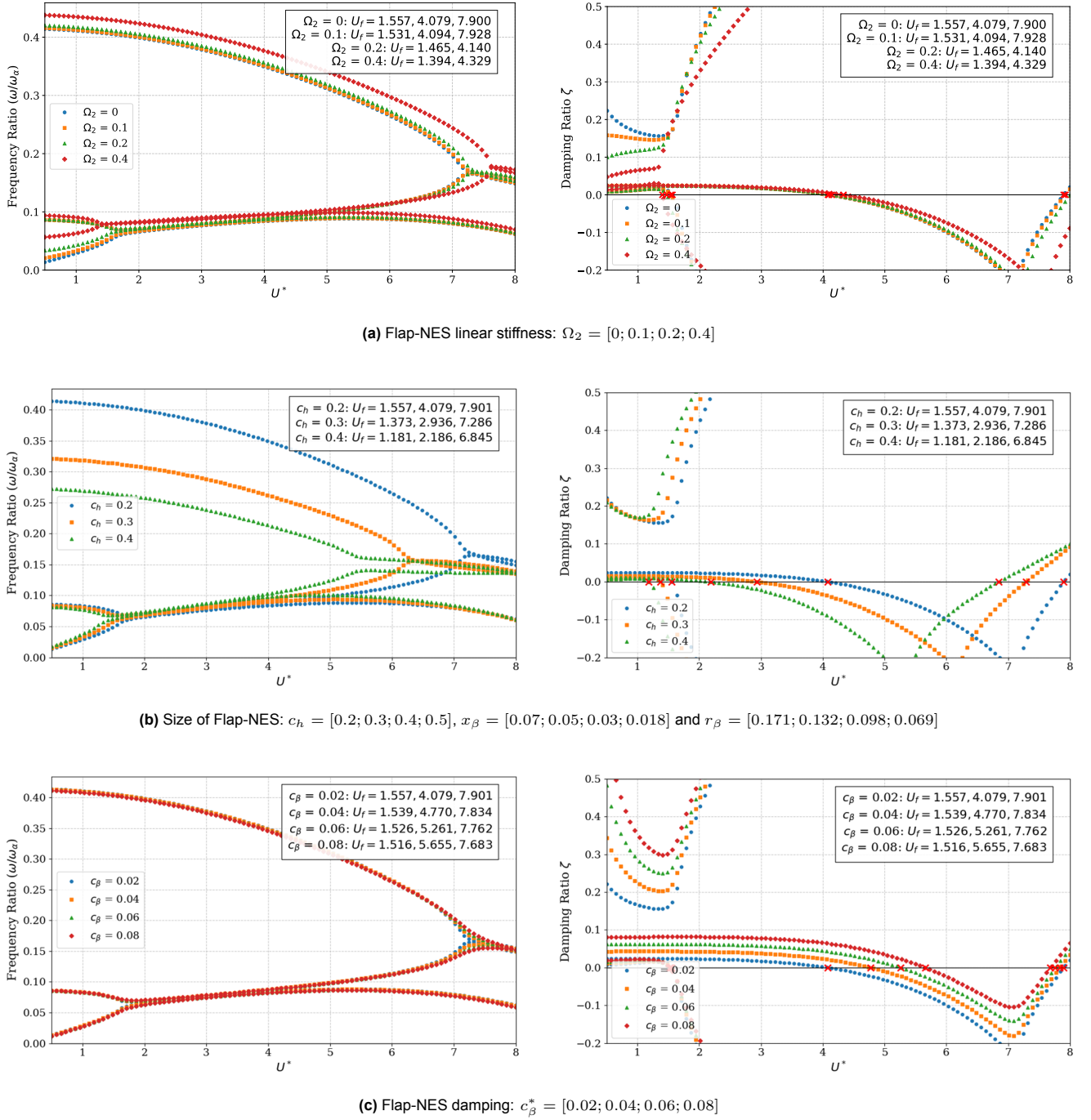


Figure D.1: Flutter diagram (parameter sweep): Low linear stiffness

Fixed parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.2$, $x_\alpha = 0.25$, $x_\beta = 0.070$, $r_\alpha = 0.5$, $r_\beta = 0.171$, $\Omega_1 = 0.5$, $\Omega_2 = 0.01$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$.

D.1.2. Post-Flutter Analysis

Figure D.2 illustrates the post-flutter dynamic response emerging from the first Hopf bifurcation for varying Flap-NES nonlinear stiffness (Γ_β), linear stiffness (Ω_2), size (c_h), and damping (c_β^*).

The Hopf bifurcation is subcritical. Modifying Γ_β , c_h , or c_β^* does not transition the bifurcation from subcritical to supercritical. At airspeeds below the 2-DOF baseline linear flutter speed ($U^* < 4.76$), the system experiences severe LCO amplitudes in both pitch and heave. Specifically, at an airspeed of $U^* = 4.33$, the pitch and heave amplitudes reach 84.29° and 1.98 , respectively.

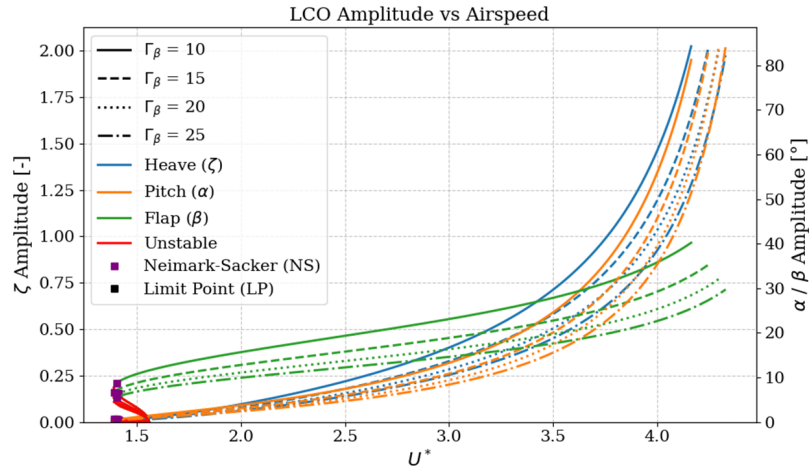
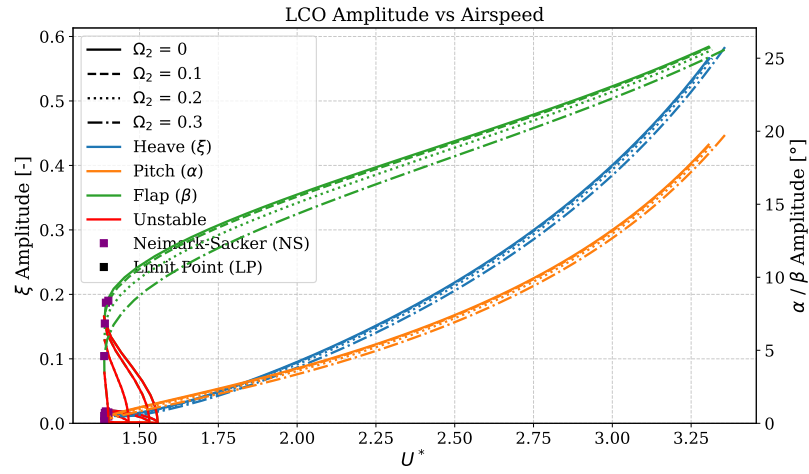
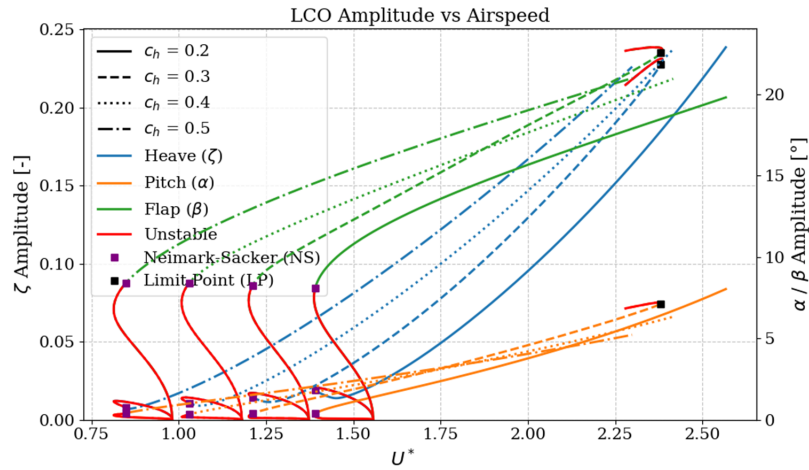
(a) Flap-NES Cubic Stiffness: $\Gamma_\beta = [10; 15; 20; 25]$ (b) Flap-NES linear stiffness: $\Omega_2 = [0; 0.1; 0.2; 0.4]$ (c) Size of Flap-NES: $c_h = [0.2; 0.3; 0.4; 0.5]$, $x_\beta = [0.07; 0.05; 0.03; 0.018]$ and $r_\beta = [0.0171; 0.132; 0.098; 0.069]$

Figure D.2: Bifurcation diagram (parameter sweep): Low linear stiffness (1/2)

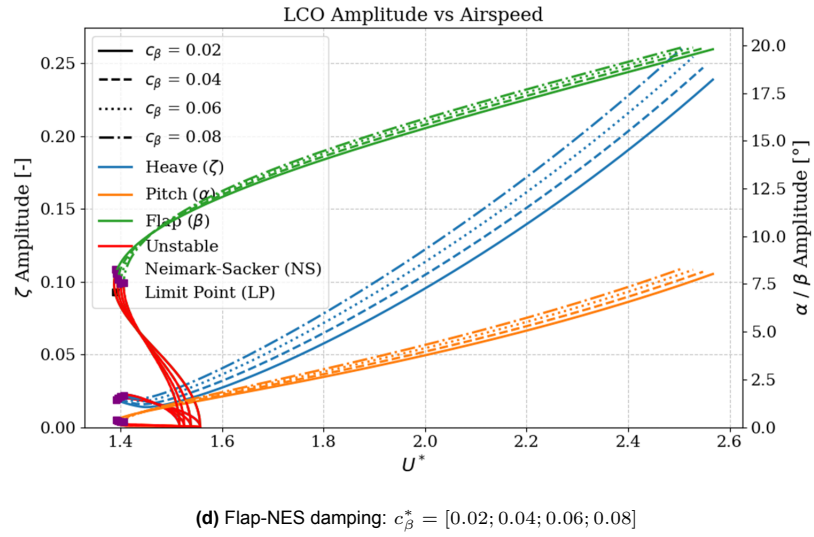


Figure D.2: Bifurcation diagram (parameter sweep): Low linear stiffness (2/2)

Fixed parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.2$, $x_{\alpha} = 0.25$, $x_{\beta} = 0.070$, $r_{\alpha} = 0.5$, $r_{\beta} = 0.171$, $\Omega_1 = 0.5$, $\Omega_2 = 0.01$, $\zeta_h = \zeta_{\alpha} = 0.002$, $c_{\beta}^* = 0.02$, $\kappa_h = \kappa_{\alpha} = 0$, $\Gamma_{\beta} = 10$.

Harmonic Balance parameters: $H = 5$, $N = 2^8$, $\Delta_{smin} = 10^{-4}$, $\Delta_{smax} = 0.01$, $\epsilon_{NR} = 10^{-9}$, $\epsilon_{stab} = 10^{-6}$, $dU^* = 0.001$, $dX = 0.02$.

D.2. Intermediate Linear Stiffness: $\Omega_1 < \Omega_2 < 1$

In the second configuration, the Flap-NES is released, but its linear stiffness is positioned intermediately between the heave and pitch modes ($\Omega_1 < \Omega_2 < 1$). While some parameter combination achieved recurring burstout and suppression TET, the resulting LCO amplitudes remain too high below the baseline liner flutter speed.

D.2.1. Linear Stability Analysis

Figure D.3 presents the flutter diagrams for the Flap-NES configured with intermediate linear stiffness ($\Omega_1 < \Omega_2 < 1$). The upper, middle, and lower branches in the frequency evolution graph correspond to the pitch, Flap-NES, and heave modes, respectively. Table D.2 summarizes the notable trends.

Table D.2: Flutter diagrams: Summary of parametric sensitivities intermediate linear stiffness

Parameter Increased	Flap-NES Ω_n^*	Flutter Mechanism	$U_{f,1}^*$	$U_{f,2}^*$
Linear Stiffness (Ω_2)	Increases	Flap/Heave	Increases	Increases
Flap Size (c_h)	No change	Flap/Heave	Increases	Increases
Damping (c_β^*)	No change	Flap/Heave	No change	Increases

The primary flutter mechanism is driven by the mode coalescence of the heave and Flap-NES modes. Consequently, this configuration reduces the linear stability envelope by at least 61% compared to the 2-DOF baseline.

As illustrated in Figure D.3a, the Flap-NES size (c_h) significantly influences the pitch natural frequency (upper branch). A larger flap (smaller c_h) further detaches the pitch eigenfrequency. Furthermore, a larger flap results in greater hopf bifurcation airspeeds.

Increasing the linear stiffness (Ω_2) increases the Flap-NES natural frequency thus increasing modal separation and results in higher linear flutter speeds as observed in Figure D.3b.

Finally, the linear damping (c_β^*) does not alter the frequency ratio evolution (Figure D.3c). While it only slightly affects the first Hopf bifurcation airspeed it does increase the second Hopf bifurcation airspeed.

D.2.2. Post-Flutter: Bifurcation Diagrams

Figure D.4 illustrates the post-flutter dynamic response emerging from the first Hopf bifurcation.

The Hopf bifurcation is supercritical. For the smallest tested Flap-NES size ($c_h = 0.4$), the system initiates the first TET mechanism (recurring burstout and suppression) as evidenced by a Neimark-Sacker (NS) bifurcations in Figure D.3a. Between $U^* = 3.753$ and 4.012 , the LCO branch destabilises; however, a stable quasi-periodic motion emerges, bounded by the two NS bifurcations. The branch then briefly regains stability before destabilising once again at $U^* = 4.041$, ultimately terminating at a Limit Point (LP) bifurcation at $U^* = 3.586$. The size of the Flap-NES serves as the primary parameter governing the onset of quasi-periodic behaviour.

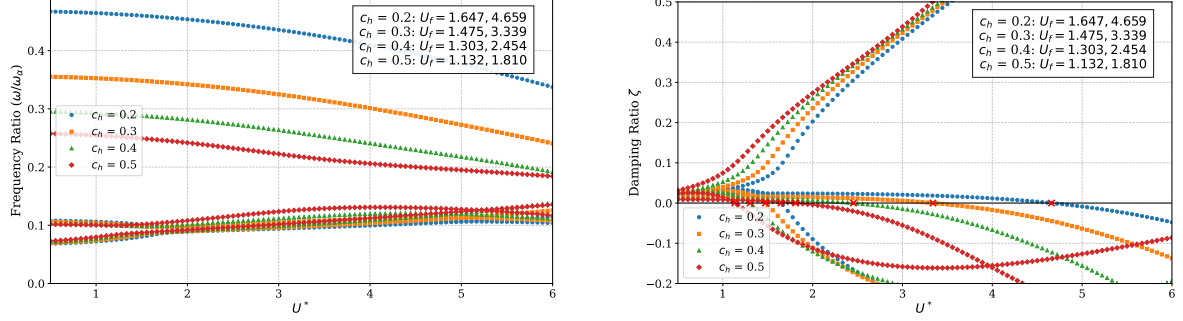
Note that for $c_h = 0.3$, an NS bifurcation is also present at $U^* = 2.528$. However, this point is reached only after the branch folds backwards at an LP bifurcation located at $U^* = 2.695$. The precedence of this limit point indicates that this specific velocity regime is strictly unstable.

Fixing $c_h = 0.4$, both the flap/pitch frequency ratio (Ω_2) and nonlinear stiffness (Γ_β) (Figures D.4c and D.4b) do not affect the post-flutter dynamics described above but only influence the resulting amplitudes. The greater Γ_β and Ω_2 , the smaller the resulting amplitudes.

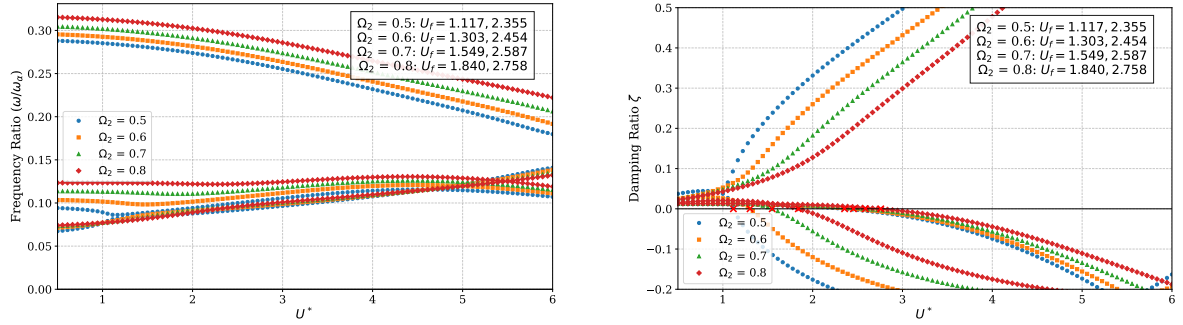
For all but one value of linear damping (c_β , Figure D.4d), the system engages in quasi-periodic motion. However, for $c_\beta^* = 0.04$, no NS bifurcation is detected; instead, an unstable zone bounded by two LP

bifurcations is recorded ($U^* = 3.926 - 3.494$). As such, there exists an optimum c_β^* to obtain quasi-periodic motion.

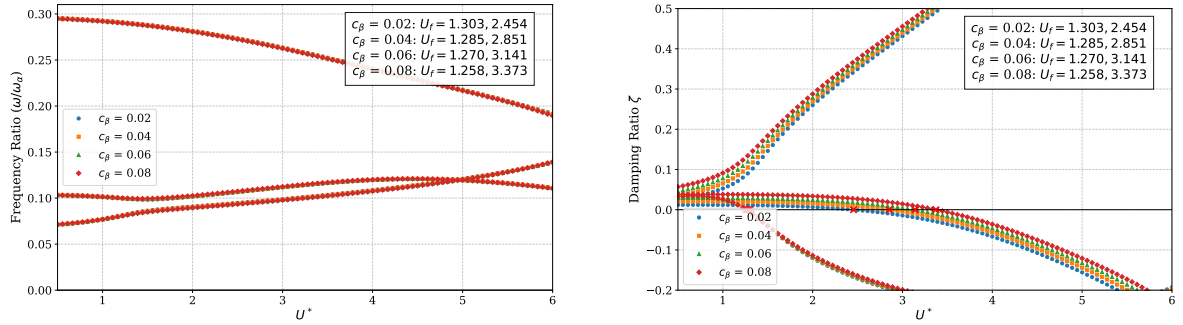
Although quasi-periodic motion indicates active TET, the resulting LCO amplitudes remain too severe for effective flutter mitigation. Even in the optimal case ($U^* = 3.81$), the flap, pitch, and heave amplitudes reach 32.45° , 24.09° , and 0.85 , respectively.



(a) Size of Flap-NES: $c_h = [0.2; 0.3; 0.4; 0.5]$, $x_\beta = [0.07; 0.05; 0.03; 0.018]$ and $r_\beta = [0.171; 0.132; 0.098; 0.069]$



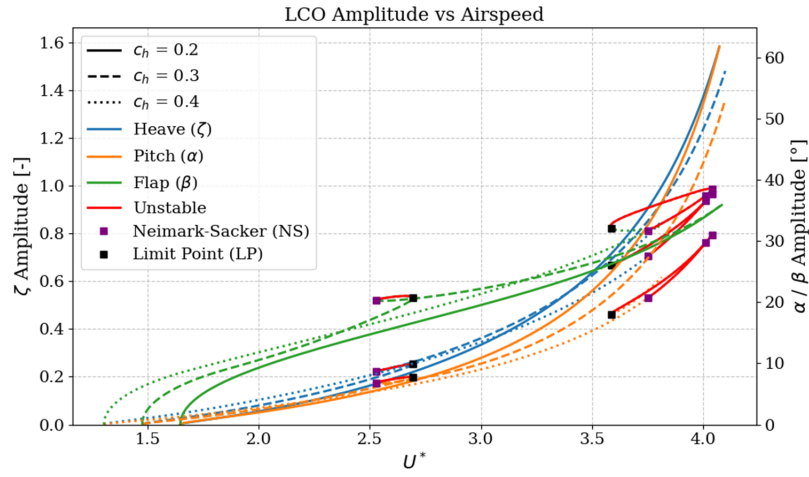
(b) Flap-NES linear stiffness: $\Omega_2 = [0.5; 0.6; 0.7; 0.8]$



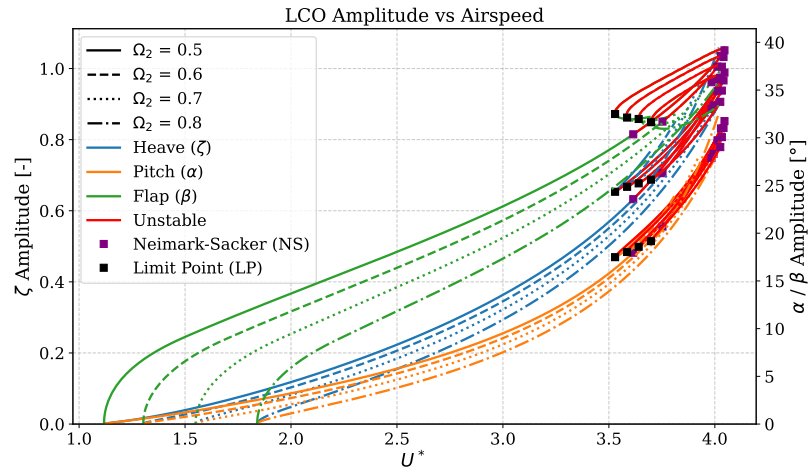
(c) Flap-NES damping: $c_\beta^* = [0.02; 0.04; 0.06; 0.08]$

Figure D.3: Flutter diagram (parameter sweep): Intermediate linear stiffness

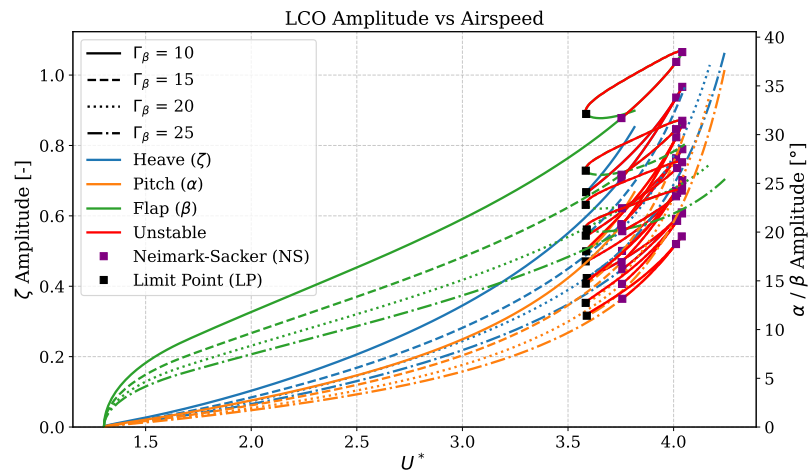
Fixed parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.4$, $x_\alpha = 0.25$, $x_\beta = 0.031$, $r_\alpha = 0.5$, $r_\beta = 0.098$, $\Omega_1 = 0.5$, $\Omega_2 = 0.6$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$.



(a) Size of Flap-NES: $c_h = [0.2; 0.3; 0.4; 0.5]$, $x_\beta = [0.07; 0.05; 0.03; 0.018]$ and $r_\beta = [0.0171; 0.132; 0.098; 0.069]$

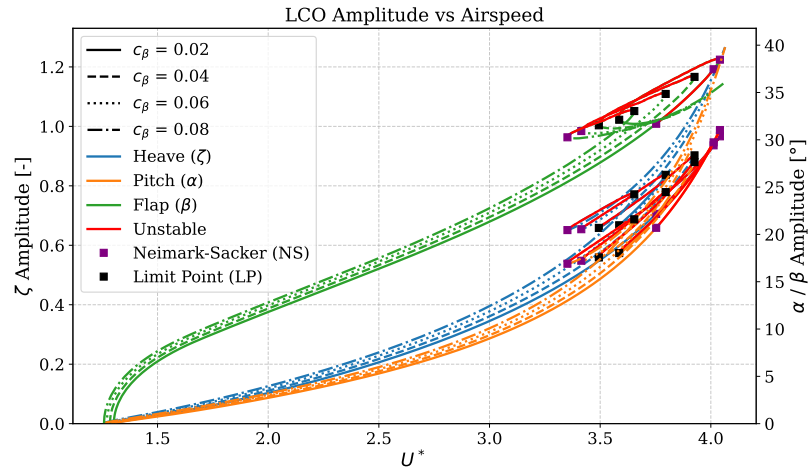


(b) Flap-NES linear stiffness: $\Omega_2 = [0.5; 0.6; 0.7; 0.8]$



(c) Flap-NES Cubic Stiffness: $\Gamma_\beta = [10; 15; 20; 25]$

Figure D.4: Bifurcation diagram (parameter sweep): Intermediate linear stiffness (1/2)

(d) Flap-NES damping: $c_\beta^* = [0.02; 0.04; 0.06; 0.08]$ **Figure D.4:** Bifurcation diagram (parameter sweep): Intermediate linear stiffness (2/2)

Fixed parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.4$, $x_\alpha = 0.25$, $x_\beta = 0.031$, $r_\alpha = 0.5$, $r_\beta = 0.098$, $\Omega_1 = 0.5$, $\Omega_2 = 0.6$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$, $\kappa_h = \kappa_\alpha = 0$, $\Gamma_\beta = 10$.

Harmonic Balance parameters: $H = 5$, $N = 2^8$, $\Delta s_{min} = 10^{-4}$, $\Delta s_{max} = 0.05$, $\epsilon_{NR} = 10^{-9}$, $\epsilon_{stab} = 10^{-6}$, $dU^* = 0.001$, $dX = 0.02$.

D.3. High Linear Stiffness: $\Omega_2 > 1$

The Flap-NES with a greater linear stiffness $\Omega_1 > 1$ exhibits, with proper tuning of its size, c_h , linear stiffness Ω_1 and linear damping c_β^* , achieves a Hopf-Hopf bifurcation which drives the post-flutter dynamics, including bounded pitch and heave LCO amplitudes and quasi-periodic motion onset. The nonlinear cubic stiffness Γ_β acts purely as a scalar, with higher values directly translating to lower LCO amplitudes.

D.3.1. Linear Stability Analysis

Figure D.5 presents the flutter diagrams for the Flap-NES with increased linear stiffness ($\Omega_2 > 1$). The top, middle and bottom frequency branches correspond to flap, pitch and heave, respectively. Table D.3 summarizes the primary trends.

Table D.3: Flutter diagrams: Summary of parametric sensitivities higher linear stiffness

Parameter Increased	Flap-NES Ω_n^*	Flutter Mechanism	$U_{f,1}^*$	ΔU_f^*
Linear Stiffness (Ω_2)	Increases	Heave/Pitch	Increases	Minimal for $\Omega_2 = 1.1$ and 1.2
Flap Size (c_h)	Increases	No frequency coalescence for $c_h = 0.5$	Increases	Minimal for $c_h = 0.5$
Damping (c_β^*)	No change	Heave/Pitch	No change	No change

When decreasing the size of the Flap-NES (increasing c_h), (Figure D.5a), the natural frequency of the flap decreases. A classic heave-pitch coalescence is observed for $c_h < 0.5$ due to a highly detached Flap-NES eigenfrequency. However, at $c_h = 0.5$ the heave, pitch and flap modes are decoupled. The heave and flap simultaneously become unstable at two distinct linear flutter speeds through SDOF flutter mechanisms ($U_{f,1}^* = 2.735$ and $U_{f,2}^* = 2.765$).

As established in Section 3.1.1, this simultaneous destabilisation is a primary indicator of a Hopf-Hopf bifurcation, which precipitates the onset of quasi-periodic behaviour. Accordingly, the size parameter is fixed at $c_h = 0.5$.

Figure D.5b illustrates that a greater flap/pitch frequency ratio (Ω_2) increases the decoupling of the three modes. Recall, the $\Omega_2 = \frac{\omega_\beta}{\omega_\alpha}$, therefore the greater Ω_2 , the greater the flap and the lower the pitch natural frequency.

More importantly, the values of $\Omega_2 = 1.1$ and 1.2 yield the smallest difference between the two successive Hopf bifurcation airspeeds ($\Delta U_f^* = 0.03$). There exists an optimal Ω_2 value for inducing quasi-periodic behaviour via a Hopf-Hopf bifurcation.

Finally, while the linear damping (c_β^*) has a negligible impact on the frequency ratio evolution (Figure D.5c), it affects the difference between the Hopf Bifurcation airspeeds. The minimum difference ($\Delta U_f^* = 0.03$) is achieved at the lowest tested damping value of $c_\beta^* = 0.02$. There exists an optimal value of $c_\beta^* = 0.02$ to achieve quasi-periodic motion.

D.3.2. Post-Flutter

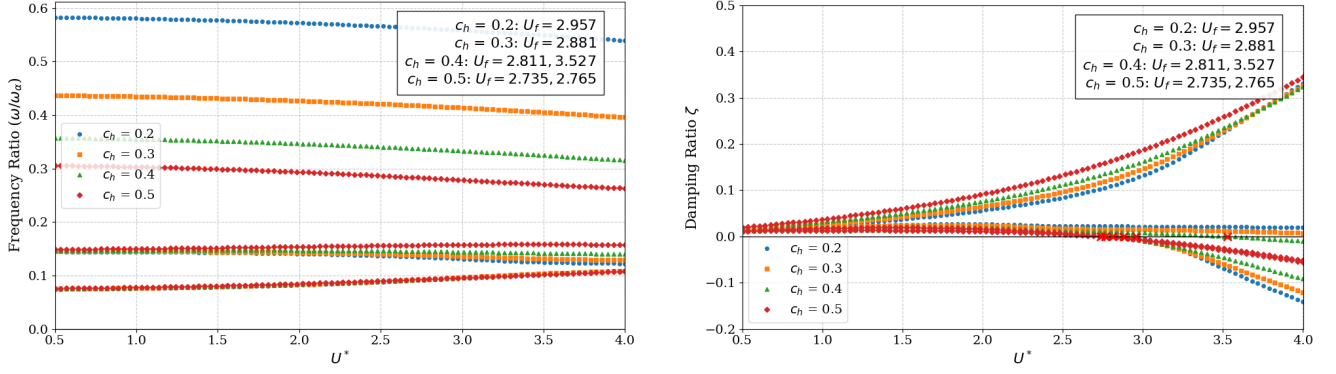
Figure D.6 depicts the post-flutter dynamic response emerging from the first Hopf bifurcation for varying Flap-NES nonlinear stiffness (Γ_β), size (c_h) and damping (c_β^*). The final tested velocities, resulting LCO amplitudes and NS bifurcation airspeed are presented in Table ??.

The Hopf Bifurcation is supercritical, and for a set of parameter combinations, the Flap-NES successfully bounds the LCO amplitudes of pitch and heave below 1.67 and 0.02 respectively before engaging in quasi-periodic motion as marked by the NS bifurcation at $U_{NS}^* = 4.17$ or $U_{NS}^* = 3.99$.

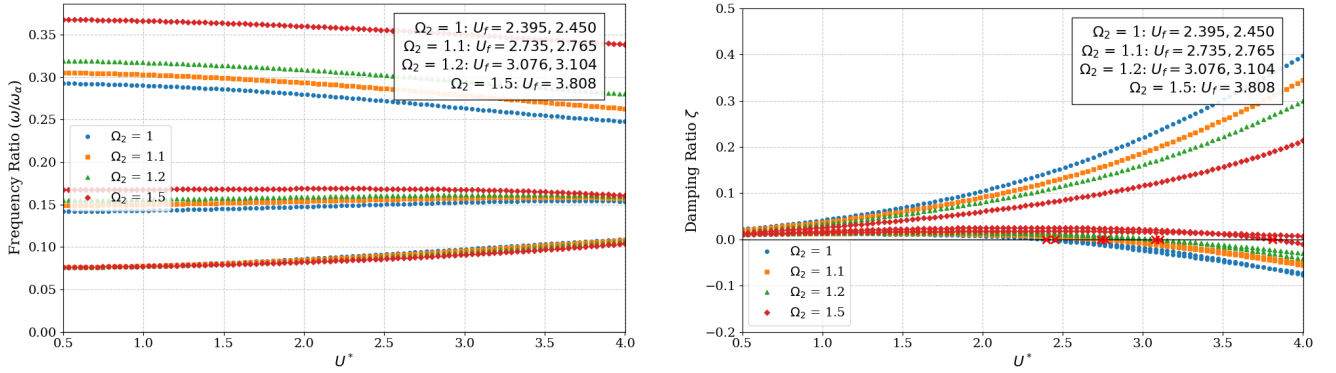
As illustrated in Figures D.6b and D.6c, there exists an optimal Ω_2 and c_β^* to achieve the above-mentioned dynamics. These correspond to $\Omega_2 = 1.1, 1.2$ and $c_\beta^* = 0.02$. Recall from the linear stability

analysis, these values were noticed to have the smallest difference between the two successive Hopf bifurcation airspeeds, confirming that this response is driven by a Hopf-Hopf bifurcation as documented by Dimitriadis [4] and discussed in Section 3.1.1.

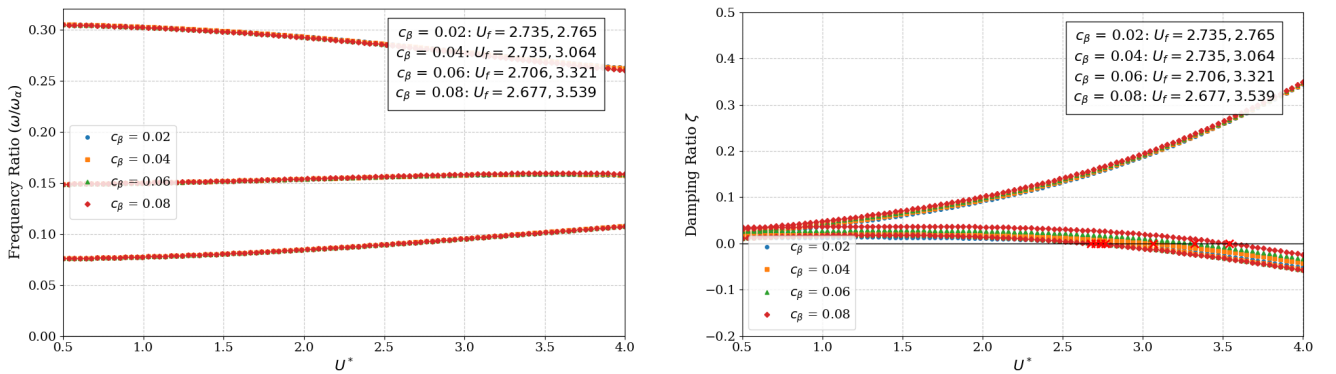
The nonlinear stiffness of the Flap-NES Γ_β (Figure D.6a) does not affect the previously described fundamental dynamics and NS bifurcation airspeed. However, the greater Γ_β , the lower the resulting LCO amplitudes. Increasing Γ_β from 10 to 25 decreases the Flap-NES amplitude by 34%, the pitch by 36% and the heave by 50%.



(a) Size of Flap-NES: $c_h = [0.2; 0.3; 0.4; 0.5]$, $x_\beta = [0.07; 0.05; 0.03; 0.018]$ and $r_\beta = [0.171; 0.132; 0.098; 0.069]$



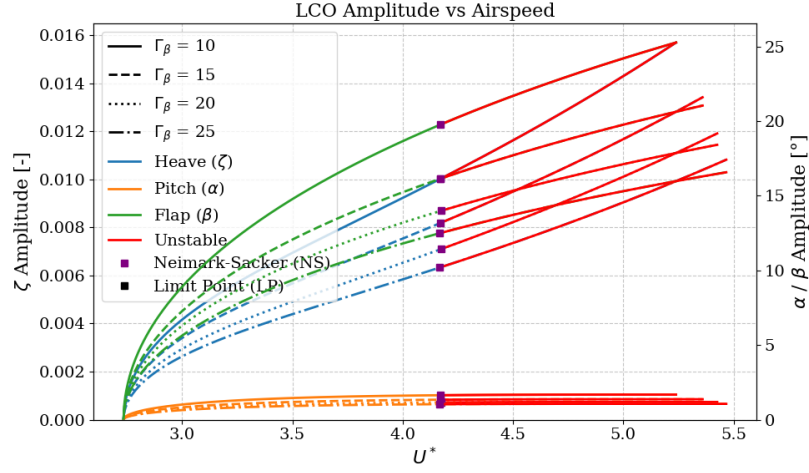
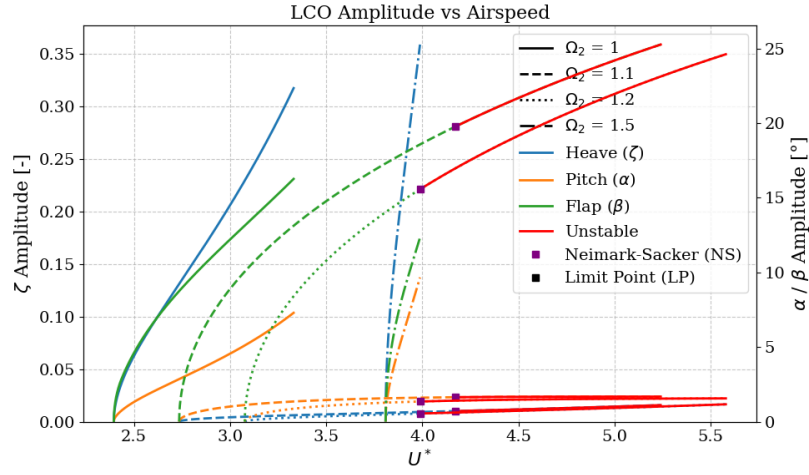
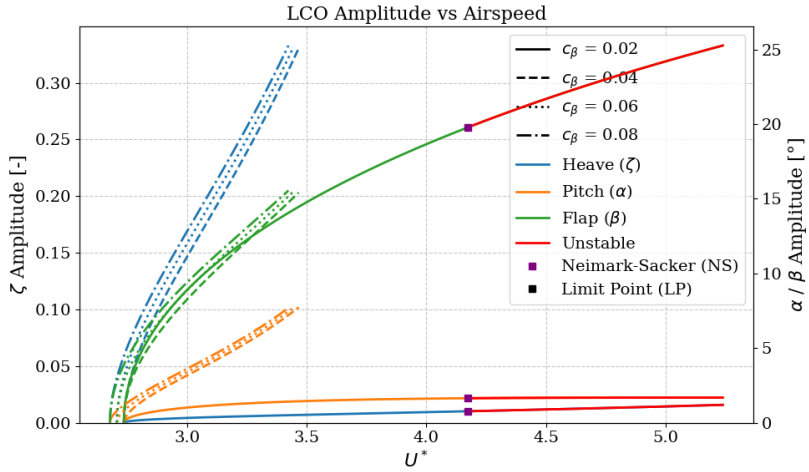
(b) Flap-NES linear stiffness: $\Omega_2 = [1; 1.1; 1.2; 1.5]$



(c) Flap-NES damping: $c_\beta^* = [0.02; 0.04; 0.06; 0.08]$

Figure D.5: Flutter diagram (parameter sweep): High linear stiffness

Fixed parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.5$, $x_\alpha = 0.25$, $x_\beta = 0.018$, $r_\alpha = 0.5$, $r_\beta = 0.069$, $\Omega_1 = 0.5$, $\Omega_2 = 1.1$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$.

(a) Flap-NES Cubic Stiffness: $\Gamma_\beta = [10; 15; 20; 25]$ (b) Flap/Pitch Frequency Ratio: $\Omega_2 = [1; 1.1; 1.2; 1.5]$ (c) Flap-NES damping: $c_\beta^* = [0.02; 0.04; 0.06; 0.08]$ **Figure D.6:** Bifurcation diagram (parameter sweep): High linear stiffness

Fixed parameters: $\mu = 100$, $a_h = -0.5$, $c_h = 0.5$, $x_\alpha = 0.25$, $x_\beta = 0.018$, $r_\alpha = 0.5$, $r_\beta = 0.069$, $\Omega_1 = 0.5$, $\Omega_2 = 1.1$, $\zeta_h = \zeta_\alpha = 0.002$, $c_\beta^* = 0.02$, $\kappa_h = \kappa_\alpha = 0$, $\Gamma_\beta = 10$.

Harmonic Balance parameters: $H = 5$, $N = 2^8$, $\Delta_{smin} = 10^{-4}$, $\Delta_{smax} = 0.05$, $\epsilon_{NR} = 10^{-9}$, $\epsilon_{stab} = 10^{-6}$, $dU^* = 0.001$, $dX = 0.02$.